

SAE / RDA exploratory LFP atlas

Physiological references and completed experiments

95 completed numbered states are shown, one per page. Each page contains the whole available LFP and the final 30 seconds.

- 0-3 Physiological references: wake, N2, N3, REM
- 5-33 Wake-derived slowing experiments
- 34-59 N3-centered perturbations
- 60-89 Targeted pathway and intrinsic-current tests
- 90-97 Thirty-second causal pathway pilots
- 98-106 Closed-loop gain screen

Representative-run convention

States 0-89 use seed 1. States 90-106 use the matched seed-2 series with a ramped -0.30 uA/cm^2 TC current. States 103 and 106 also have seeds 1 and 3, but seed 2 is shown here for direct comparison with the full causal screen.

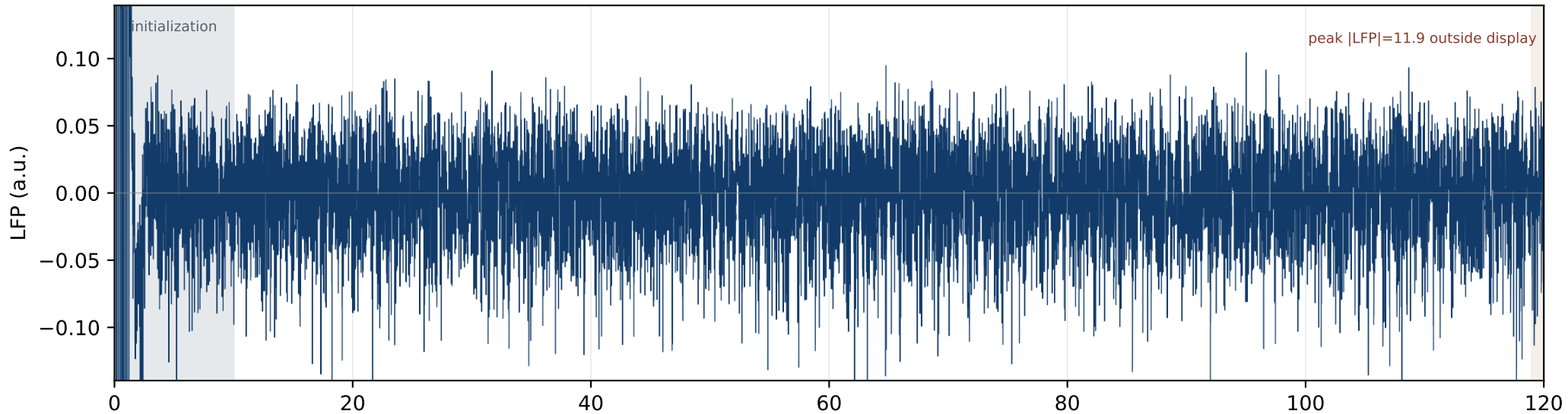
State 4 is reserved. Defined experiments not run: 63, 64, 65, 66, 67, 72, 73, 81, 83, 85, 87.

Y-scales are chosen independently for each panel. Use the printed RMS and p99.9 values for amplitude comparisons. This atlas is an exploratory visual record, not a declaration that any state is GRDA.

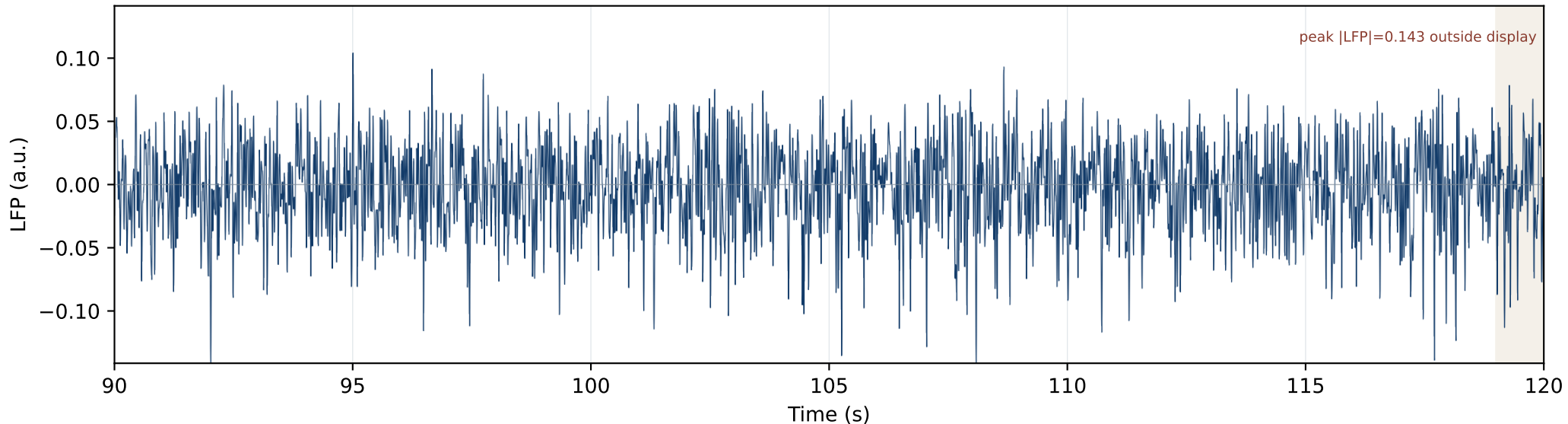
state_0

wake reference (Fink et al.)

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.14 a.u.



Final 30 seconds independent scale: ± 0.141 a.u.



10-119 s metrics: RMS 0.0343 a.u. | p99.9 $|LFP|$ 0.127 a.u. | delta 16.1% | theta 14.3% | centroid 13.54 Hz

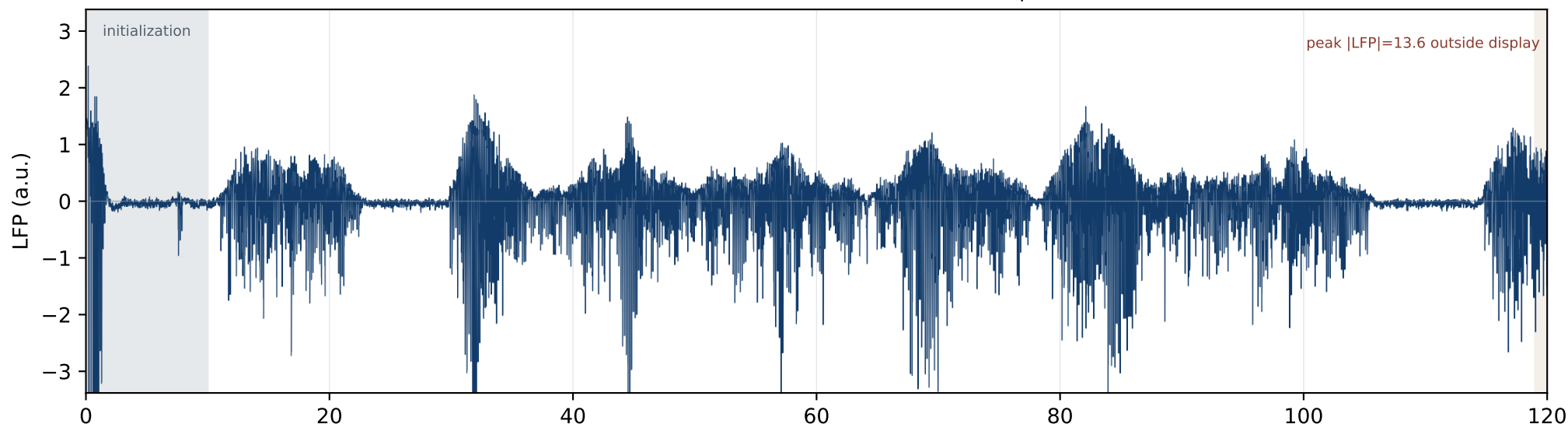
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state0_nhost=8.txt

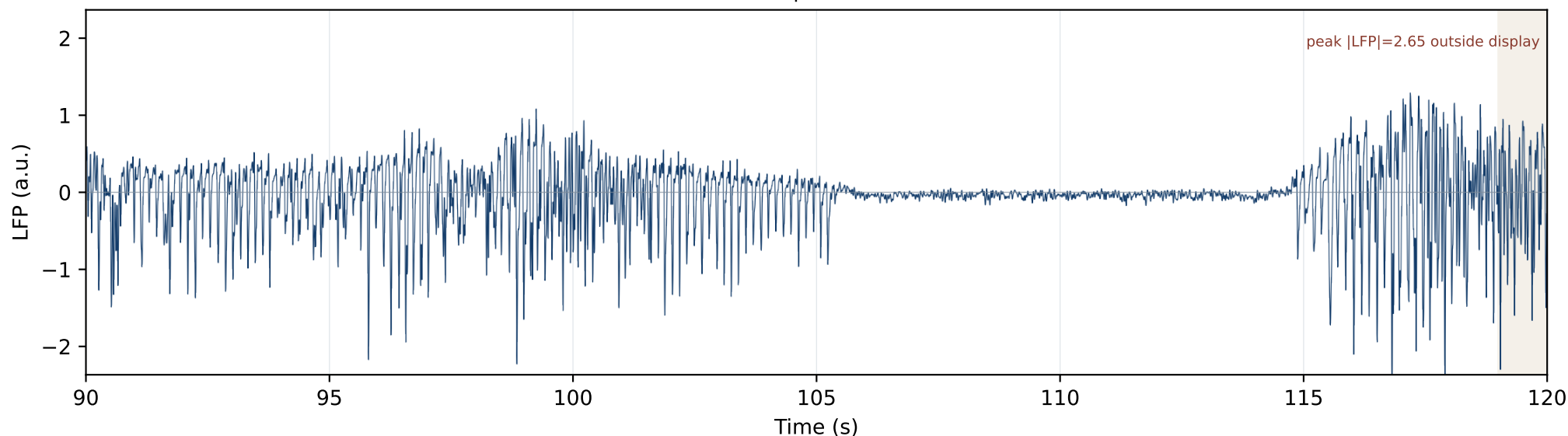
state_1

N2 reference (Fink et al.)

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.38 a.u.



Final 30 seconds independent scale: ± 2.37 a.u.



10-119 s metrics: RMS 0.537 a.u. | p99.9 |LFP| 3.07 a.u. | delta 6.2% | theta 56.1% | centroid 10.13 Hz

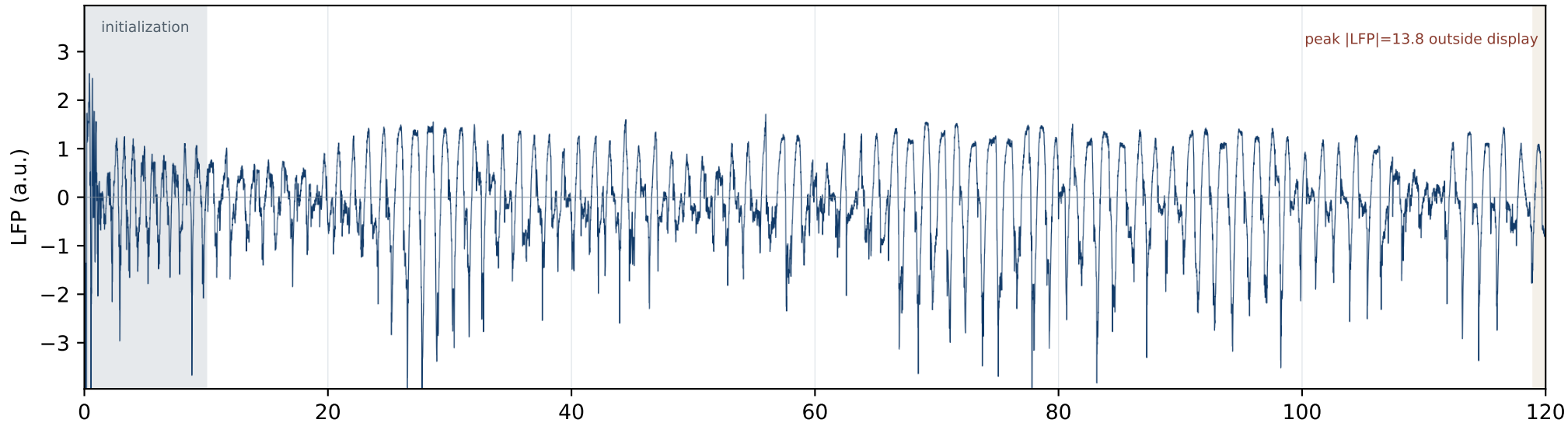
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state1_seed1_nhost=8.txt

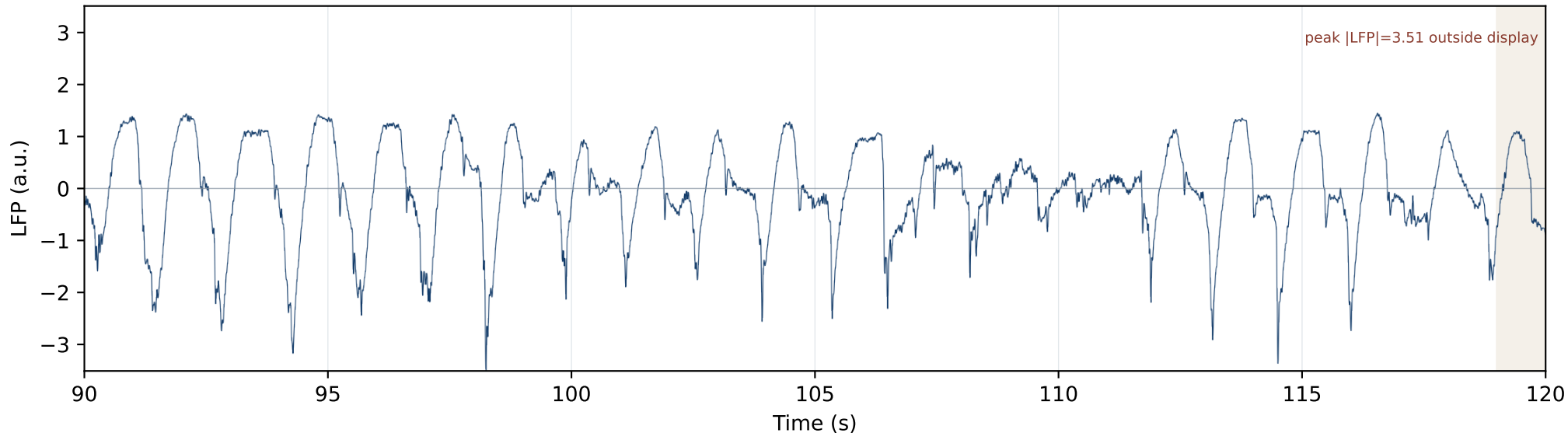
state_2

N3 reference (Fink et al.)

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.95 a.u.



Final 30 seconds independent scale: ± 3.51 a.u.



10-119 s metrics: RMS 0.911 a.u. | p99.9 $|LFP|$ 3.59 a.u. | delta 96.1% | theta 1.7% | centroid 1.34 Hz

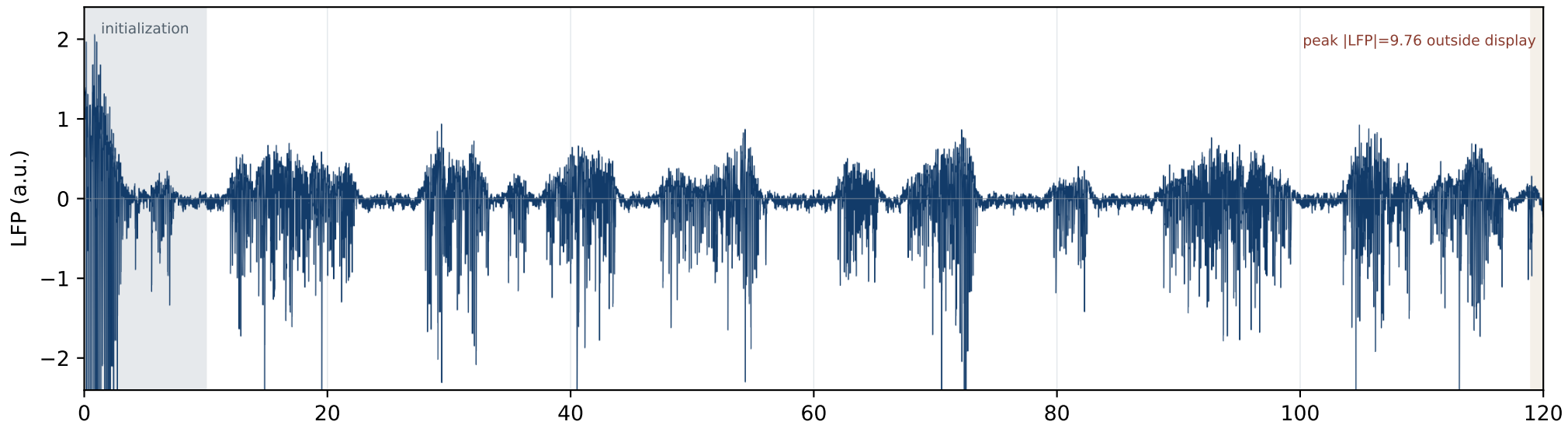
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state2_seed1_nhost=8.txt

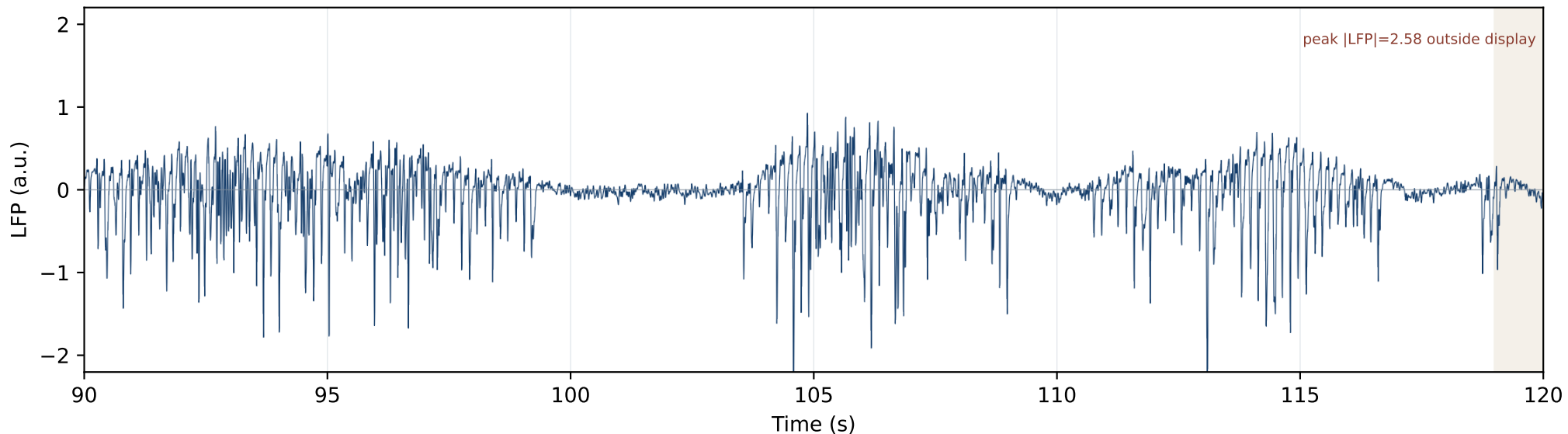
state_3

REM reference (Fink et al.)

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 2.4 a.u.



Final 30 seconds independent scale: ± 2.2 a.u.



10-119 s metrics: RMS 0.327 a.u. | p99.9 $|LFP|$ 2.18 a.u. | delta 12.6% | theta 45.4% | centroid 9.91 Hz

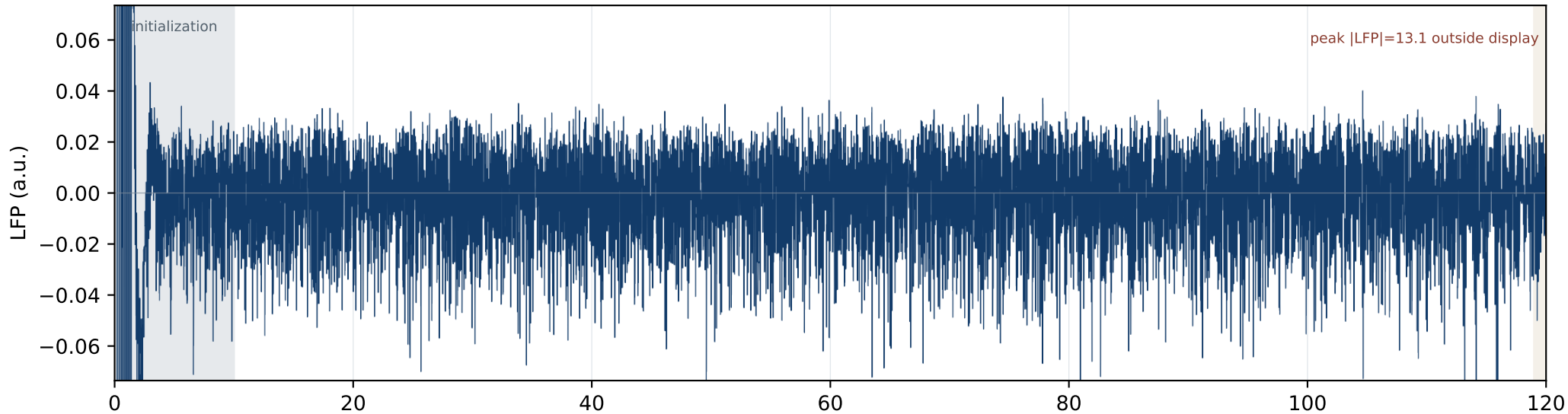
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state3_seed1_nhost=8.txt

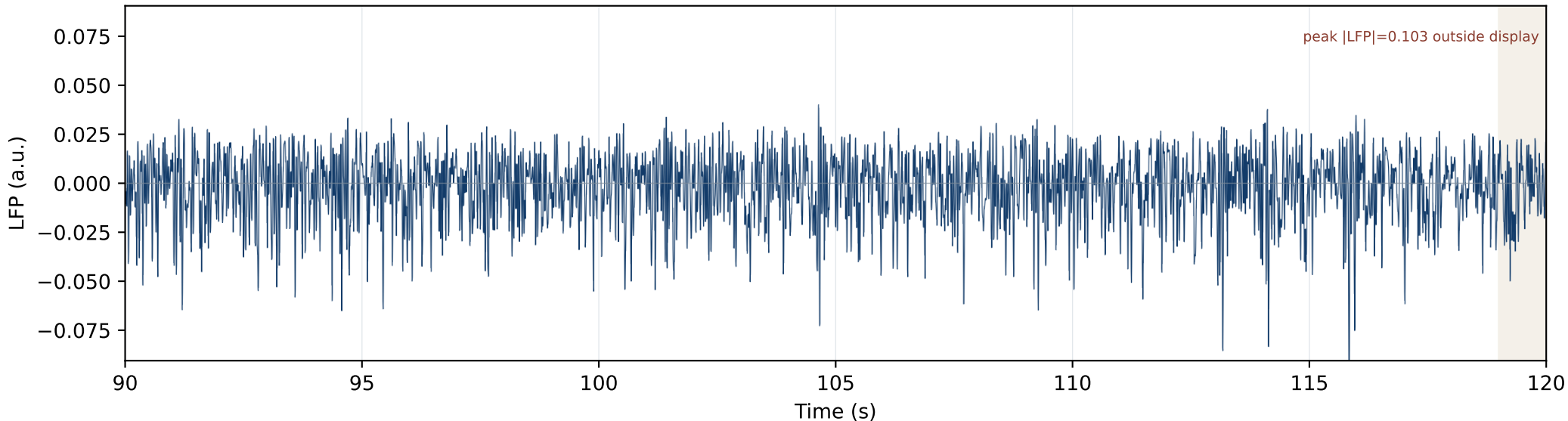
state_5

SAE-theta starter

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.0735 a.u.



Final 30 seconds independent scale: ± 0.0905 a.u.



10-119 s metrics: RMS 0.017 a.u. | p99.9 |LFP| 0.0669 a.u. | delta 10.6% | theta 12.6% | centroid 14.61 Hz

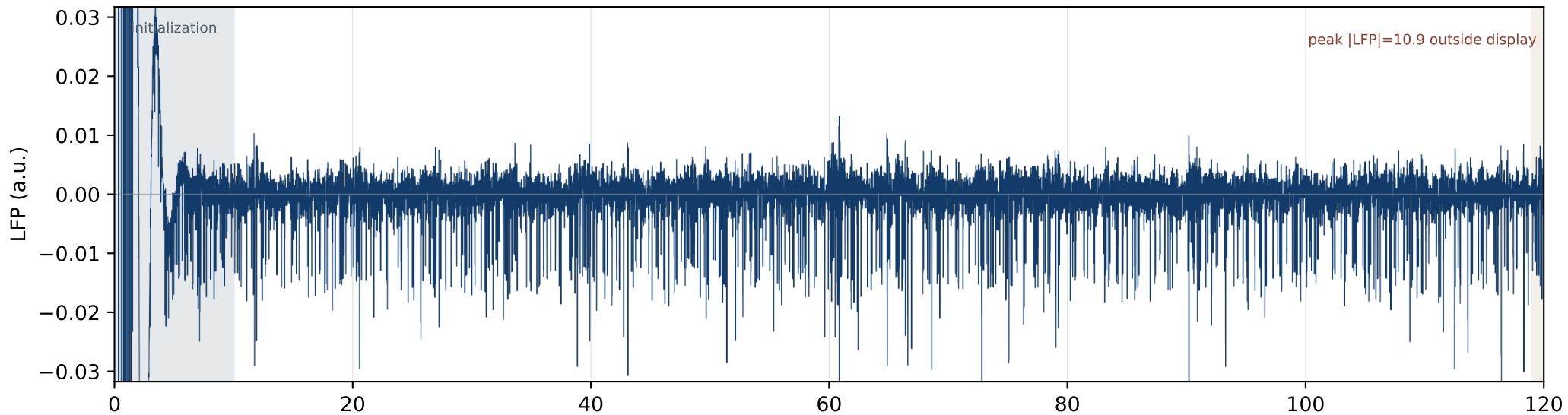
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state5_nhost=8.txt

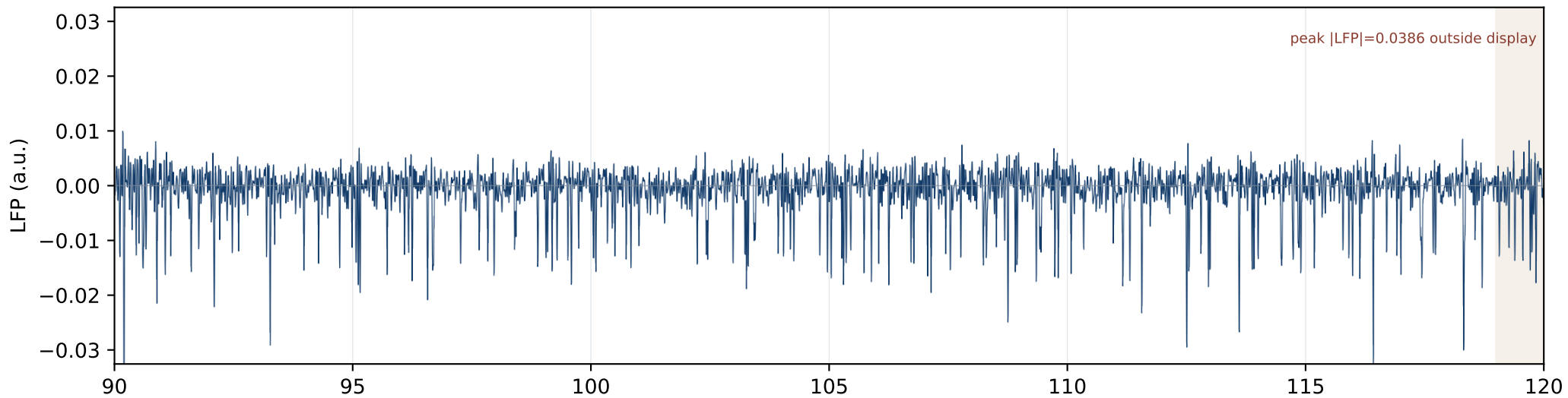
state_6

SAE-delta starter

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.0317 a.u.



Final 30 seconds independent scale: ± 0.0326 a.u.



10-119 s metrics: RMS 0.00461 a.u. | p99.9 |LFP| 0.0289 a.u. | delta 10.0% | theta 13.7% | centroid 14.48 Hz

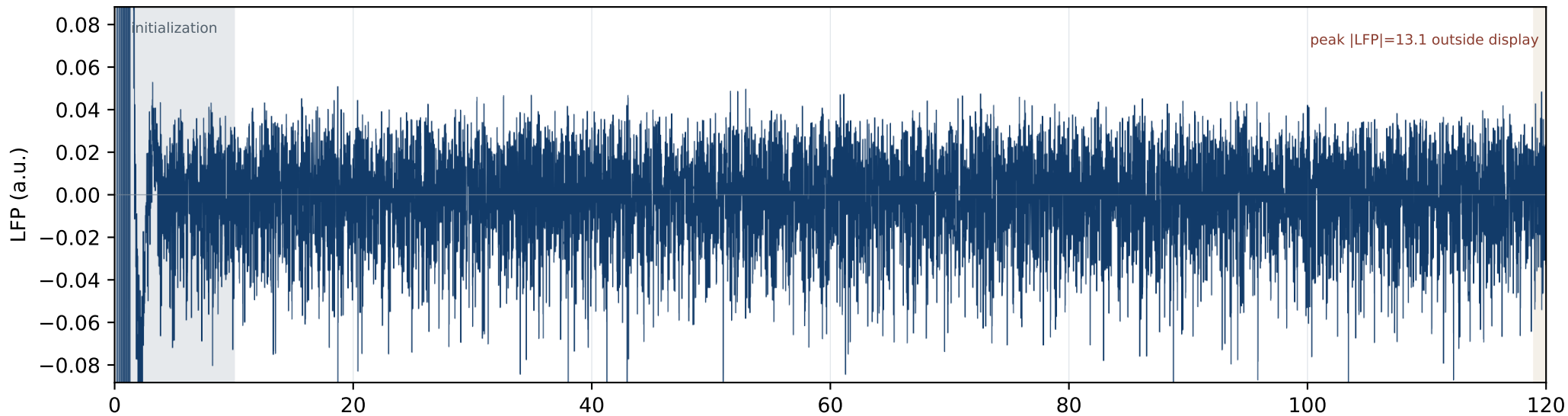
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state6_nhost=8.txt

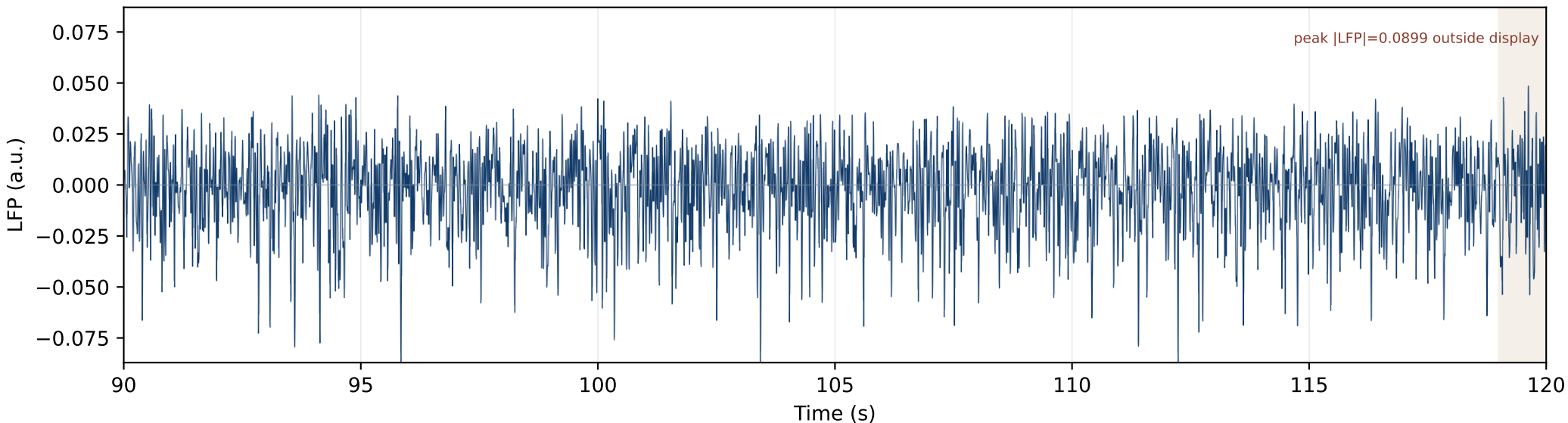
state_7

theta starter with recurrent cortical drive restored

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.0883 a.u.



Final 30 seconds independent scale: ± 0.0871 a.u.



10-119 s metrics: RMS 0.0208 a.u. | p99.9 |LFP| 0.0802 a.u. | delta 13.2% | theta 13.4% | centroid 14.00 Hz

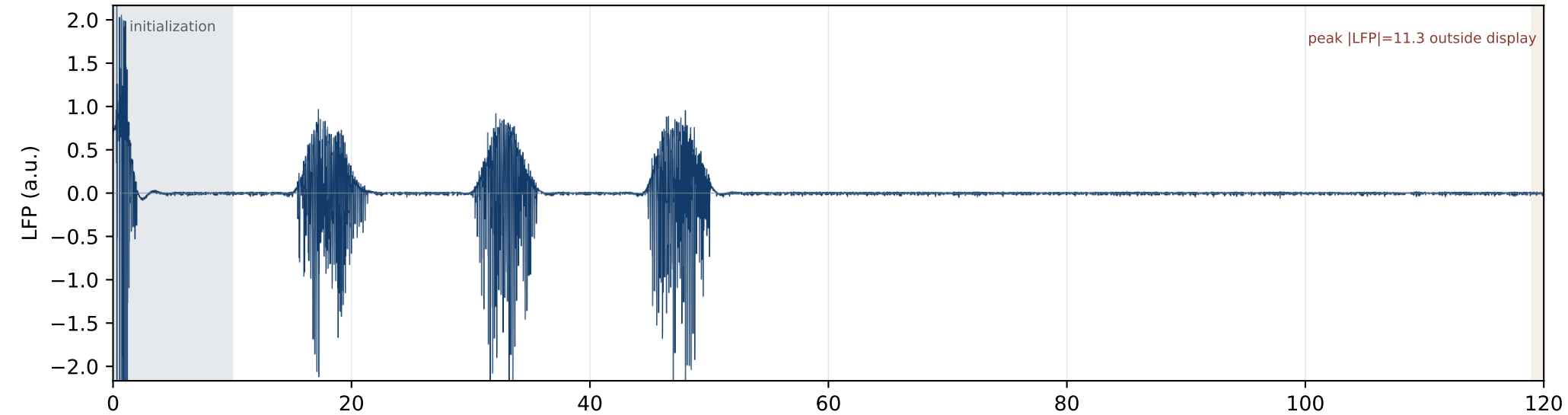
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state7_nhost=8.txt

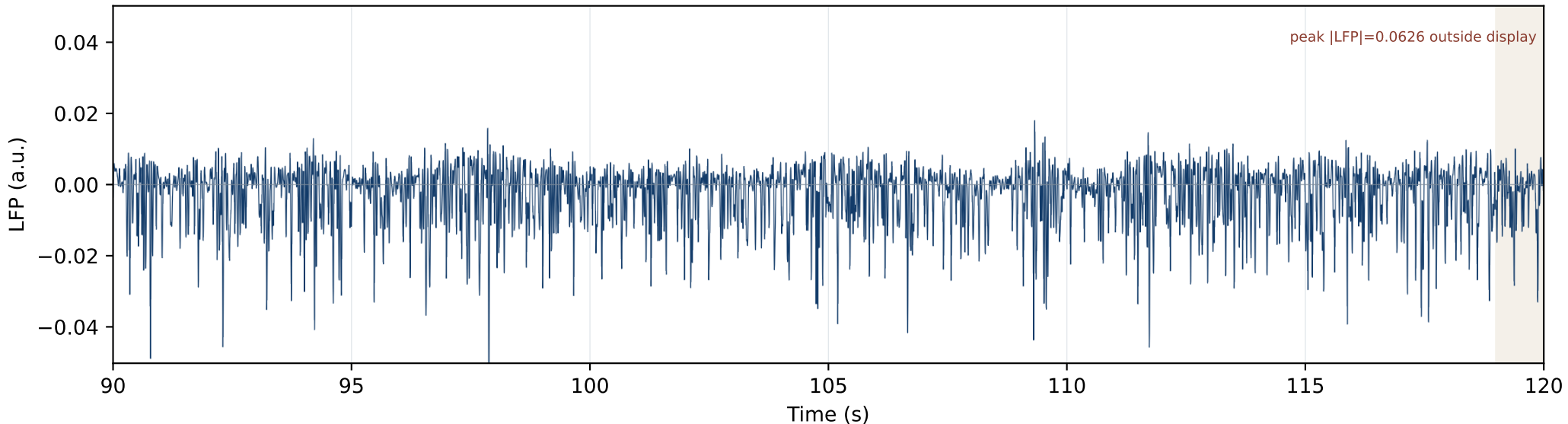
state_8

delta starter with recurrent cortical drive restored

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 2.17 a.u.



Final 30 seconds independent scale: ± 0.0502 a.u.



10-119 s metrics: RMS 0.213 a.u. | p99.9 $|LFP|$ 1.97 a.u. | delta 1.9% | theta 60.2% | centroid 10.07 Hz

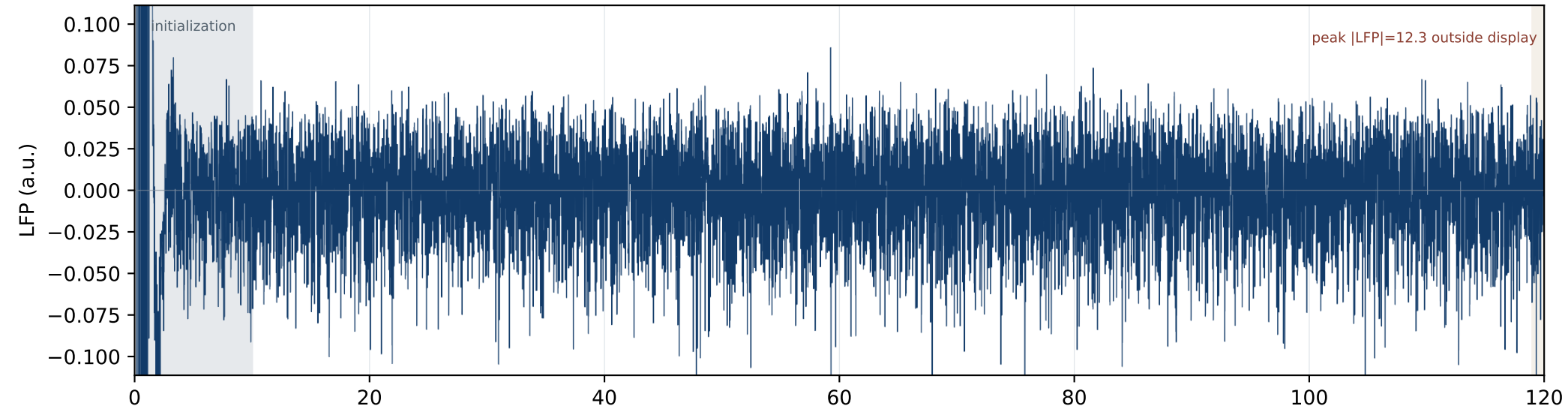
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state8_nhost=8.txt

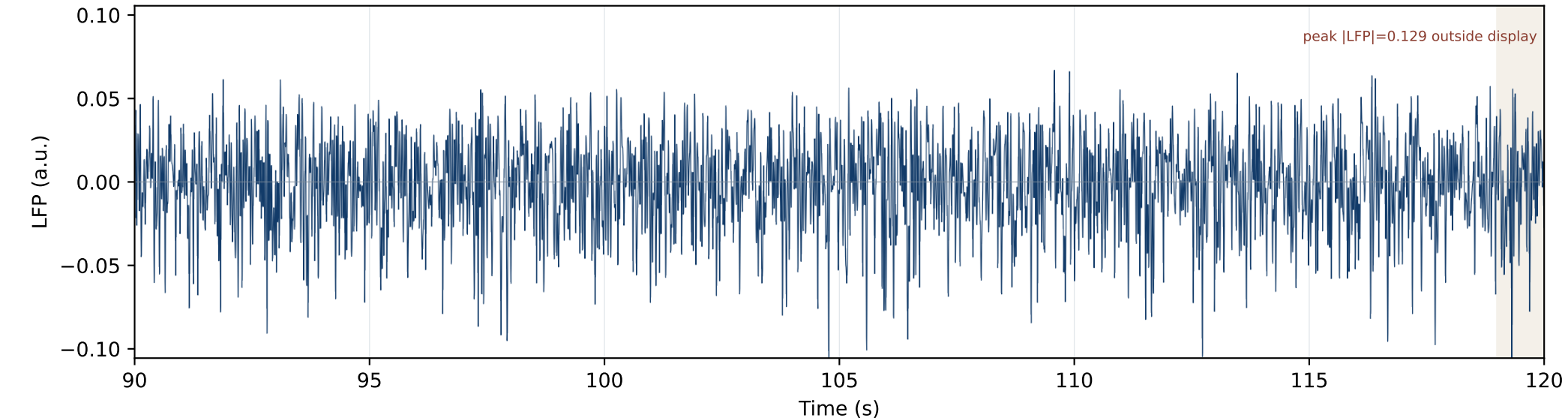
state_9

wakeful-slow N1-like bridge

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.111 a.u.



Final 30 seconds independent scale: ± 0.106 a.u.



10-119 s metrics: RMS 0.0272 a.u. | p99.9 $|LFP|$ 0.101 a.u. | delta 15.2% | theta 12.2% | centroid 13.76 Hz

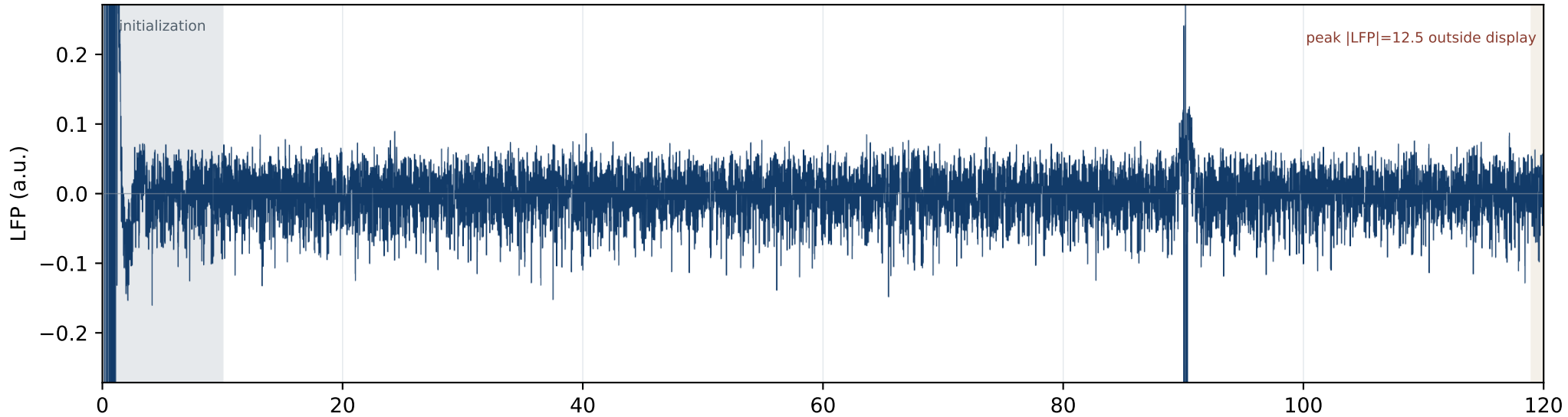
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state9_nhost=8.txt

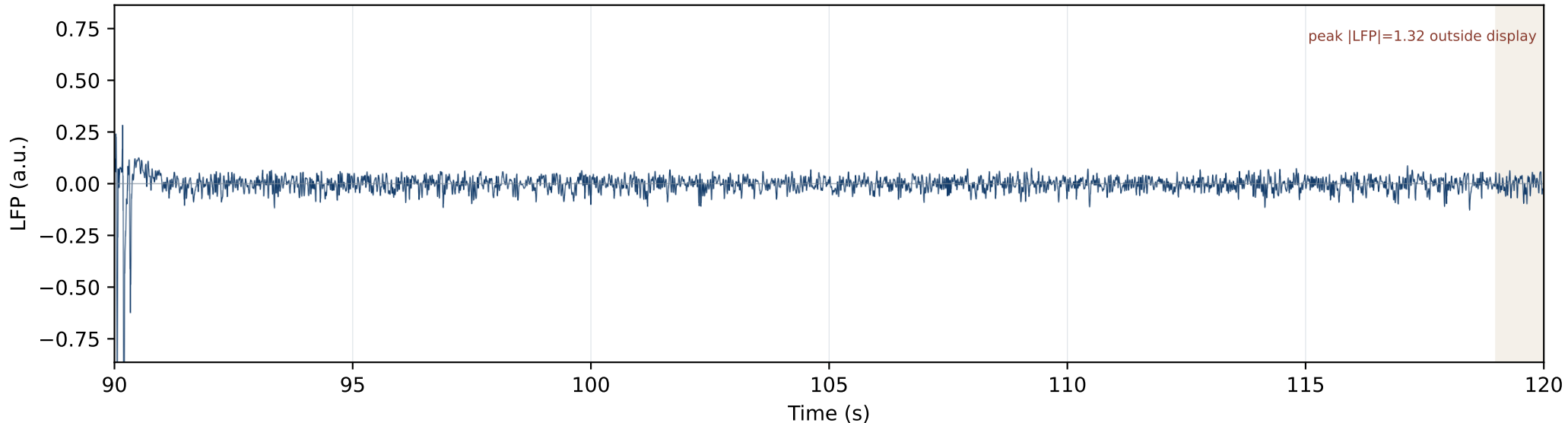
state_10

recurrence-edge probe

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.271 a.u.



Final 30 seconds independent scale: ± 0.864 a.u.



10-119 s metrics: RMS 0.0397 a.u. | p99.9 $|LFP|$ 0.247 a.u. | delta 16.9% | theta 20.6% | centroid 12.50 Hz

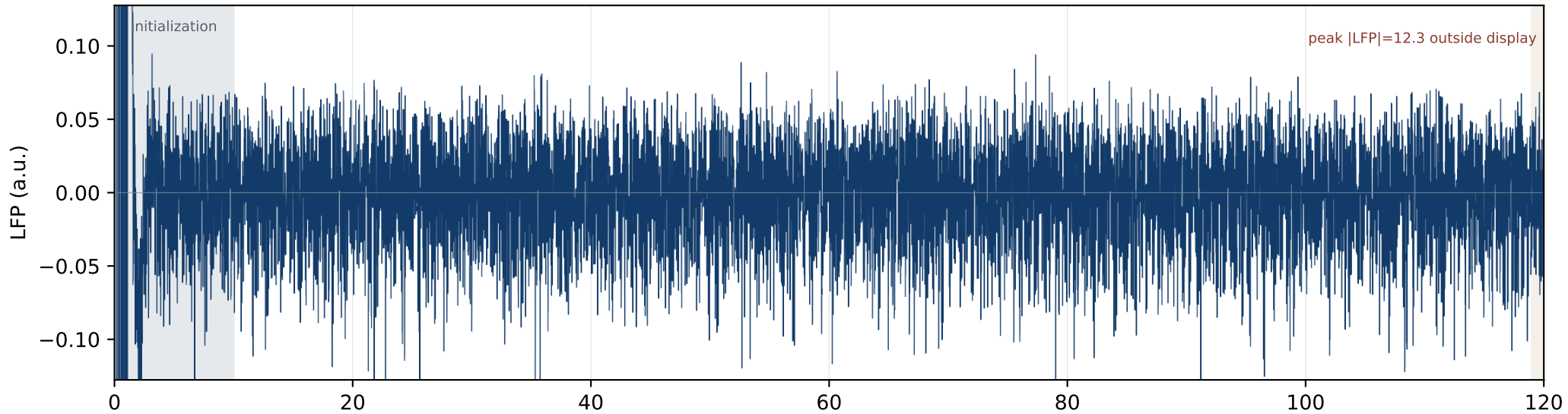
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state10_nhost=8.txt

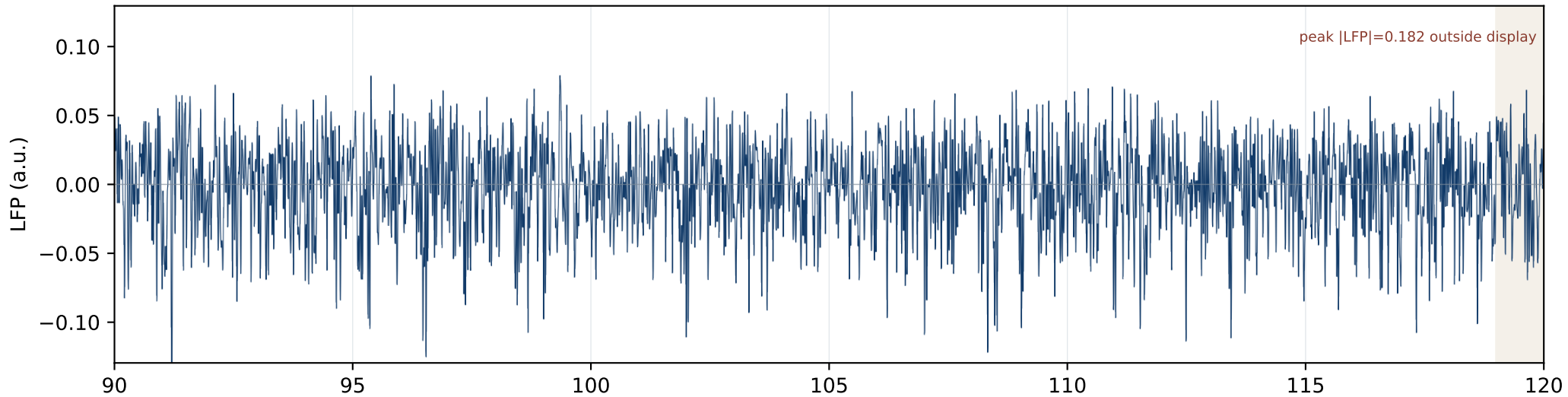
state_11

cortical E/I rebalance

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.128 a.u.



Final 30 seconds independent scale: ± 0.13 a.u.



10-119 s metrics: RMS 0.0321 a.u. | p99.9 |LFP| 0.116 a.u. | delta 15.7% | theta 14.4% | centroid 13.49 Hz

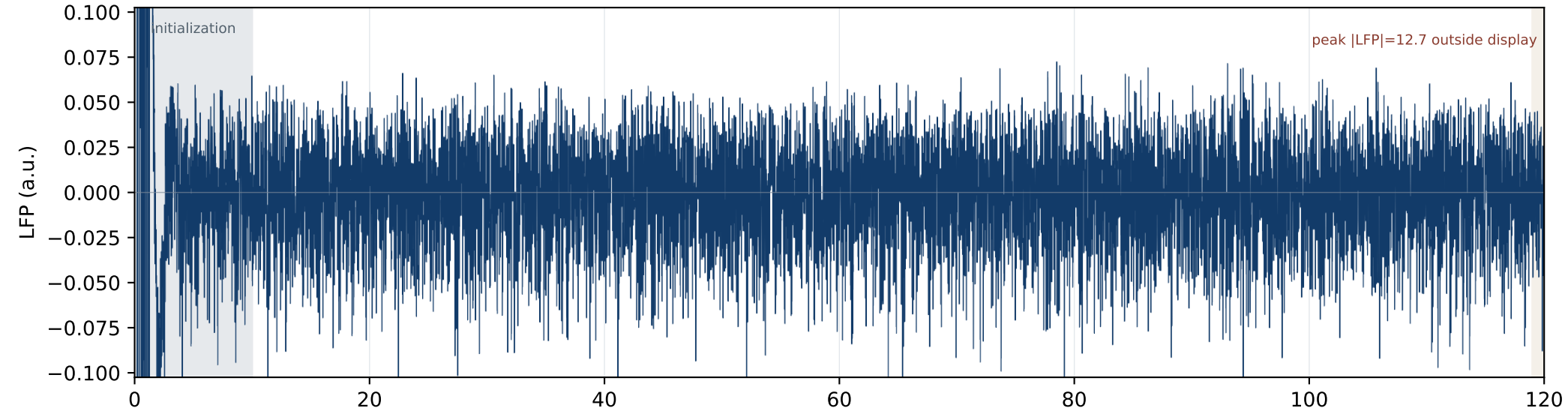
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state11_nhost=8.txt

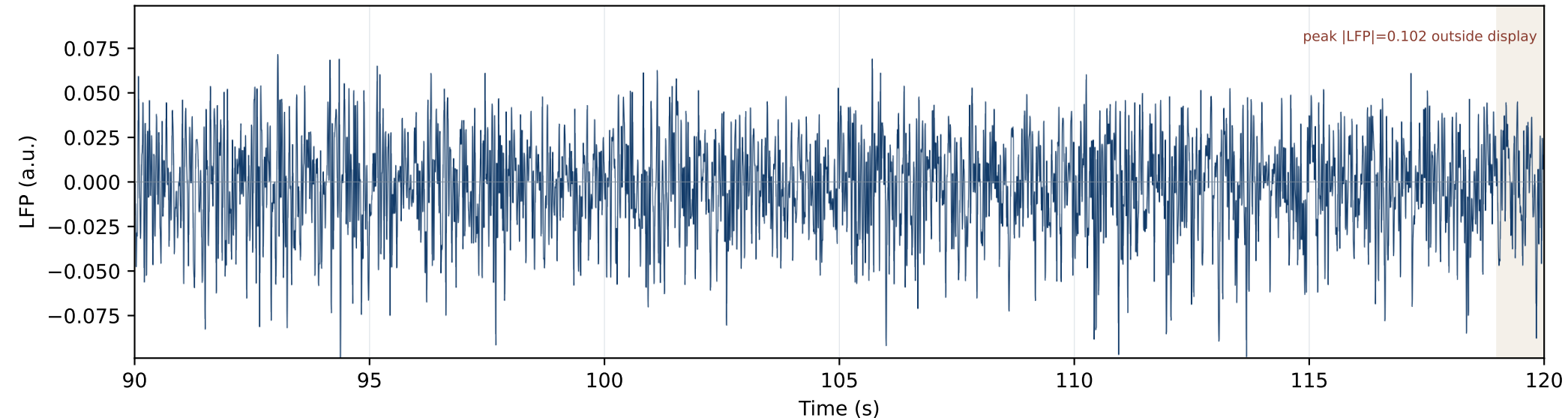
state_12

thalamic recruitment probe

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.102 a.u.



Final 30 seconds independent scale: ± 0.0989 a.u.



10-119 s metrics: RMS 0.0265 a.u. | p99.9 |LFP| 0.0932 a.u. | delta 14.1% | theta 14.5% | centroid 13.88 Hz

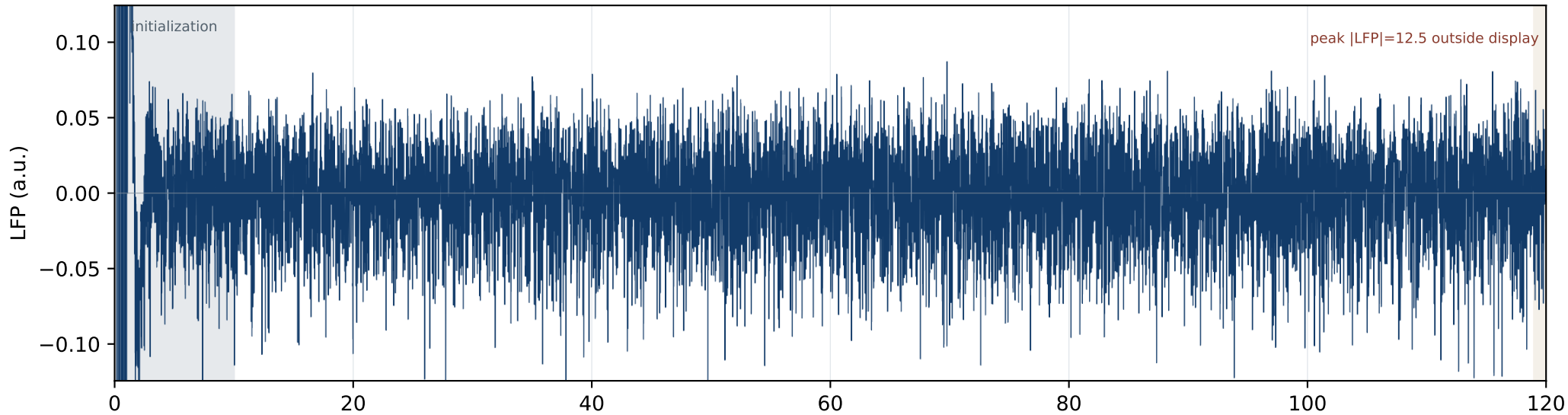
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state12_nhost=8.txt

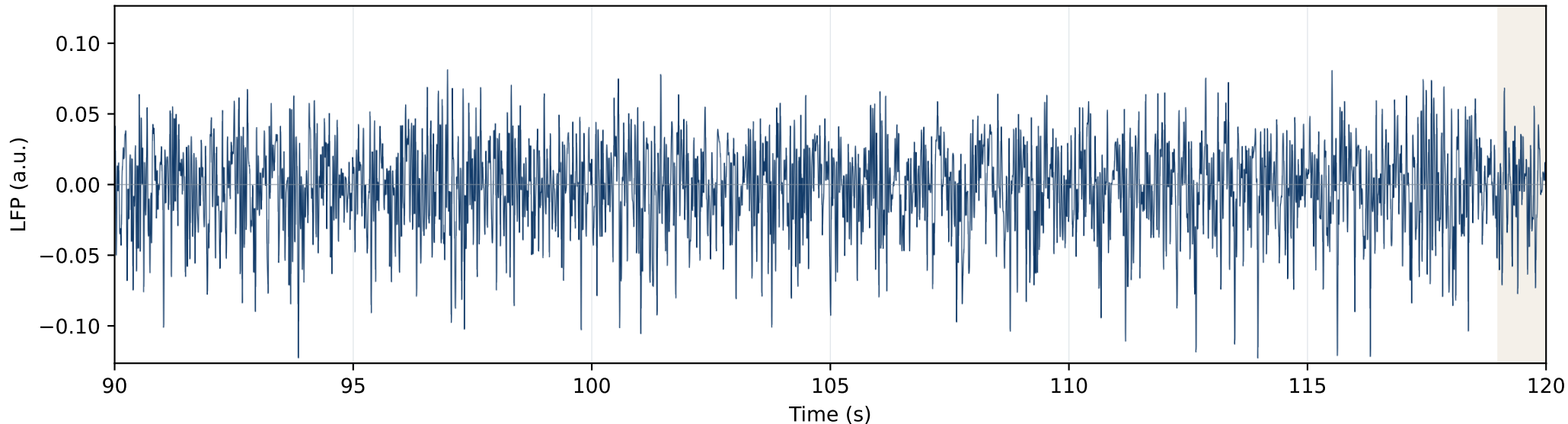
state_13

cortical GABA-A decay 1.5x

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.124 a.u.



Final 30 seconds independent scale: ± 0.126 a.u.



10-119 s metrics: RMS 0.0306 a.u. | p99.9 |LFP| 0.113 a.u. | delta 15.1% | theta 14.5% | centroid 13.58 Hz

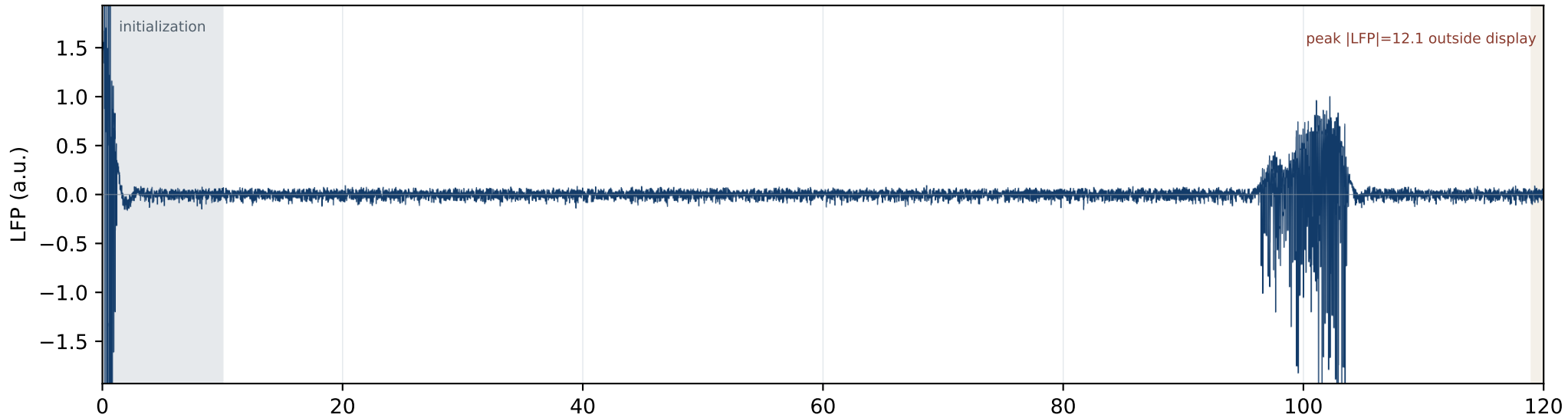
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state13_seed1_nhost=8.txt

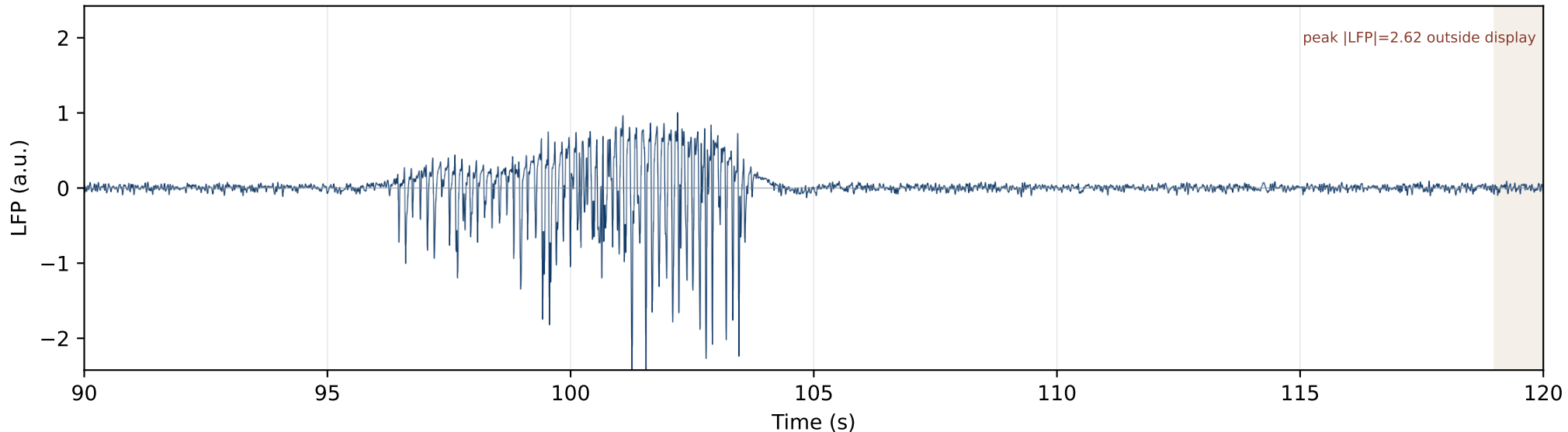
state_14

thalamic GABA-A decay 1.5x

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 1.93 a.u.



Final 30 seconds independent scale: ± 2.42 a.u.



10-119 s metrics: RMS 0.151 a.u. | p99.9 |LFP| 1.76 a.u. | delta 7.0% | theta 58.0% | centroid 10.14 Hz

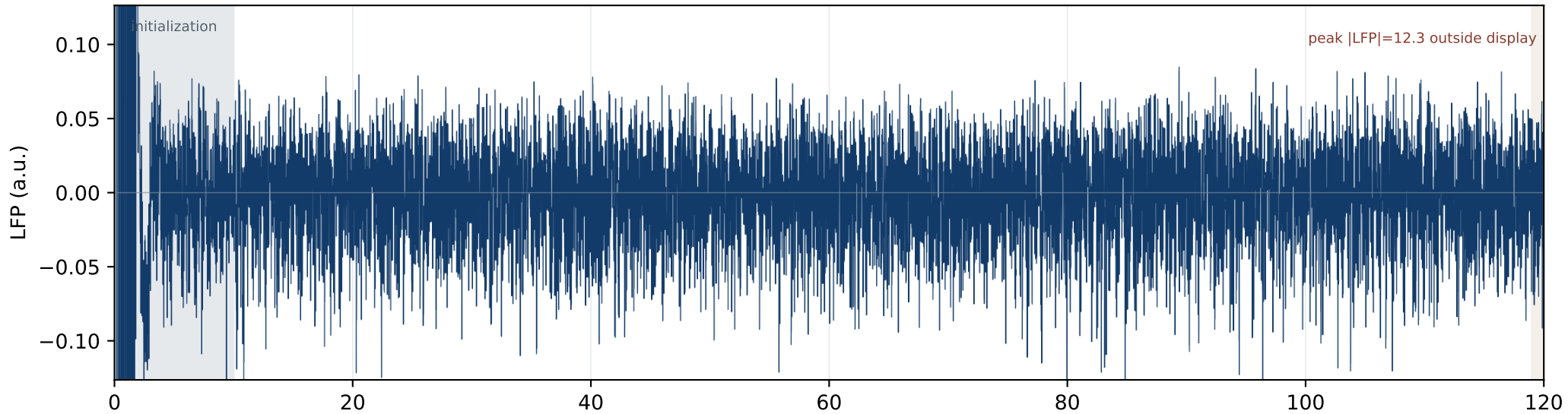
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Source: lfp_state14_seed1_nhost=8.txt

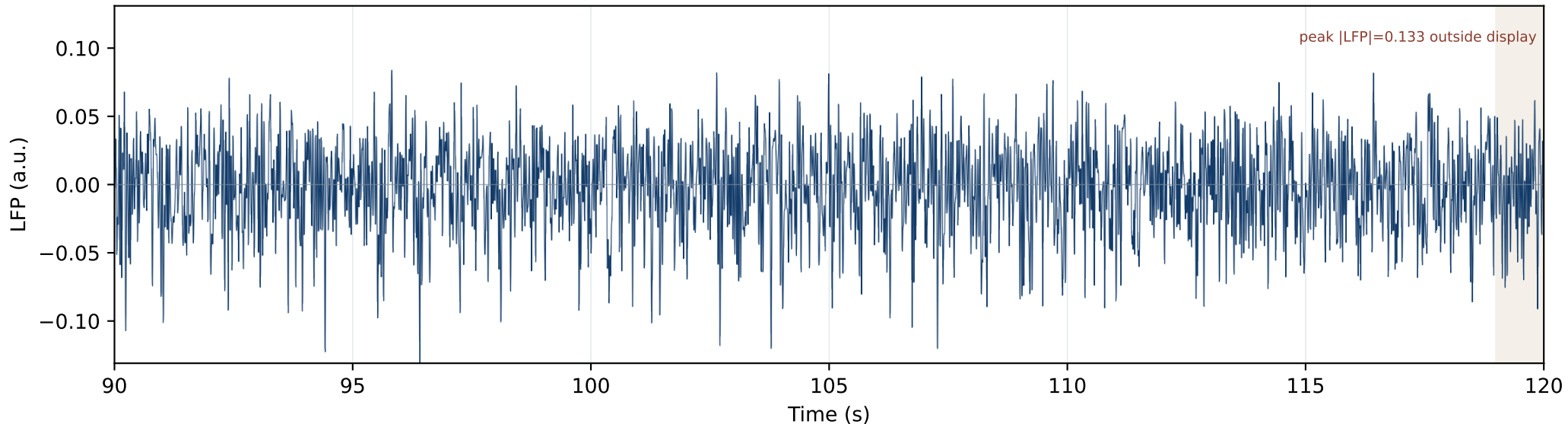
state_15

combined GABA-A decay 1.5x

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: +/-0.126 a.u.



Final 30 seconds independent scale: +/-0.131 a.u.



10-119 s metrics: RMS 0.031 a.u. | p99.9 |LFP| 0.115 a.u. | delta 14.8% | theta 15.1% | centroid 13.75 Hz

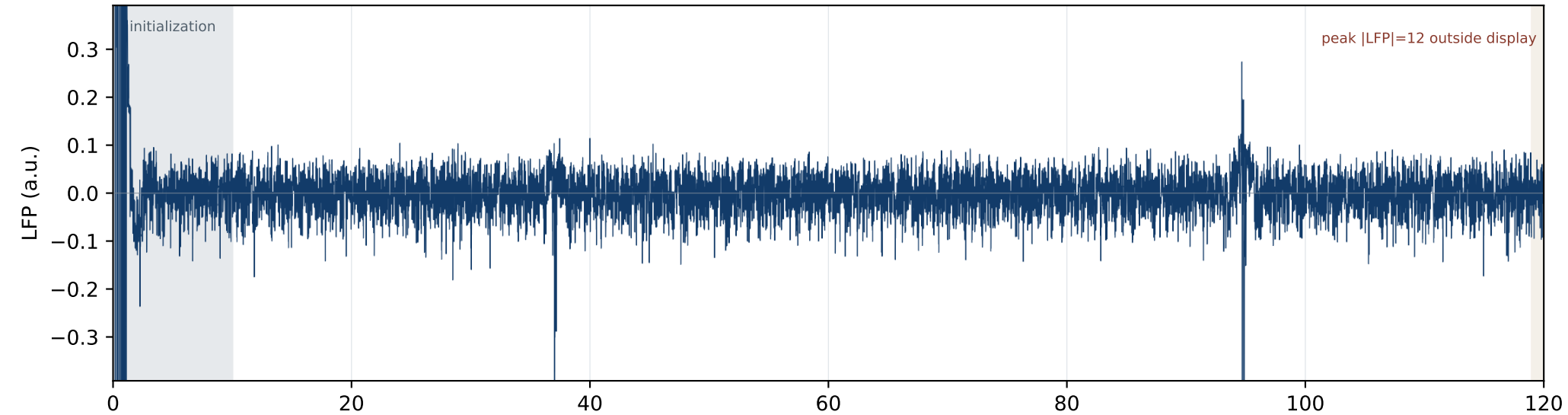
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state15_seed1_nhost=8.txt

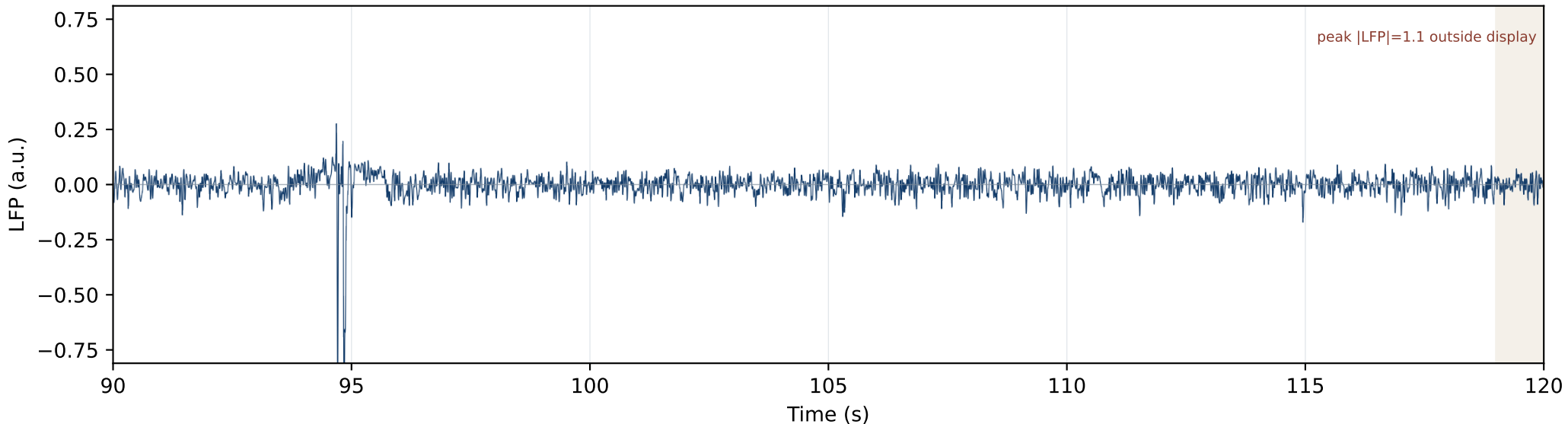
state_16

balanced cortical wake-to-N2 trajectory 50%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.391 a.u.



Final 30 seconds independent scale: ± 0.811 a.u.



10-119 s metrics: RMS 0.0449 a.u. | p99.9 |LFP| 0.356 a.u. | delta 23.0% | theta 21.2% | centroid 11.37 Hz

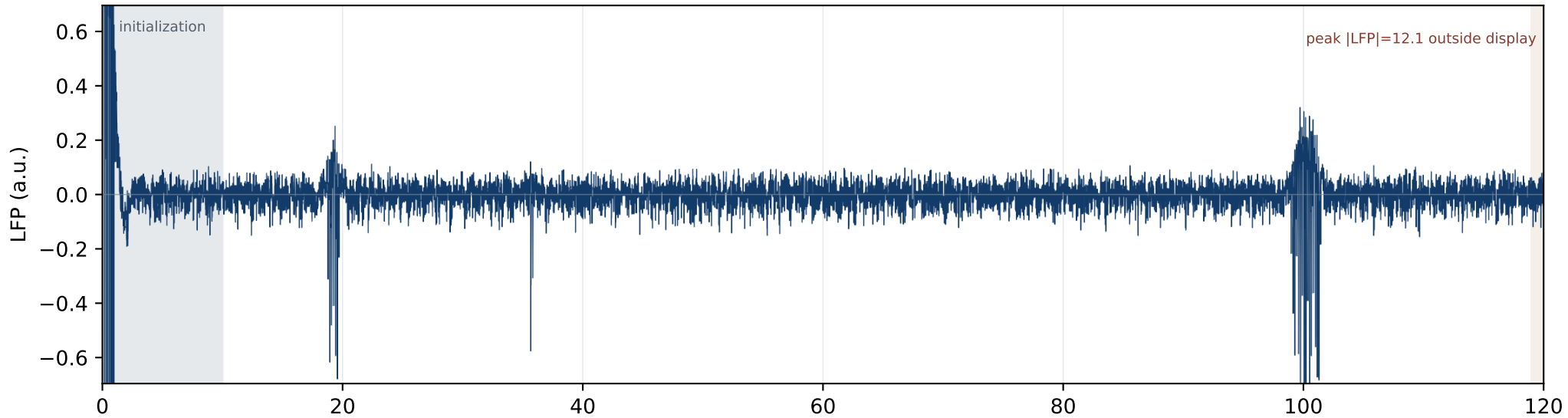
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state16_seed1_nhost=8.txt

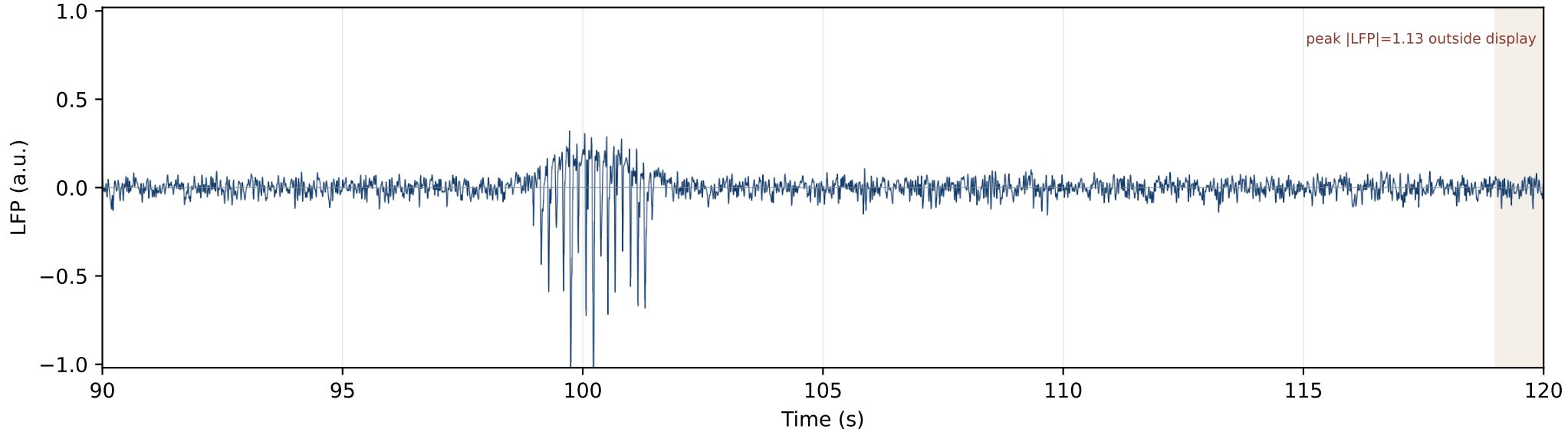
state_17

balanced cortical wake-to-N2 trajectory 62.5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.696 a.u.



Final 30 seconds independent scale: ± 1.02 a.u.



10-119 s metrics: RMS 0.0587 a.u. | p99.9 $|LFP|$ 0.632 a.u. | delta 15.0% | theta 33.7% | centroid 10.98 Hz

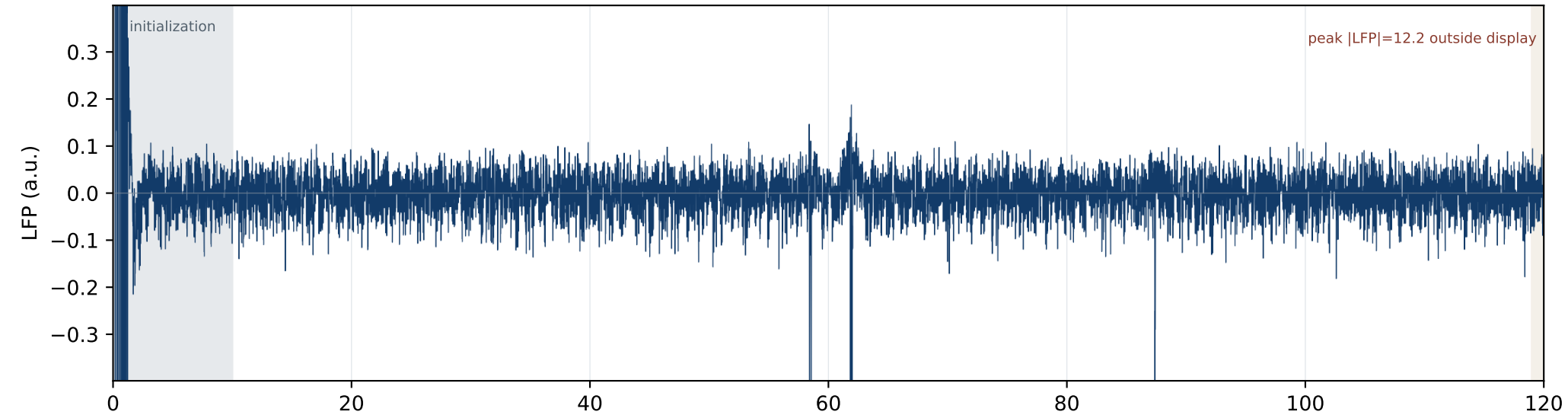
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Source: lfp_state17_seed1_nhost=8.txt

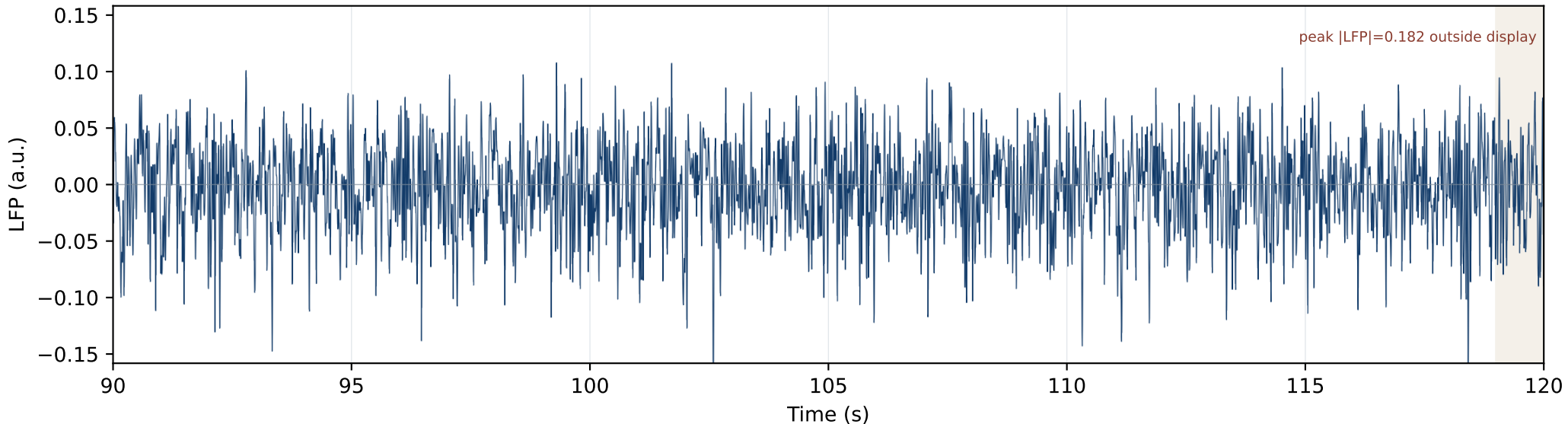
state_18

balanced cortical wake-to-N2 trajectory 75%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.399 a.u.



Final 30 seconds independent scale: ± 0.158 a.u.



10-119 s metrics: RMS 0.0456 a.u. | p99.9 |LFP| 0.363 a.u. | delta 23.2% | theta 20.3% | centroid 11.24 Hz

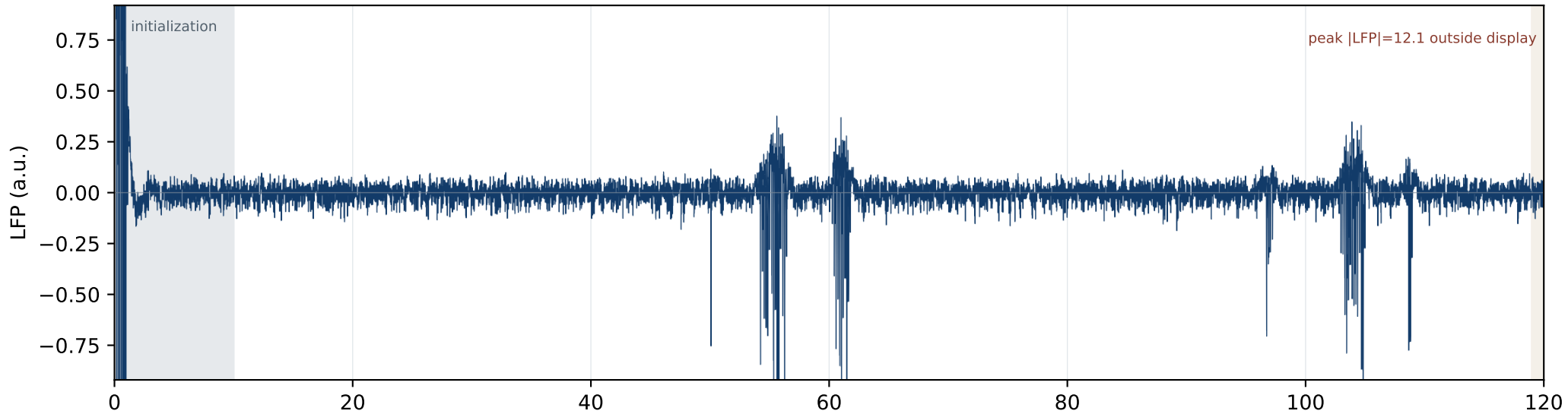
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state18_seed1_nhost=8.txt

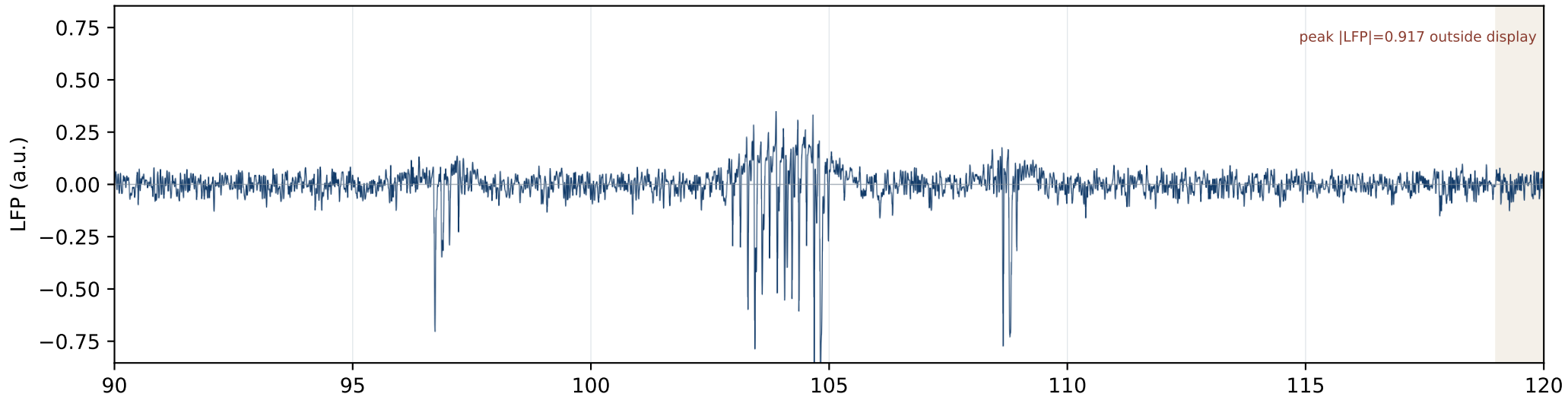
state_19

RE->TC GABA-B +20%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: +/-0.92 a.u.



Final 30 seconds independent scale: +/-0.854 a.u.



10-119 s metrics: RMS 0.076 a.u. | p99.9 |LFP| 0.836 a.u. | delta 13.8% | theta 35.5% | centroid 11.09 Hz

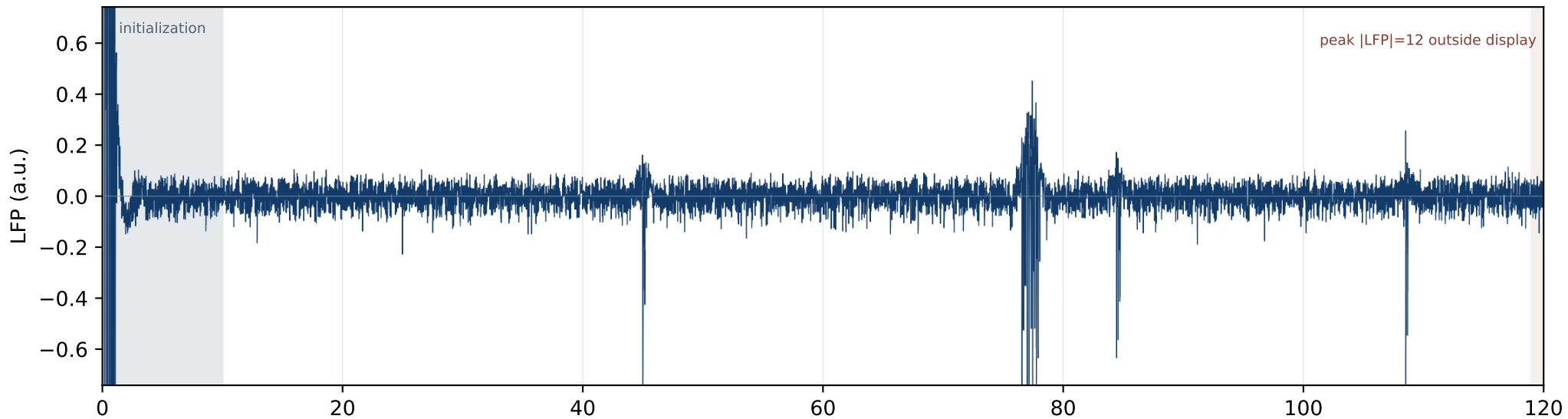
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state19_seed1_nhost=8.txt

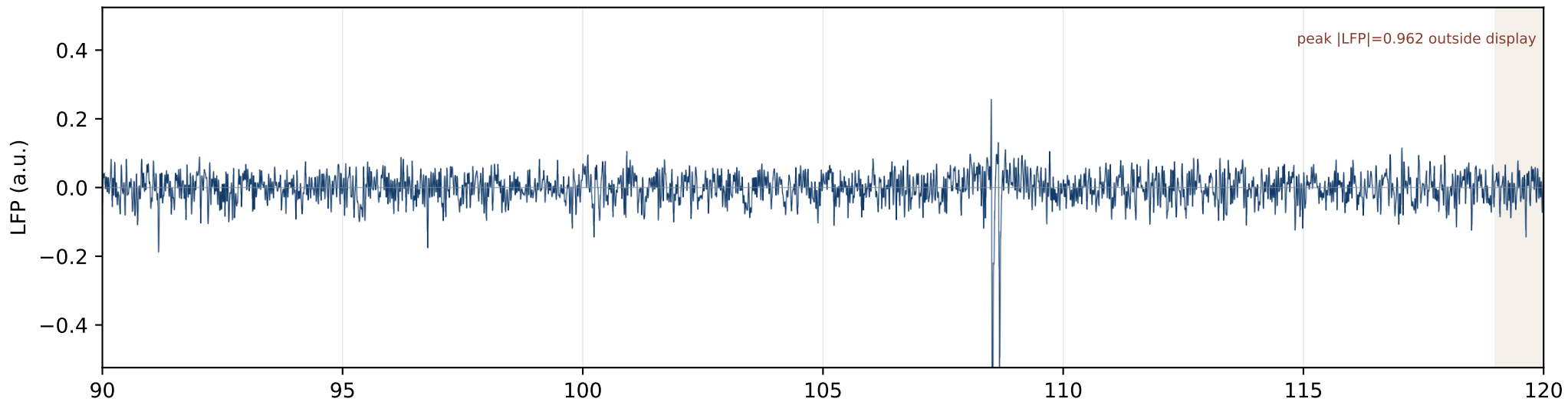
state_20

RE->RE GABA-A -25%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.742 a.u.



Final 30 seconds independent scale: ± 0.524 a.u.



10-119 s metrics: RMS 0.0592 a.u. | p99.9 |LFP| 0.674 a.u. | delta 16.8% | theta 31.8% | centroid 11.02 Hz

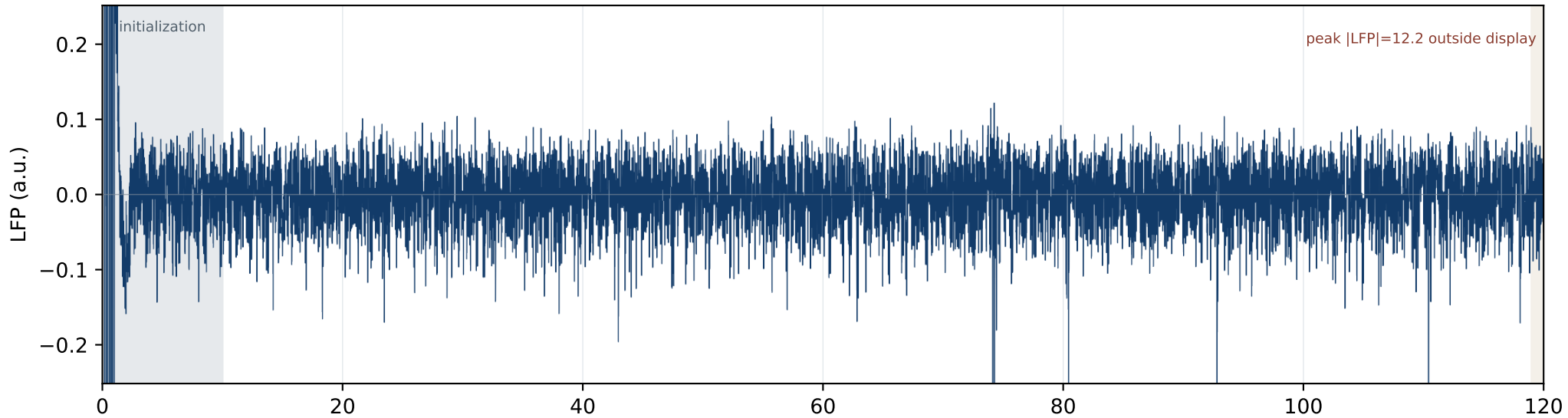
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state20_seed1_nhost=8.txt

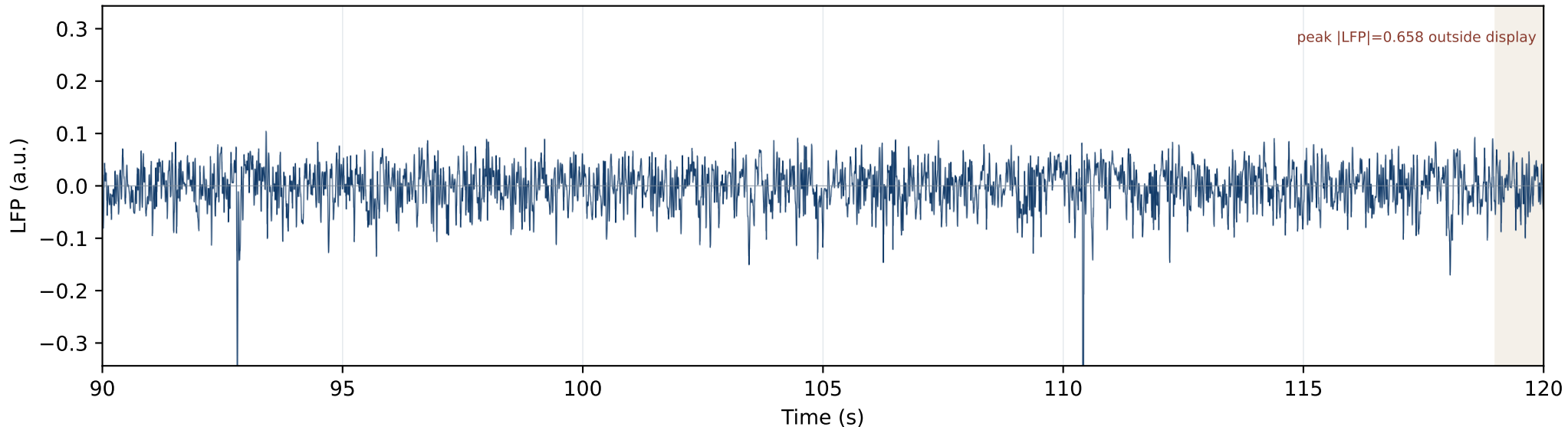
state_21

RE->TC GABA-A -20%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.252 a.u.



Final 30 seconds independent scale: ± 0.344 a.u.



10-119 s metrics: RMS 0.0406 a.u. | p99.9 |LFP| 0.229 a.u. | delta 18.9% | theta 16.4% | centroid 12.64 Hz

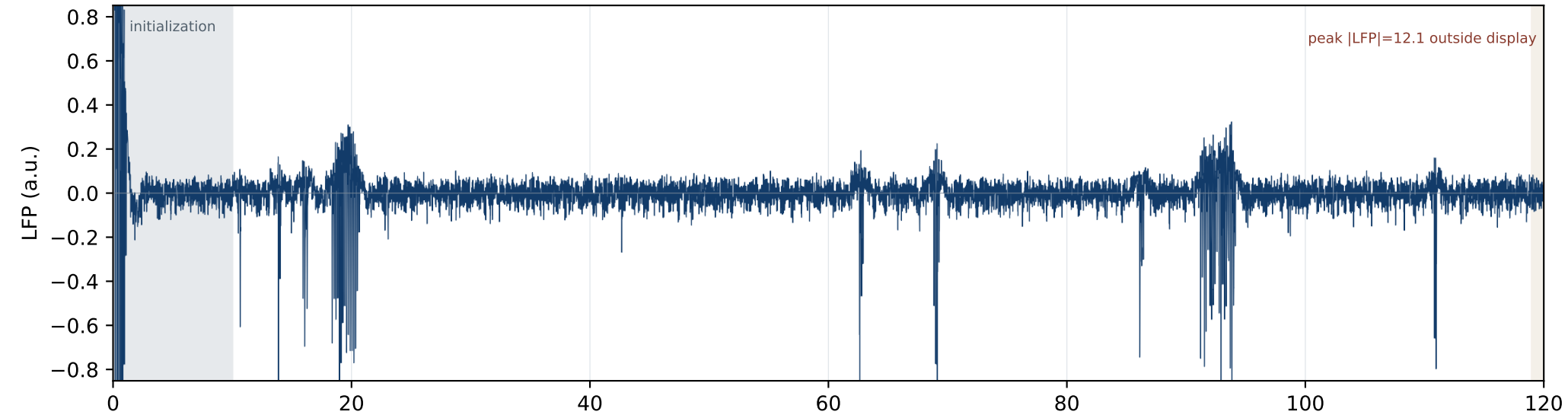
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state21_seed1_nhost=8.txt

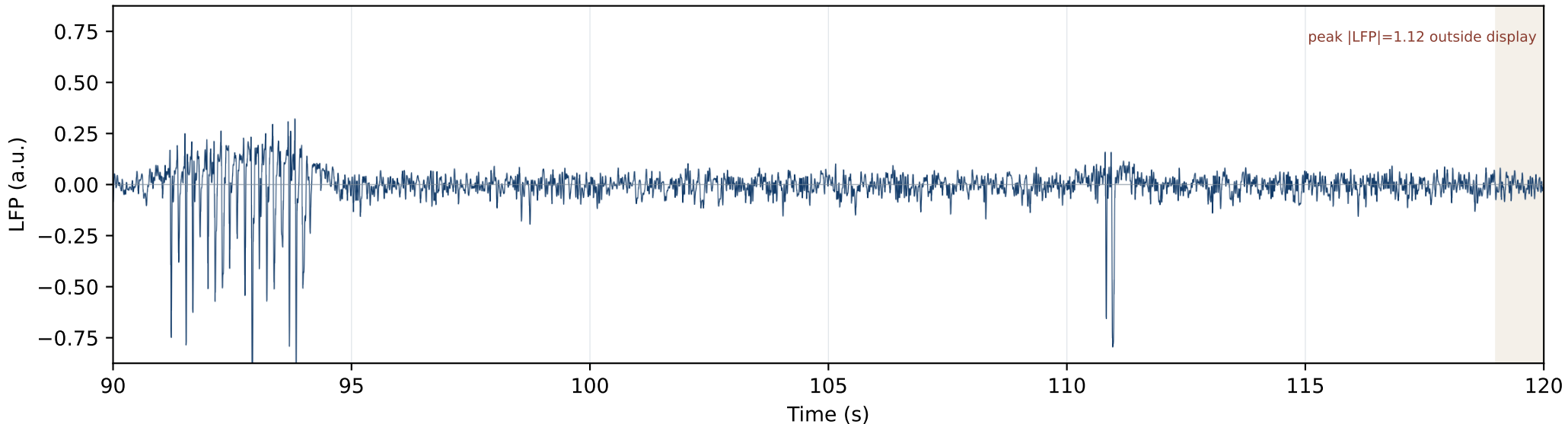
state_22

PYR->RE AMPA +10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: +/-0.852 a.u.



Final 30 seconds independent scale: +/-0.875 a.u.



10-119 s metrics: RMS 0.0771 a.u. | p99.9 |LFP| 0.774 a.u. | delta 13.7% | theta 37.0% | centroid 10.98 Hz

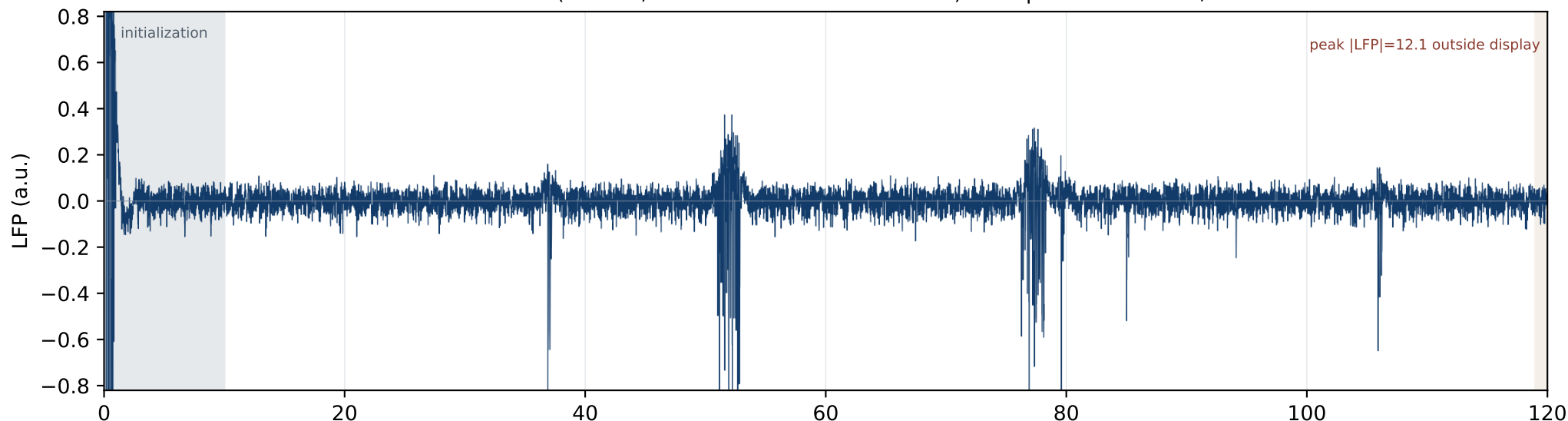
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state22_seed1_nhost=8.txt

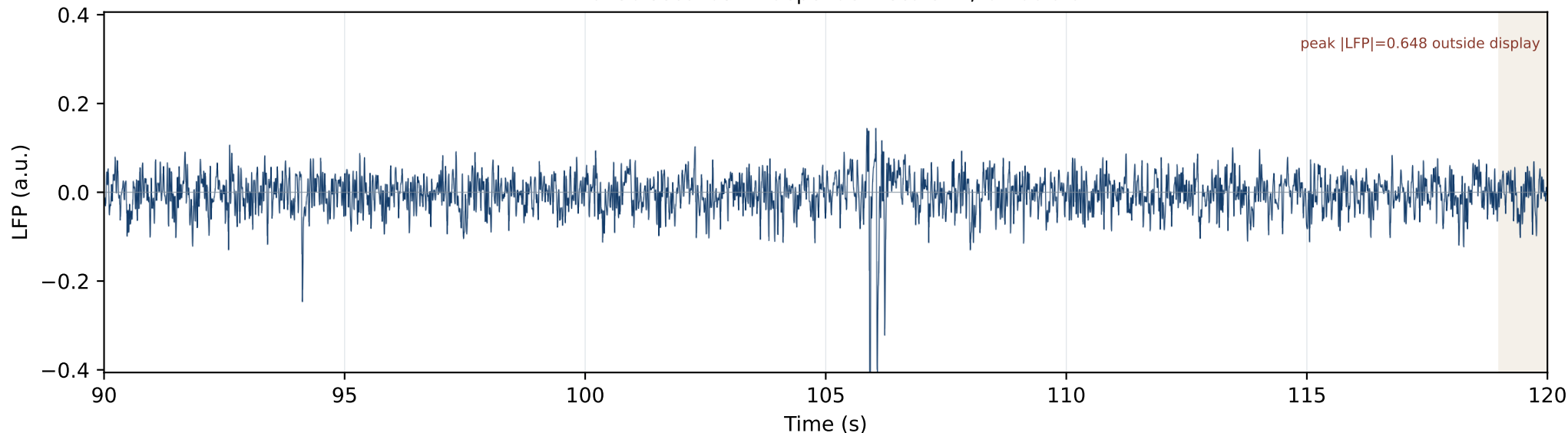
state_23

PYR->RE AMPA +10% and RE->TC GABA-B +20%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.82 a.u.



Final 30 seconds independent scale: ± 0.406 a.u.



10-119 s metrics: RMS 0.0675 a.u. | p99.9 |LFP| 0.746 a.u. | delta 16.5% | theta 35.2% | centroid 10.74 Hz

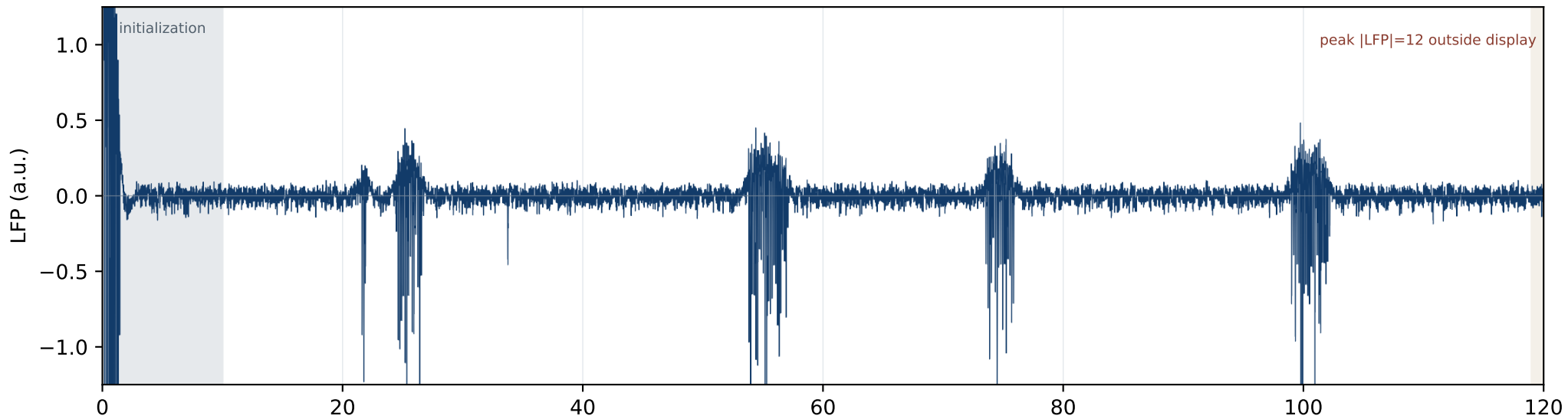
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state23_seed1_nhost=8.txt

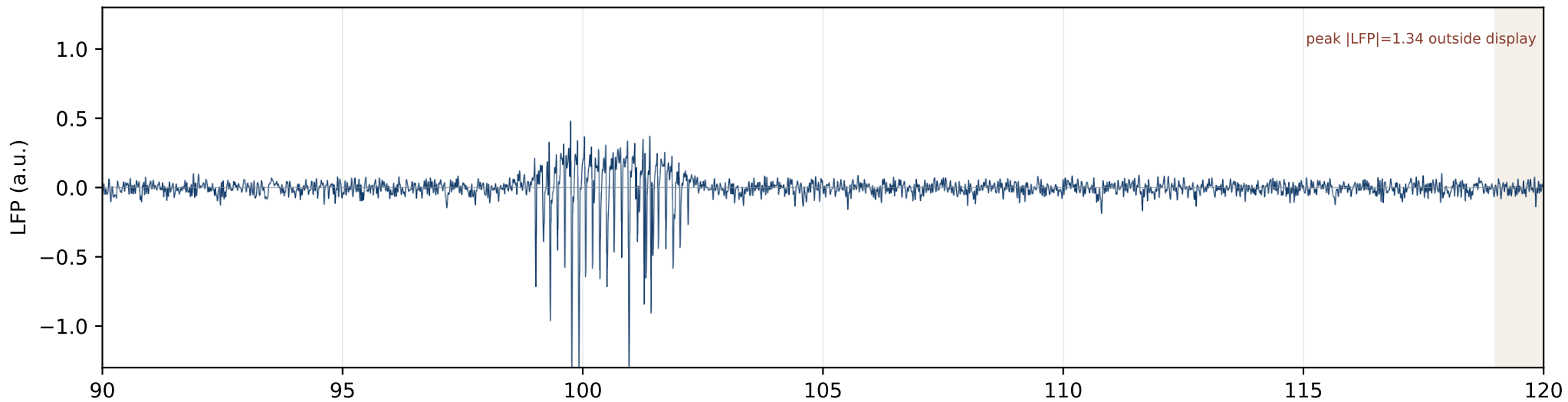
state_24

RE->RE GABA-A -25% and RE->TC GABA-B +20%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 1.25 a.u.



Final 30 seconds independent scale: ± 1.3 a.u.



10-119 s metrics: RMS 0.109 a.u. | p99.9 |LFP| 1.14 a.u. | delta 10.4% | theta 40.7% | centroid 11.28 Hz

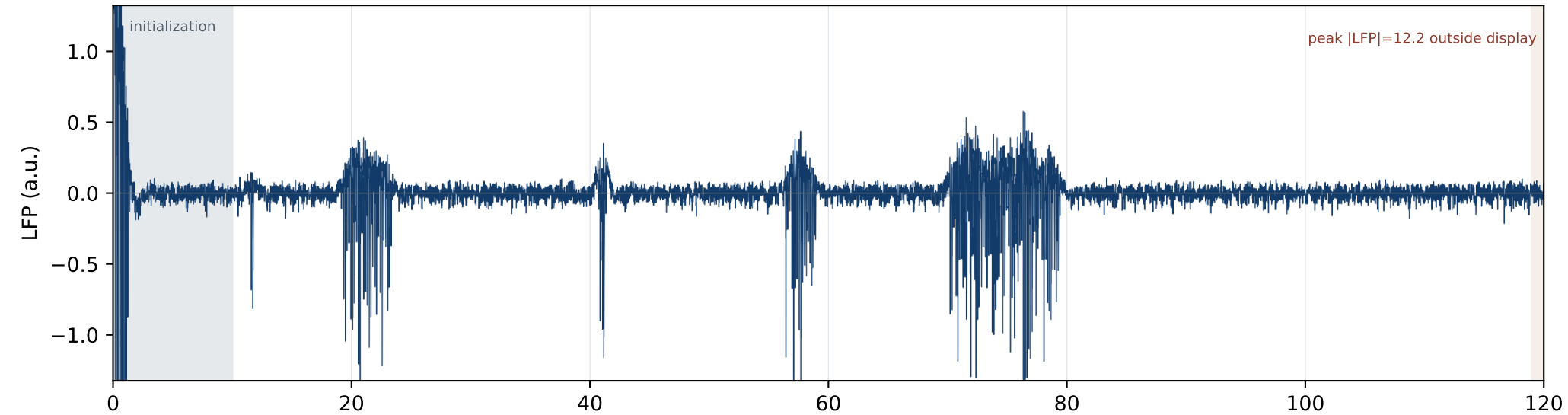
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state24_seed1_nhost=8.txt

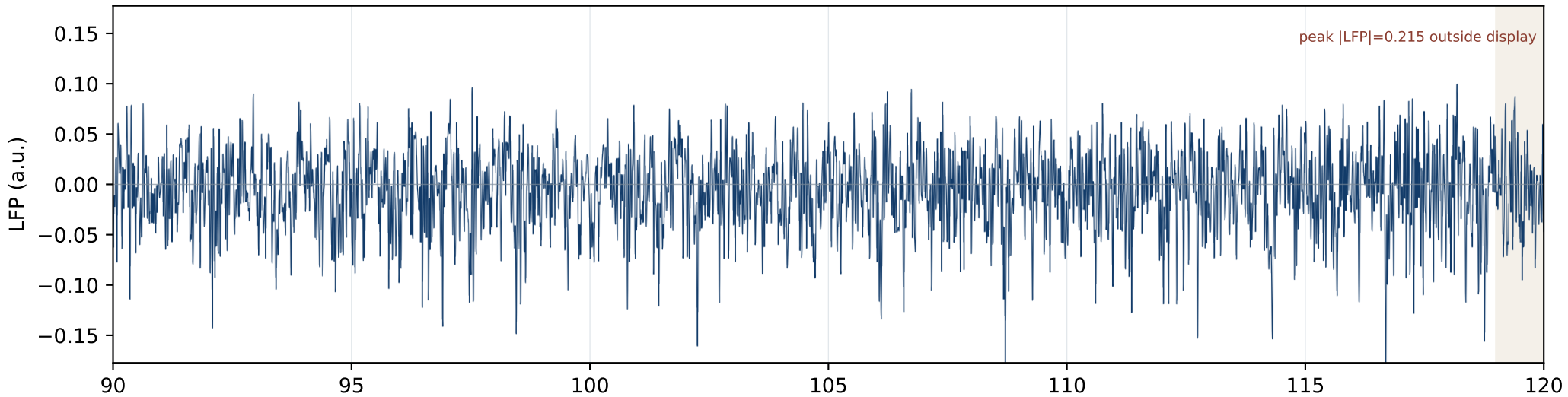
state_25

TC Ih shift -5 mV

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 1.32 a.u.



Final 30 seconds independent scale: ± 0.177 a.u.



10-119 s metrics: RMS 0.131 a.u. | p99.9 |LFP| 1.2 a.u. | delta 10.8% | theta 45.4% | centroid 10.46 Hz

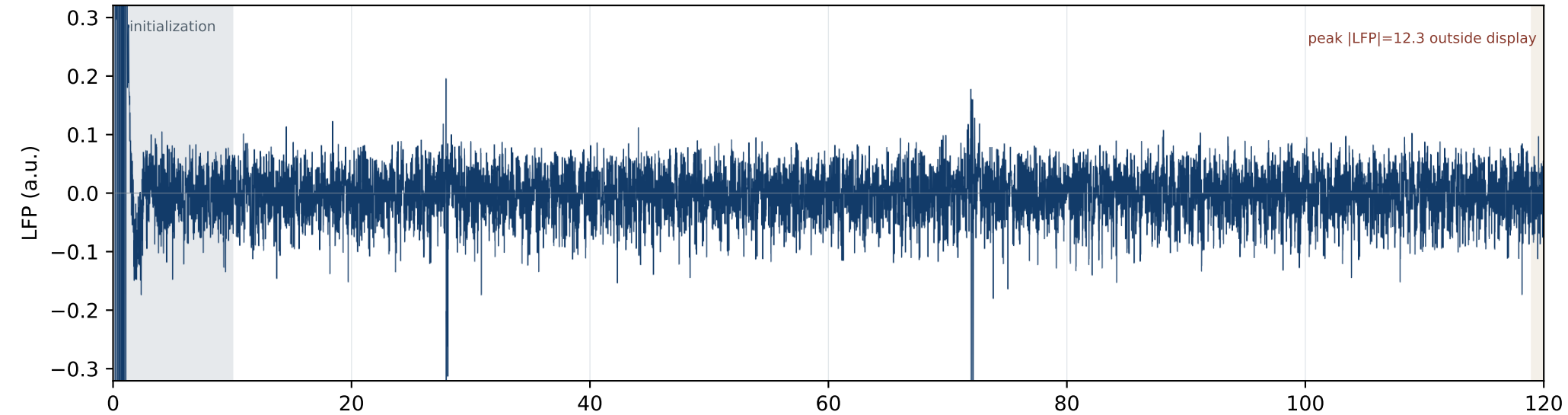
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state25_seed1_nhost=8.txt

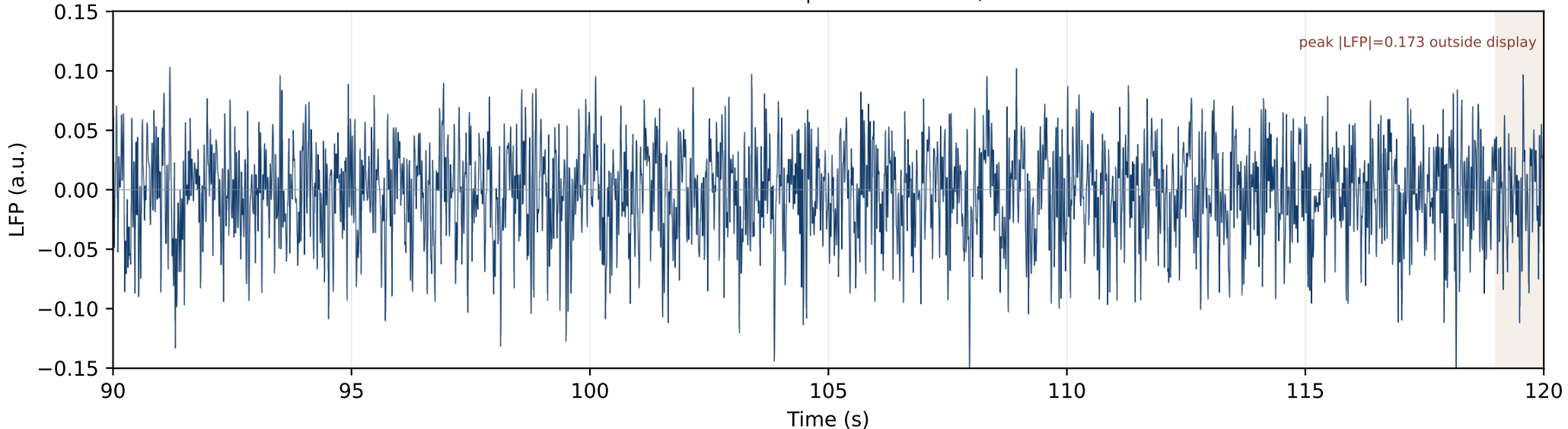
state_26

TC Ih sign control -7 mV

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.321 a.u.



Final 30 seconds independent scale: ± 0.15 a.u.



10-119 s metrics: RMS 0.0425 a.u. | p99.9 |LFP| 0.292 a.u. | delta 20.9% | theta 17.4% | centroid 12.15 Hz

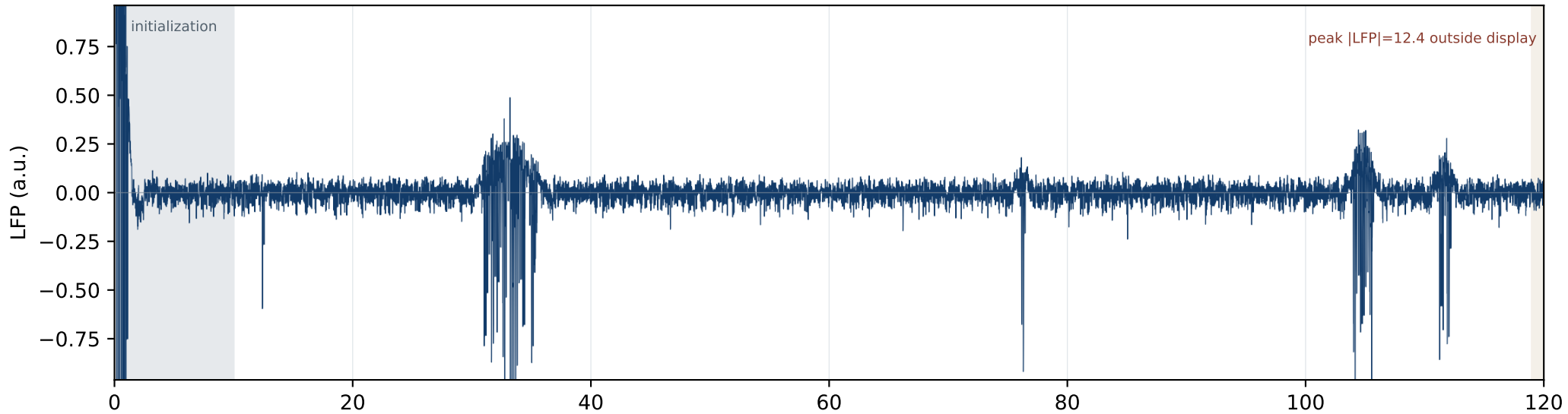
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state26_seed1_nhost=8.txt

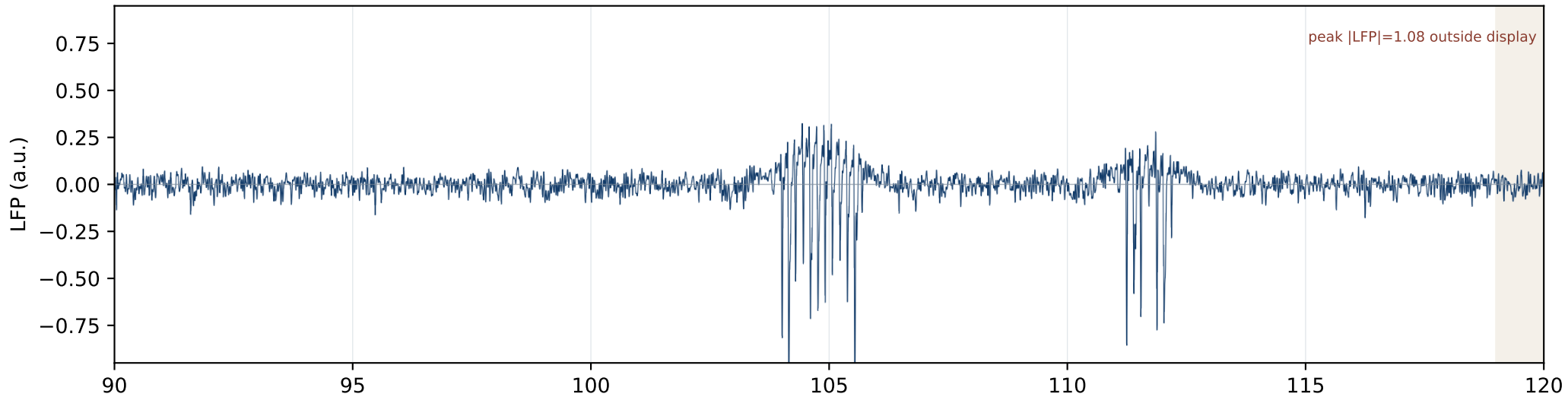
state_27

TC K leak 75% wake-to-N2

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.961 a.u.



Final 30 seconds independent scale: ± 0.95 a.u.



10-119 s metrics: RMS 0.083 a.u. | p99.9 |LFP| 0.874 a.u. | delta 15.5% | theta 36.1% | centroid 10.73 Hz

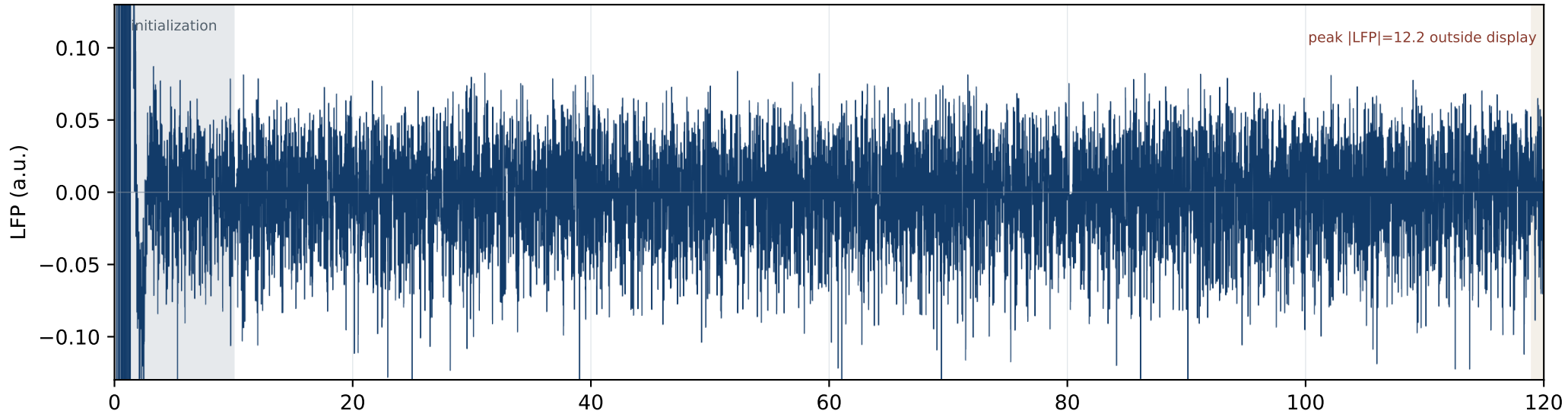
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state27_seed1_nhost=8.txt

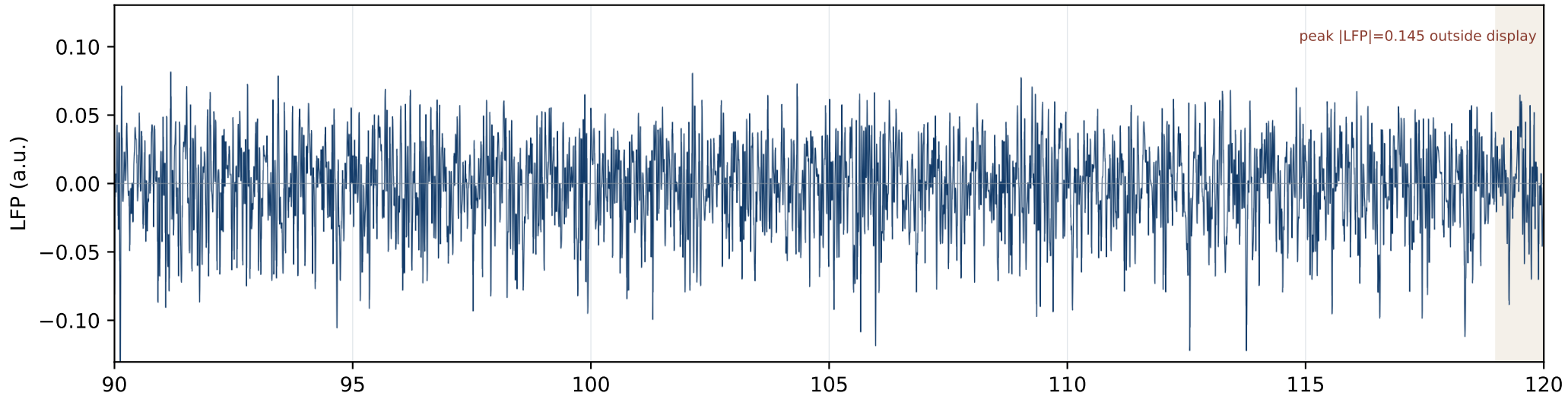
state_28

PYR dendritic Km/KCa 1.15625x

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.13 a.u.



Final 30 seconds independent scale: ± 0.13 a.u.



10-119 s metrics: RMS 0.0315 a.u. | p99.9 |LFP| 0.118 a.u. | delta 15.7% | theta 13.6% | centroid 13.66 Hz

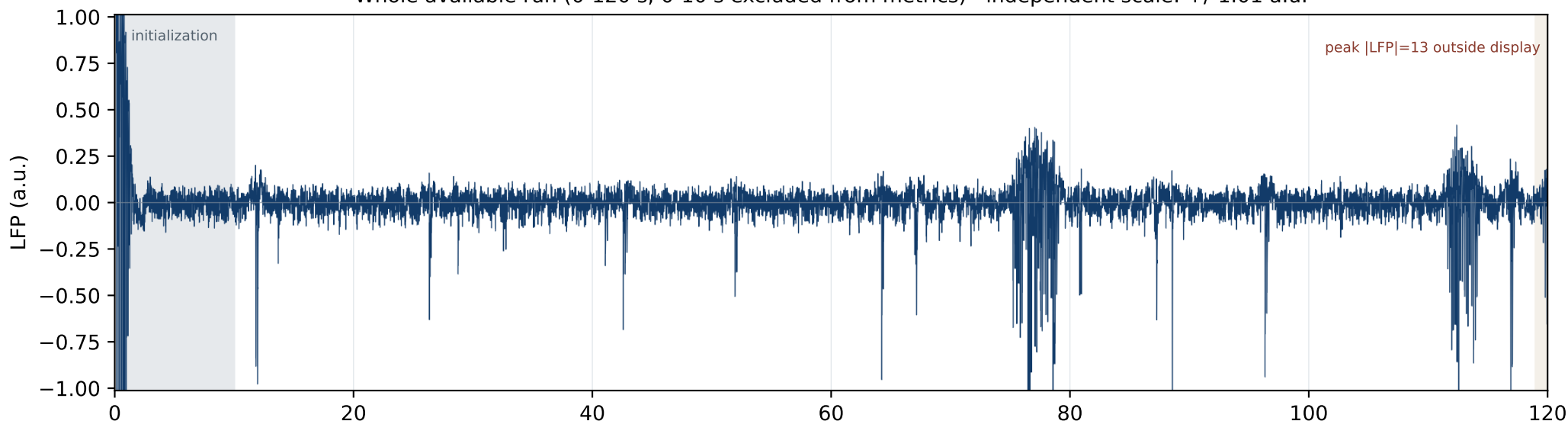
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state28_seed1_nhost=8.txt

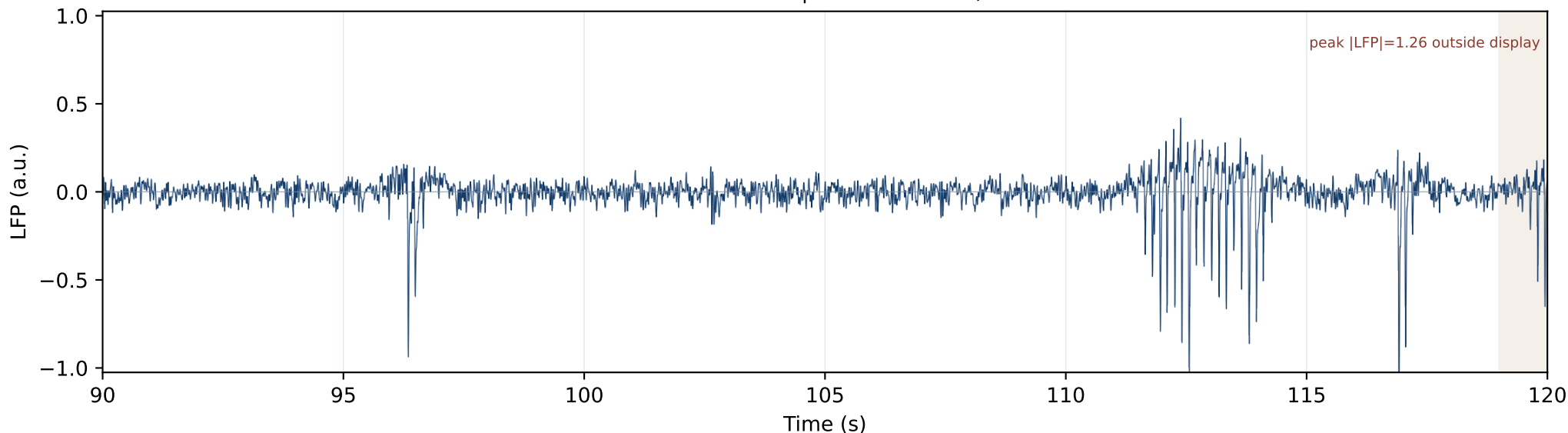
state_29

PYR dendritic NaP/Km/KCa 1.15625x

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 1.01 a.u.



Final 30 seconds independent scale: ± 1.03 a.u.



10-119 s metrics: RMS 0.0967 a.u. | p99.9 |LFP| 0.92 a.u. | delta 19.8% | theta 36.7% | centroid 9.96 Hz

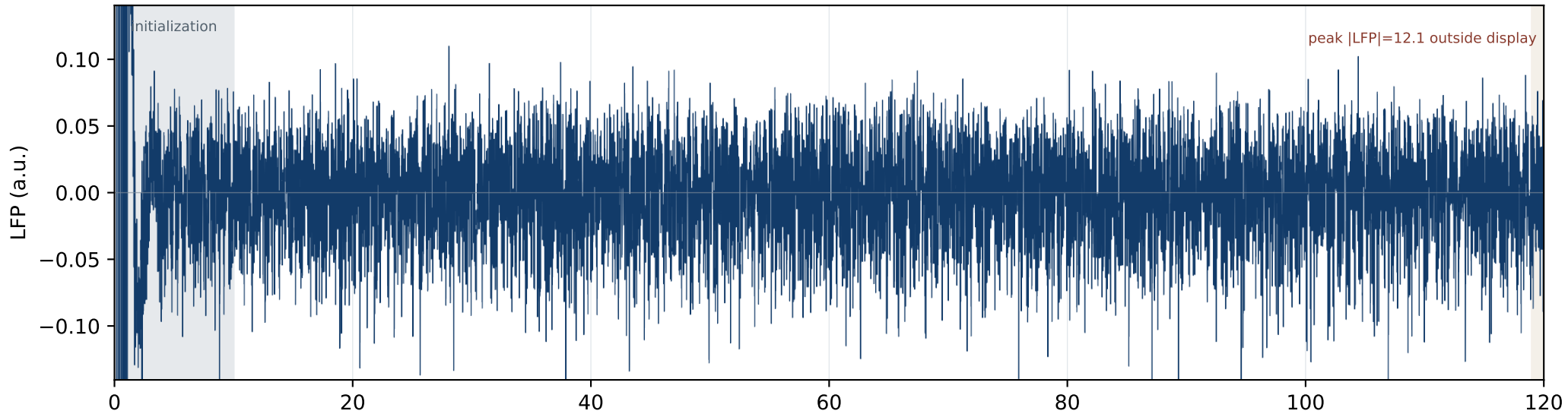
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state29_seed1_nhost=8.txt

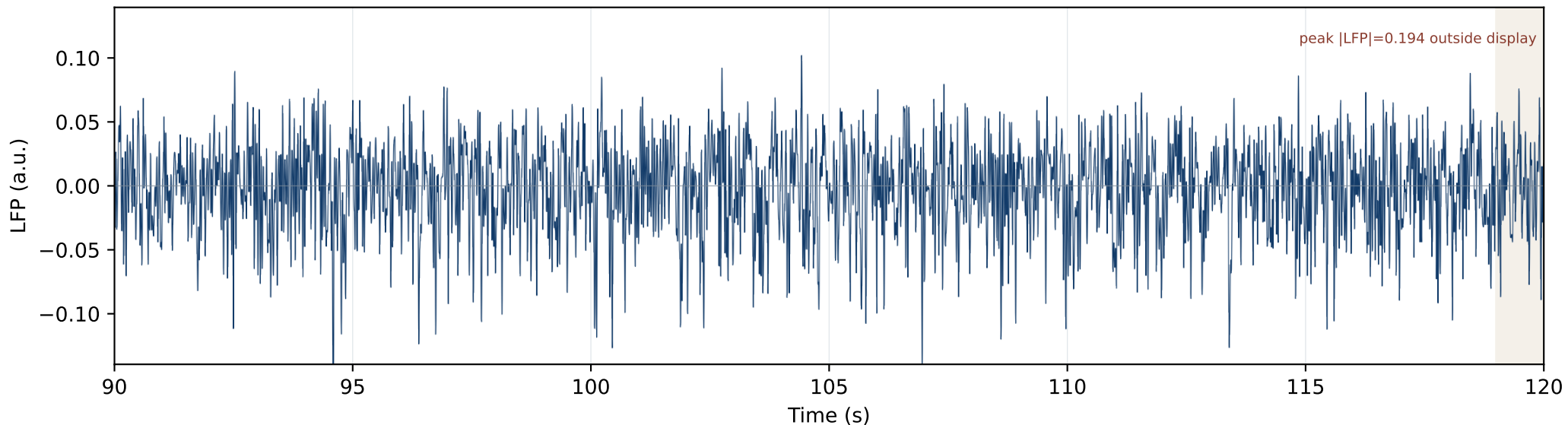
state_30

background recurrence bracket 37.5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.14 a.u.



Final 30 seconds independent scale: ± 0.14 a.u.



10-119 s metrics: RMS 0.0349 a.u. | p99.9 |LFP| 0.128 a.u. | delta 18.3% | theta 14.3% | centroid 13.06 Hz

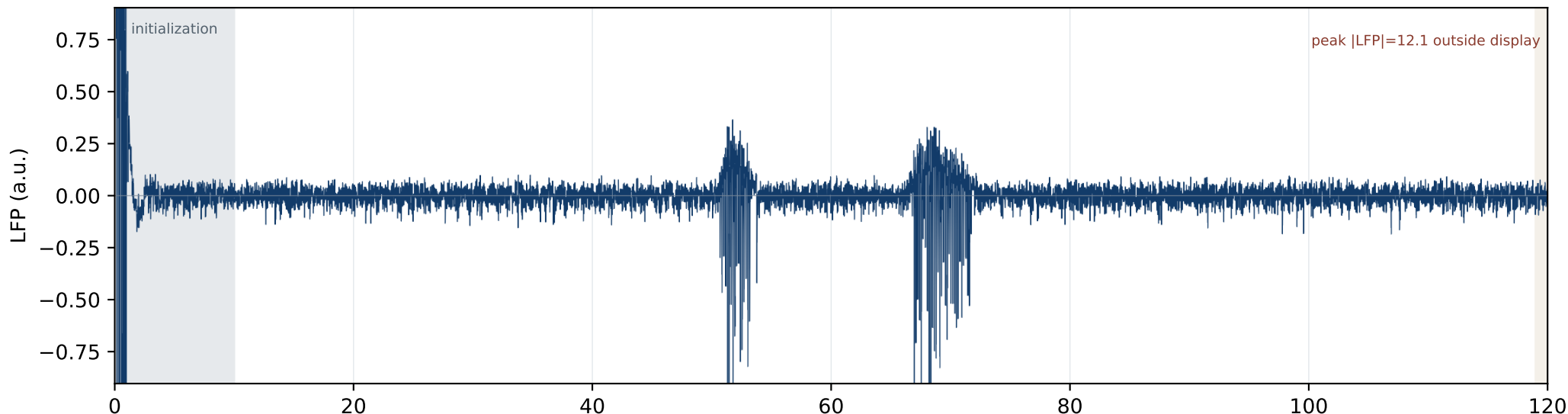
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state30_seed1_nhost=8.txt

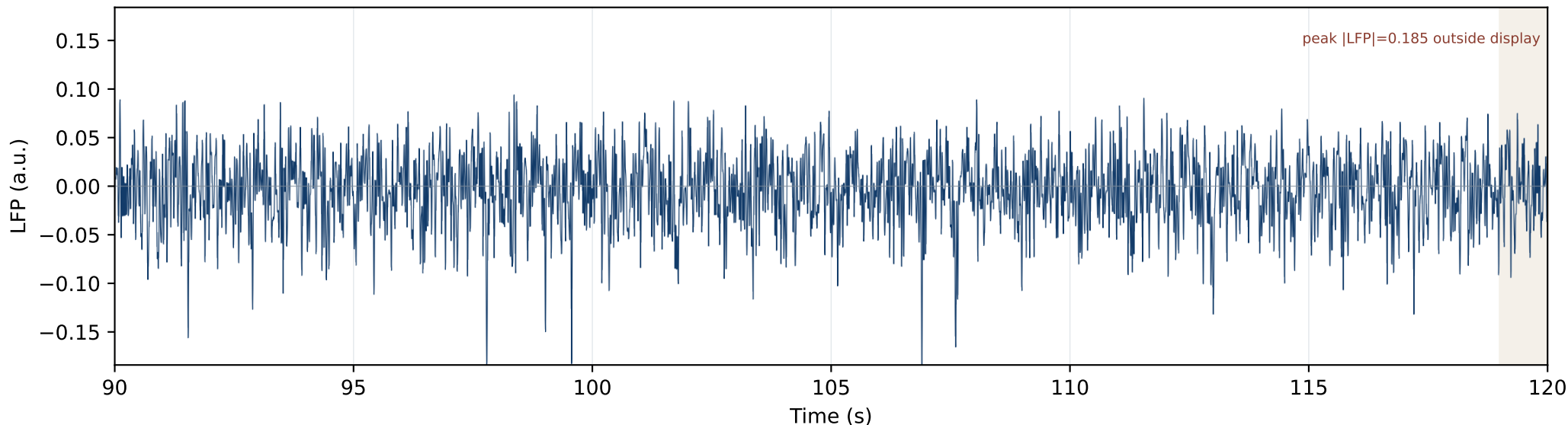
state_31

background recurrence bracket 43.75%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.904 a.u.



Final 30 seconds independent scale: ± 0.184 a.u.



10-119 s metrics: RMS 0.0756 a.u. | p99.9 $|LFP|$ 0.821 a.u. | delta 11.4% | theta 36.2% | centroid 11.44 Hz

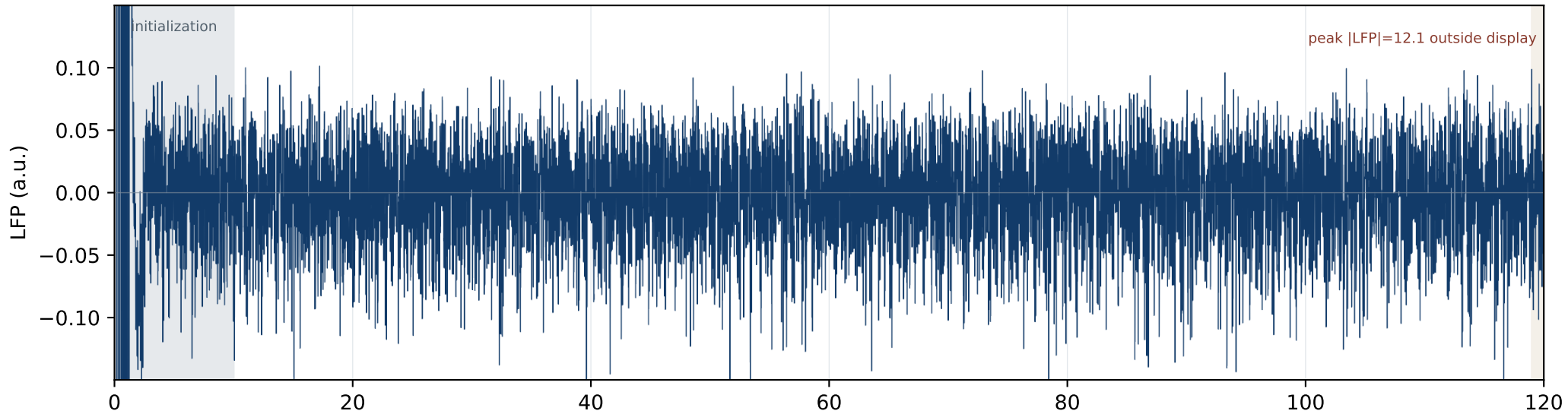
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state31_seed1_nhost=8.txt

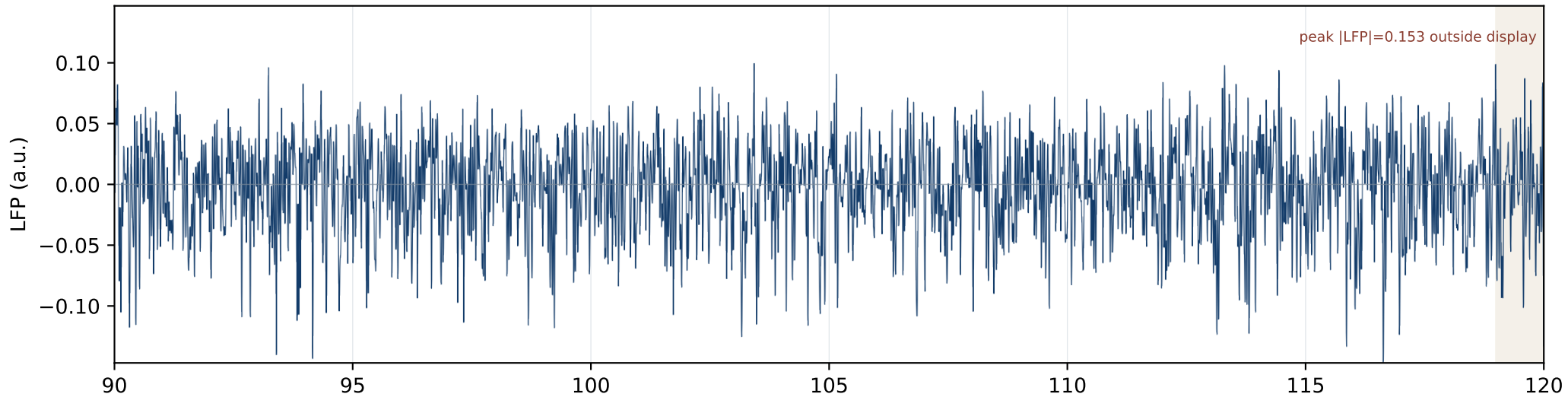
state_32

balanced cortical-axis bracket 37.5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.15 a.u.



Final 30 seconds independent scale: ± 0.147 a.u.



10-119 s metrics: RMS 0.037 a.u. | p99.9 |LFP| 0.136 a.u. | delta 19.9% | theta 14.5% | centroid 12.94 Hz

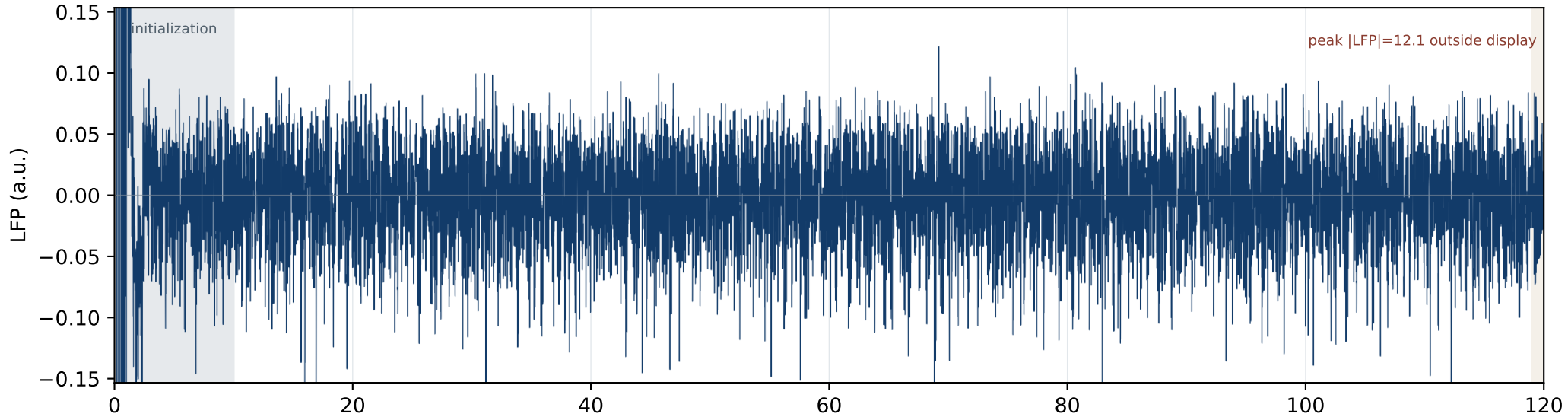
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state32_seed1_nhost=8.txt

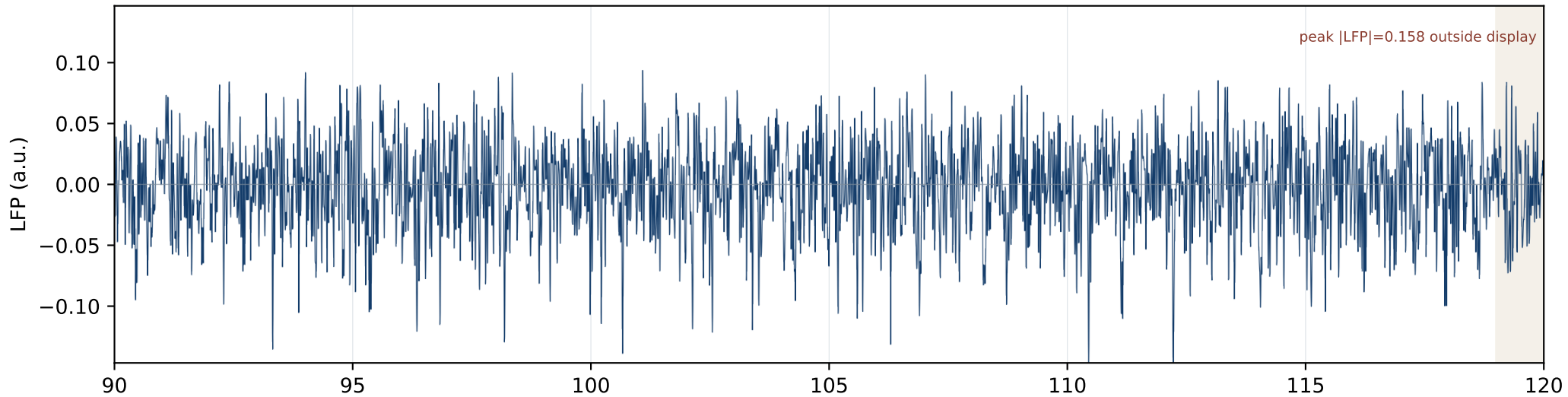
state_33

balanced cortical-axis bracket 43.75%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.153 a.u.



Final 30 seconds independent scale: ± 0.147 a.u.



10-119 s metrics: RMS 0.0368 a.u. | p99.9 |LFP| 0.139 a.u. | delta 18.7% | theta 14.6% | centroid 12.89 Hz

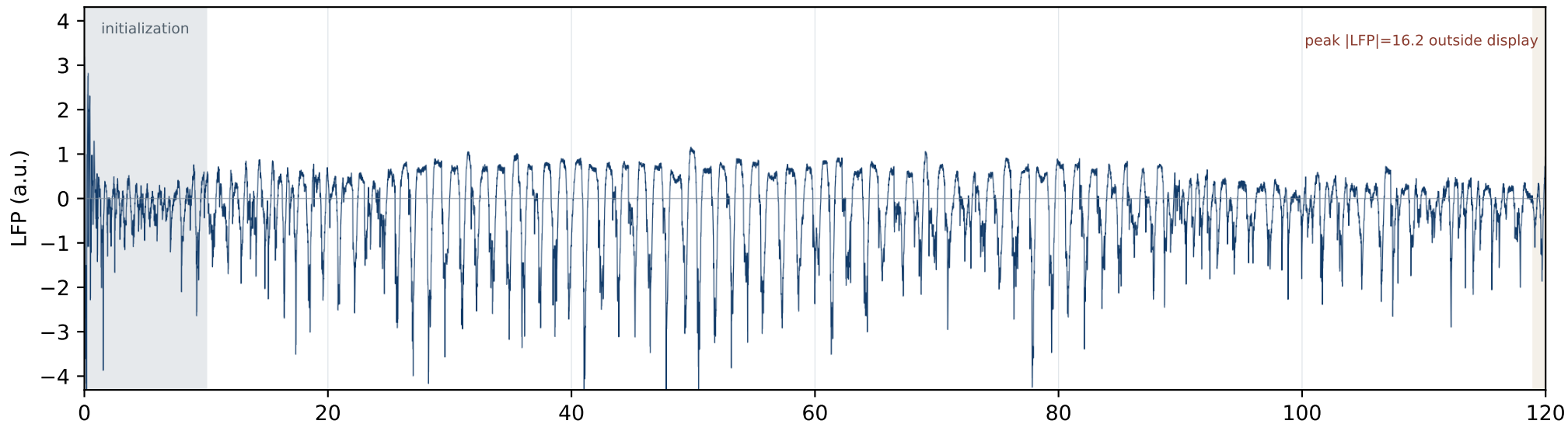
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state33_seed1_nhost=8.txt

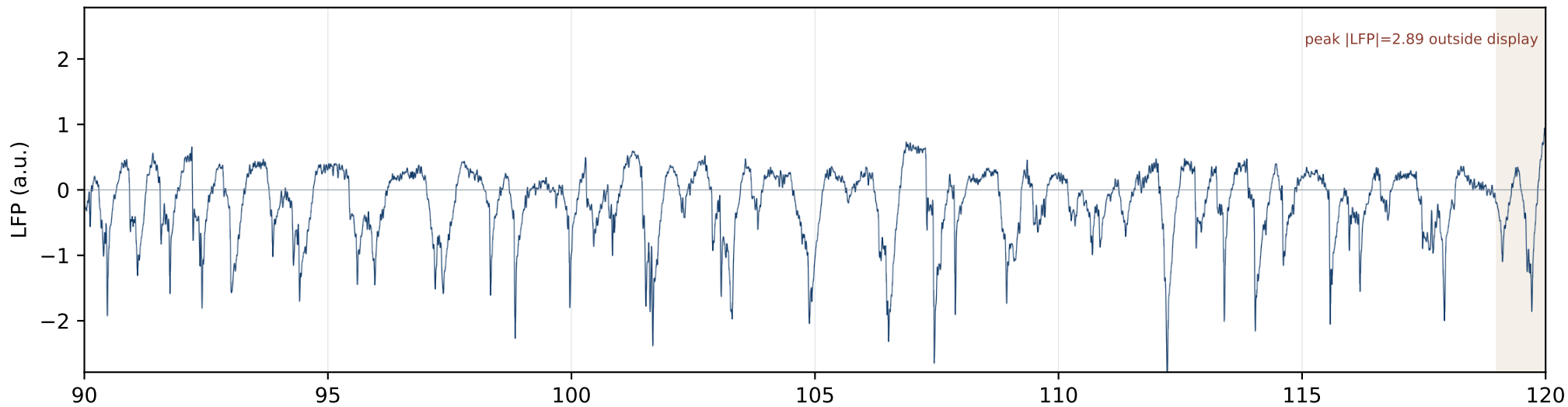
state_34

N3 minus recurrence to 2.35

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 4.31 a.u.



Final 30 seconds independent scale: ± 2.79 a.u.



10-119 s metrics: RMS 0.879 a.u. | p99.9 |LFP| 3.92 a.u. | delta 93.8% | theta 2.5% | centroid 1.66 Hz

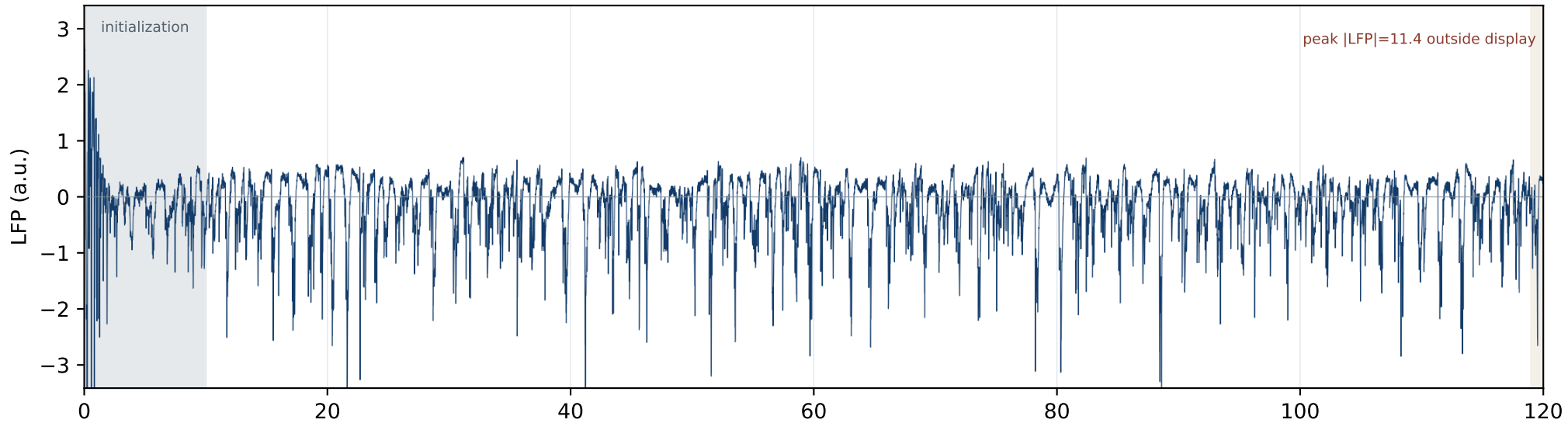
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state34_seed1_nhost=8.txt

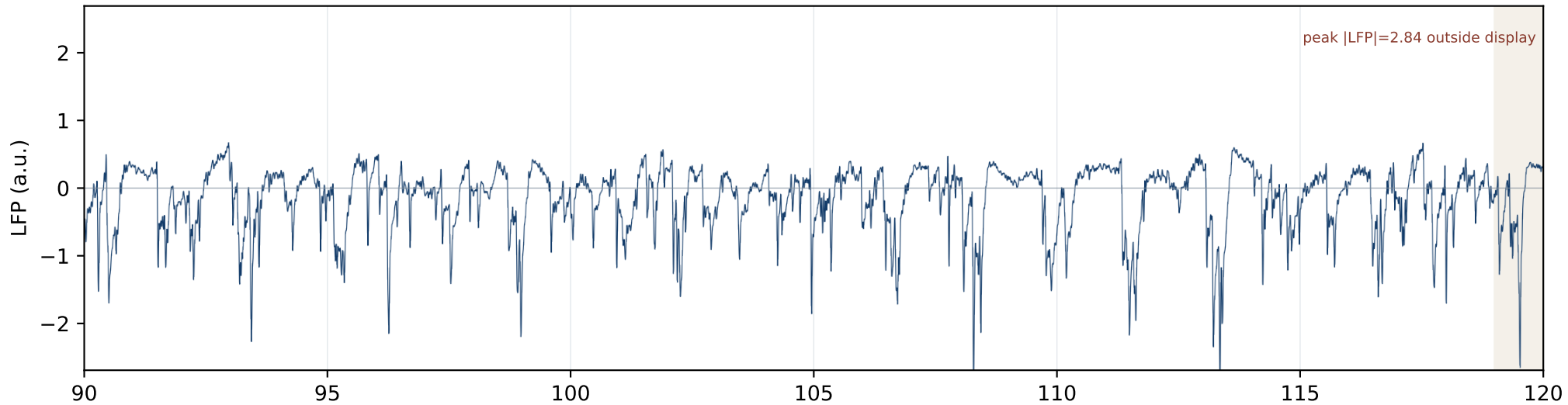
state_35

N3 minus recurrence to 2.00

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.41 a.u.



Final 30 seconds independent scale: ± 2.69 a.u.



10-119 s metrics: RMS 0.557 a.u. | p99.9 $|LFP|$ 3.1 a.u. | delta 77.5% | theta 10.2% | centroid 3.37 Hz

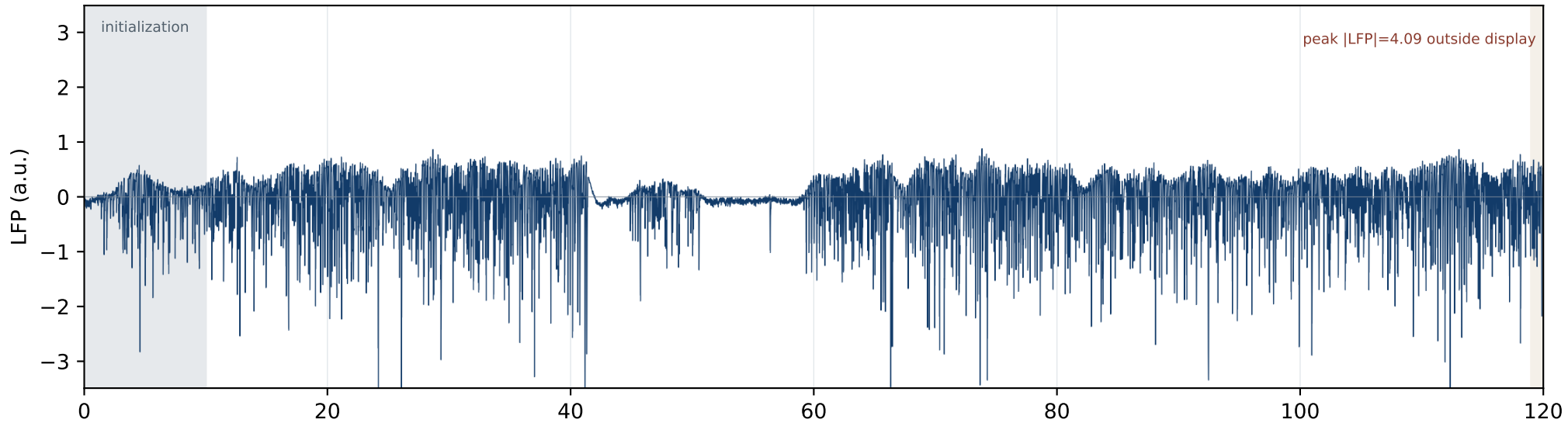
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state35_seed1_nhost=8.txt

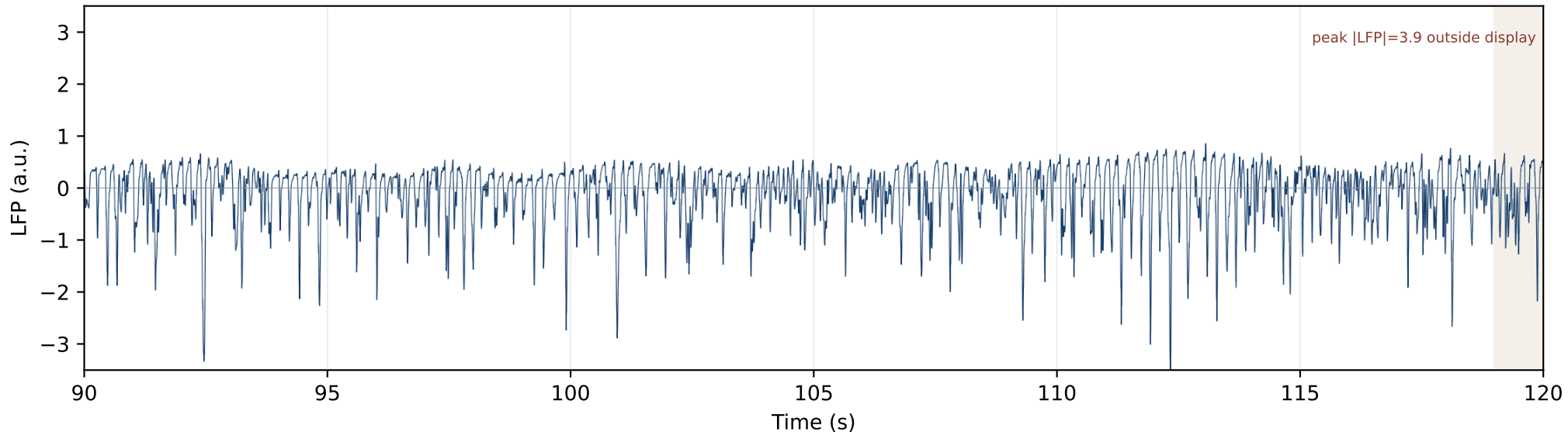
state_36

N3 minus recurrence to 1.60

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.49 a.u.



Final 30 seconds independent scale: ± 3.5 a.u.



10-119 s metrics: RMS 0.538 a.u. | p99.9 |LFP| 3.17 a.u. | delta 17.9% | theta 49.9% | centroid 7.59 Hz

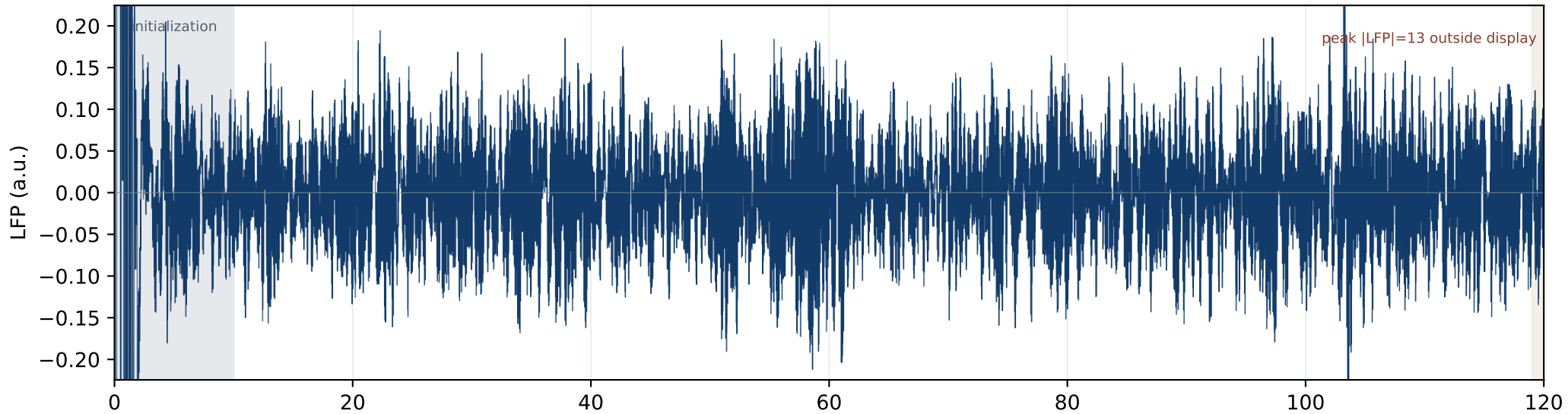
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state36_seed1_nhost=8.txt

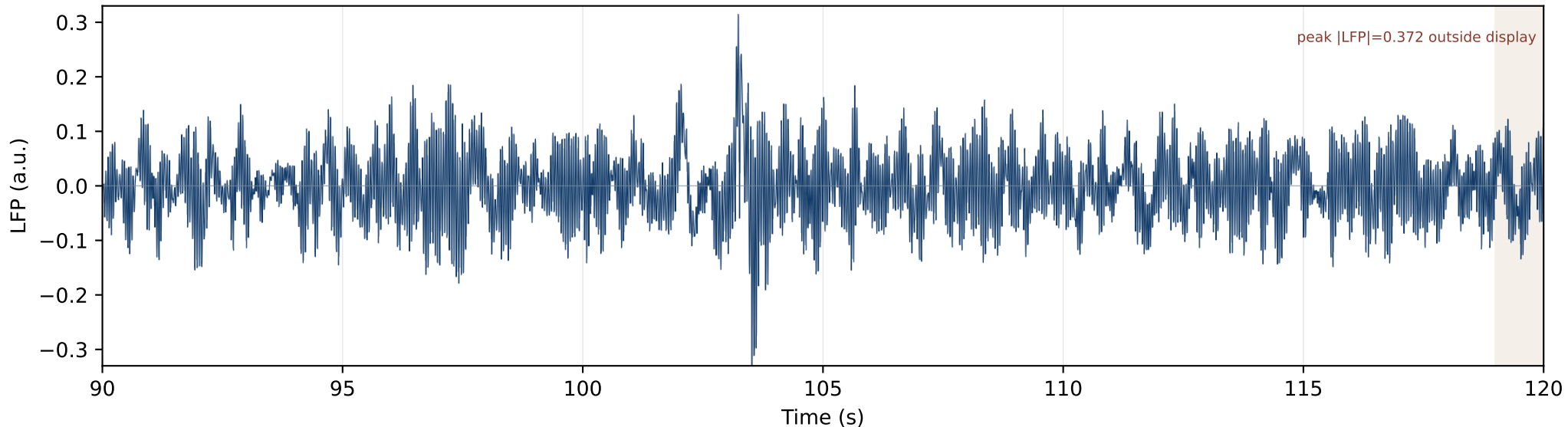
state_37

N3 cortical K leak -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 0.225 a.u.



Final 30 seconds independent scale: ± 0.33 a.u.



10-119 s metrics: RMS 0.0593 a.u. | p99.9 |LFP| 0.204 a.u. | delta 20.3% | theta 1.0% | centroid 16.87 Hz

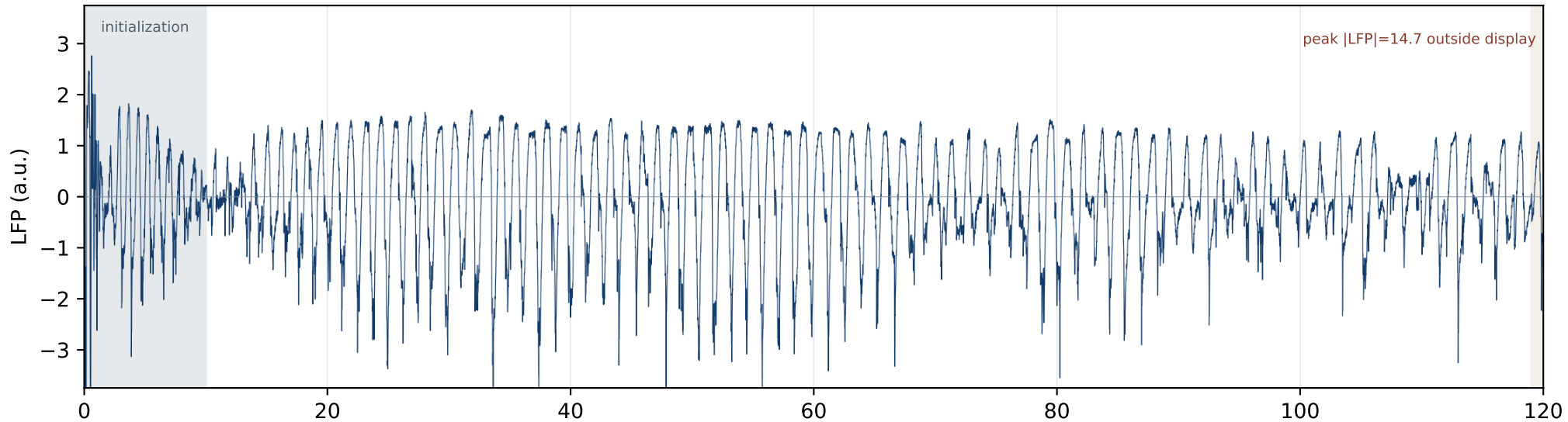
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state37_seed1_nhost=8.txt

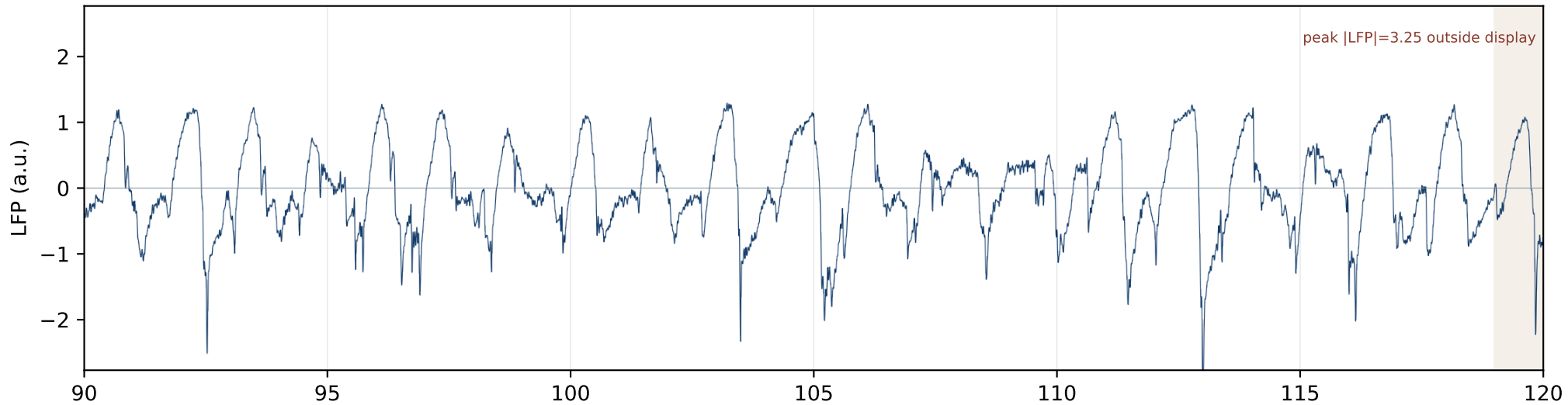
state_38

N3 RE->TC GABA-A -20%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.75 a.u.



Final 30 seconds independent scale: ± 2.77 a.u.



10-119 s metrics: RMS 1.02 a.u. | p99.9 |LFP| 3.41 a.u. | delta 97.2% | theta 1.2% | centroid 1.19 Hz

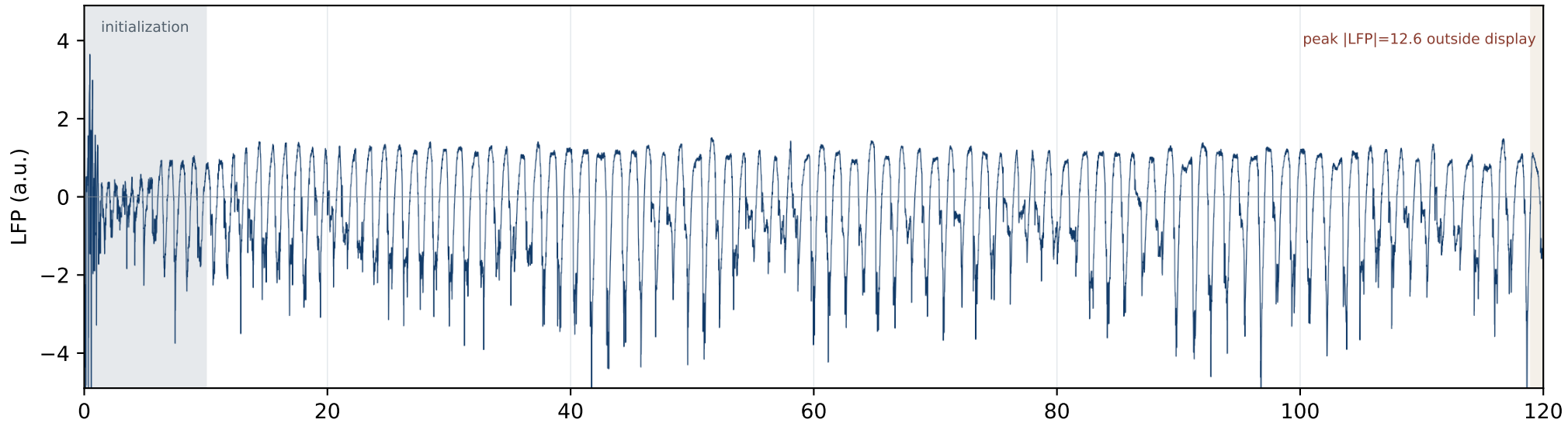
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state38_seed1_nhost=8.txt

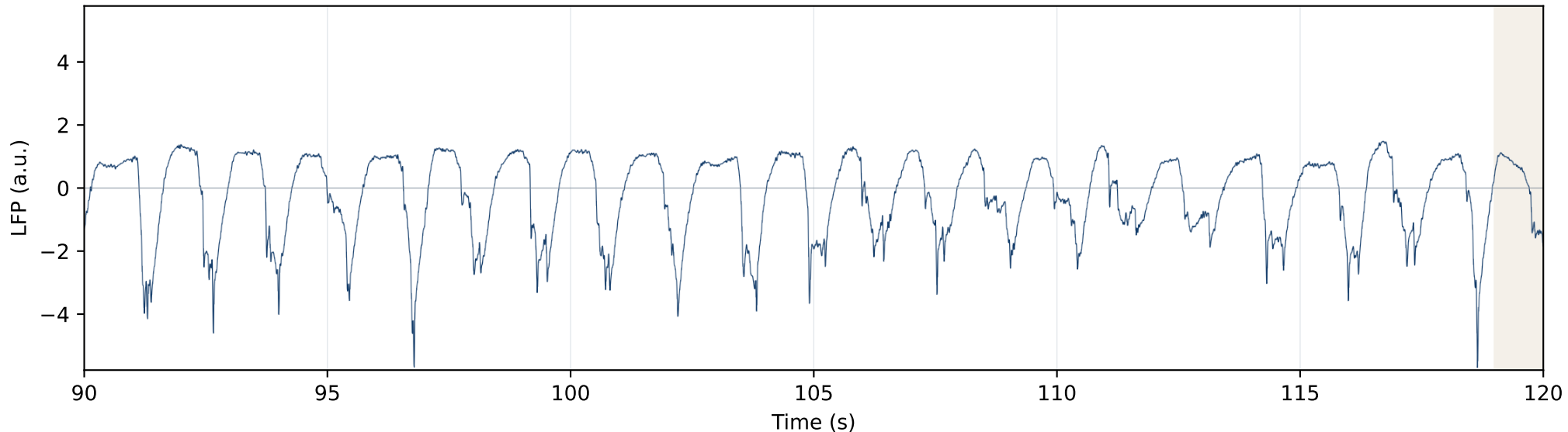
state_39

N3 with N2 thalamic intrinsic module

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: +/-4.9 a.u.



Final 30 seconds independent scale: +/-5.78 a.u.



10-119 s metrics: RMS 1.25 a.u. | p99.9 |LFP| 4.45 a.u. | delta 96.6% | theta 1.4% | centroid 1.26 Hz

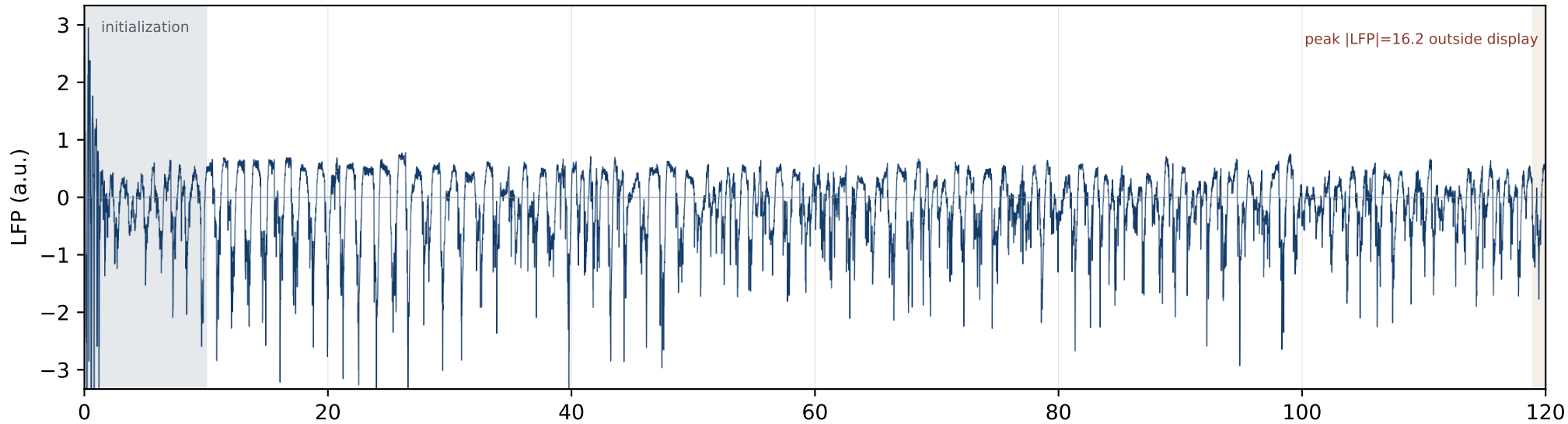
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state39_seed1_nhost=8.txt

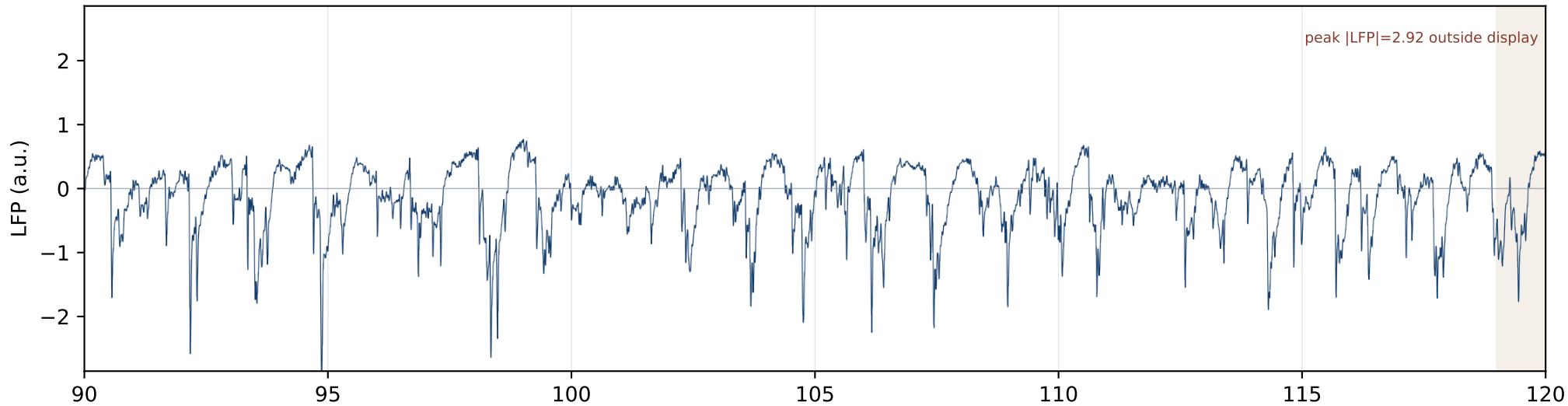
state_40

recurrence 2.00 and cortical K leak -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.34 a.u.



Final 30 seconds independent scale: ± 2.85 a.u.



10-119 s metrics: RMS 0.621 a.u. | p99.9 $|LFP|$ 3.03 a.u. | delta 84.7% | theta 6.6% | centroid 2.57 Hz

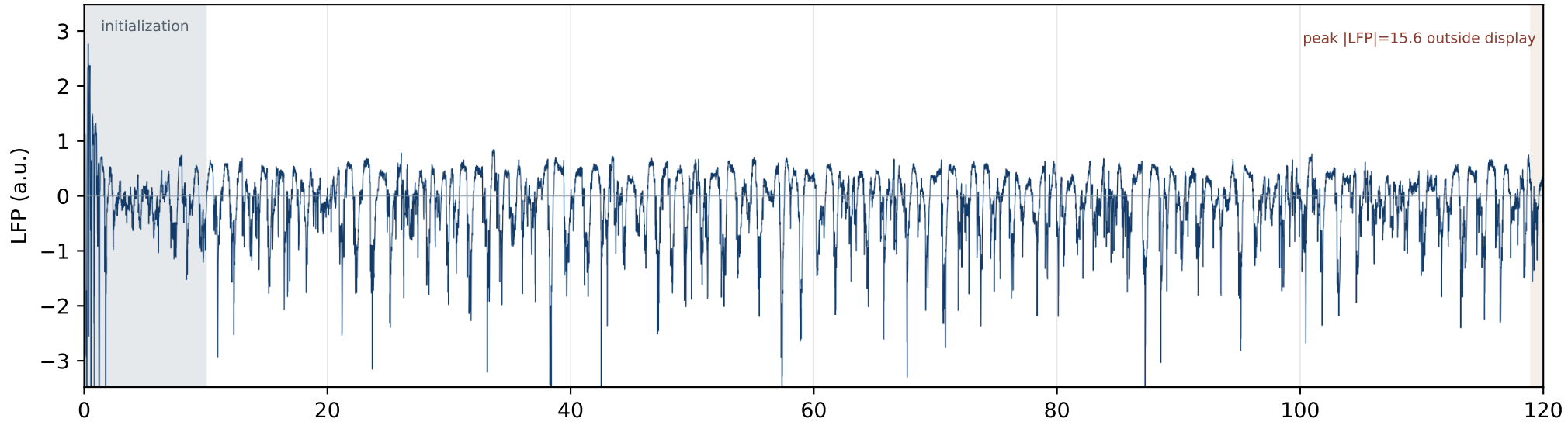
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state40_seed1_nhost=8.txt

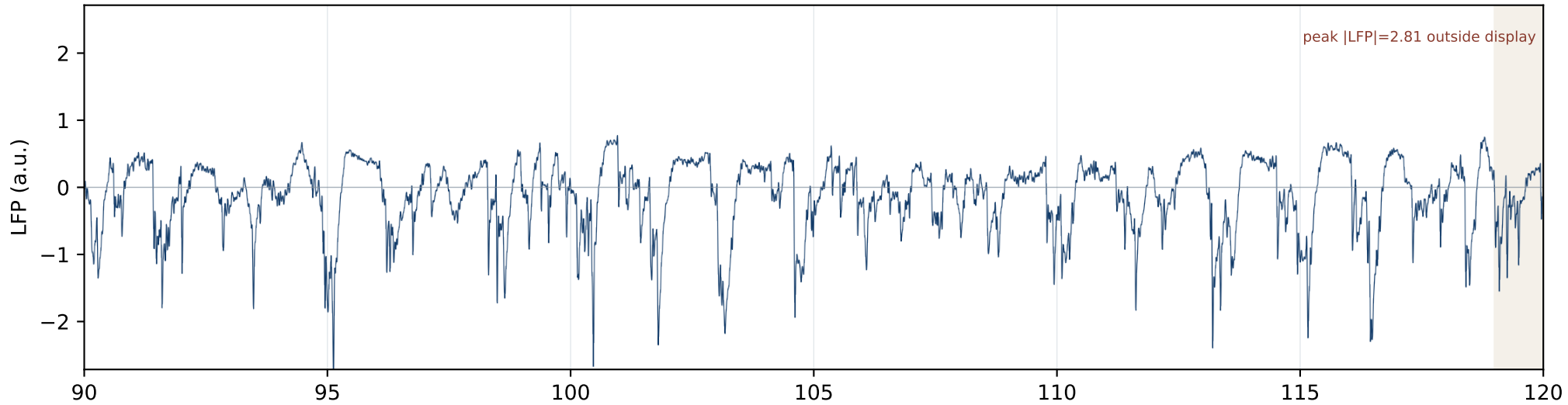
state_41

N3 recurrence 2.10

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.48 a.u.



Final 30 seconds independent scale: ± 2.71 a.u.



10-119 s metrics: RMS 0.605 a.u. | p99.9 |LFP| 3.16 a.u. | delta 85.1% | theta 6.3% | centroid 2.63 Hz

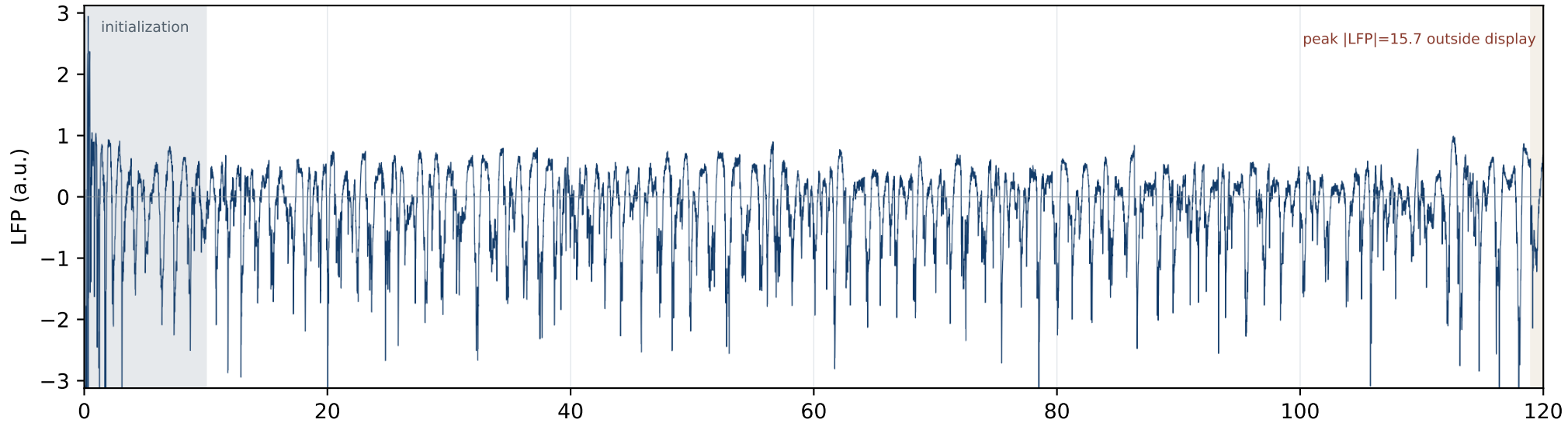
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state41_seed1_nhost=8.txt

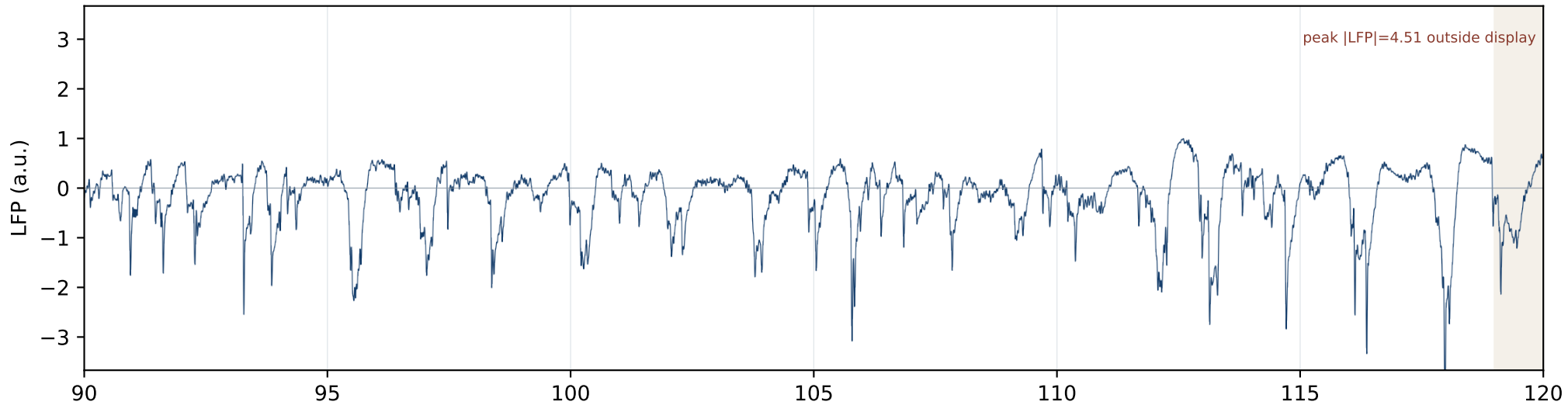
state_42

N3 recurrence 2.20

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.12 a.u.



Final 30 seconds independent scale: ± 3.67 a.u.



10-119 s metrics: RMS 0.643 a.u. | p99.9 $|LFP|$ 2.84 a.u. | delta 89.1% | theta 4.7% | centroid 2.26 Hz

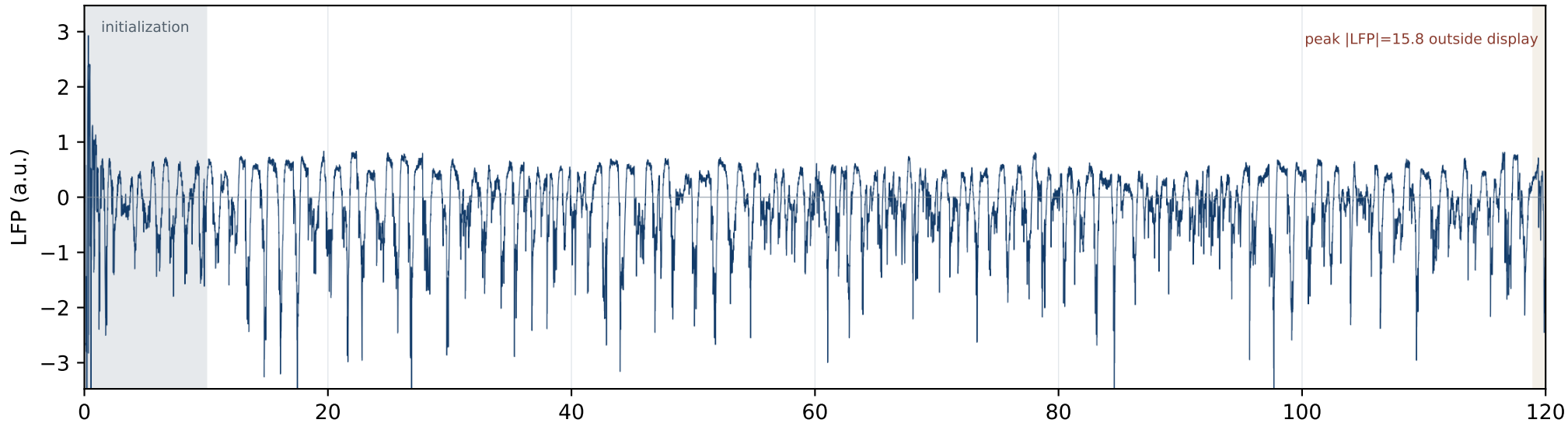
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state42_seed1_nhost=8.txt

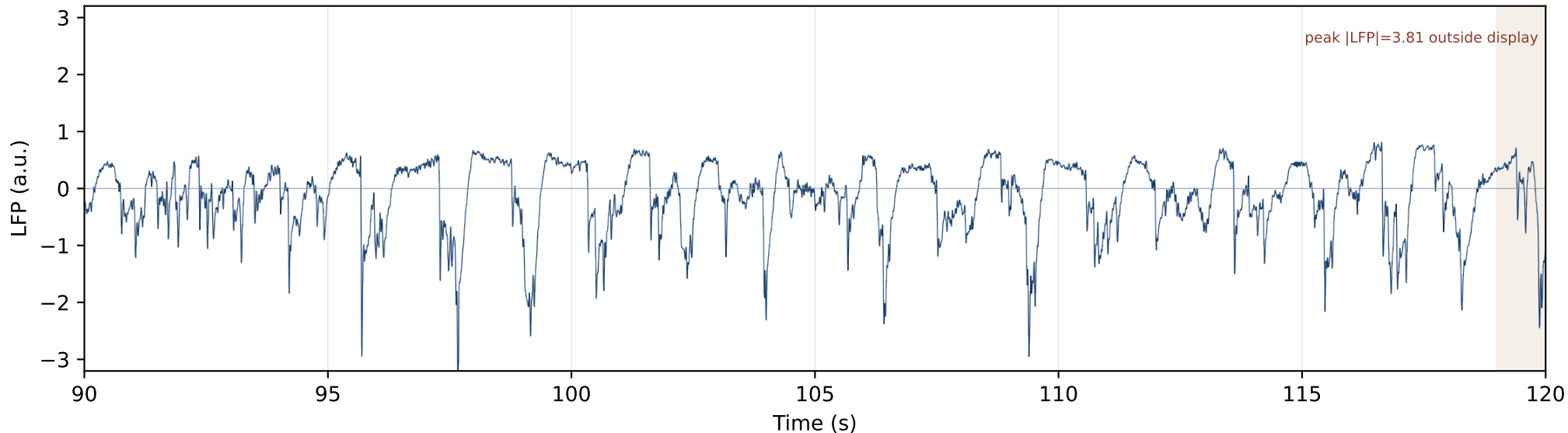
state_43

recurrence 2.10 and cortical K leak -5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.47 a.u.



Final 30 seconds independent scale: ± 3.2 a.u.



10-119 s metrics: RMS 0.655 a.u. | p99.9 |LFP| 3.16 a.u. | delta 88.5% | theta 4.7% | centroid 2.28 Hz

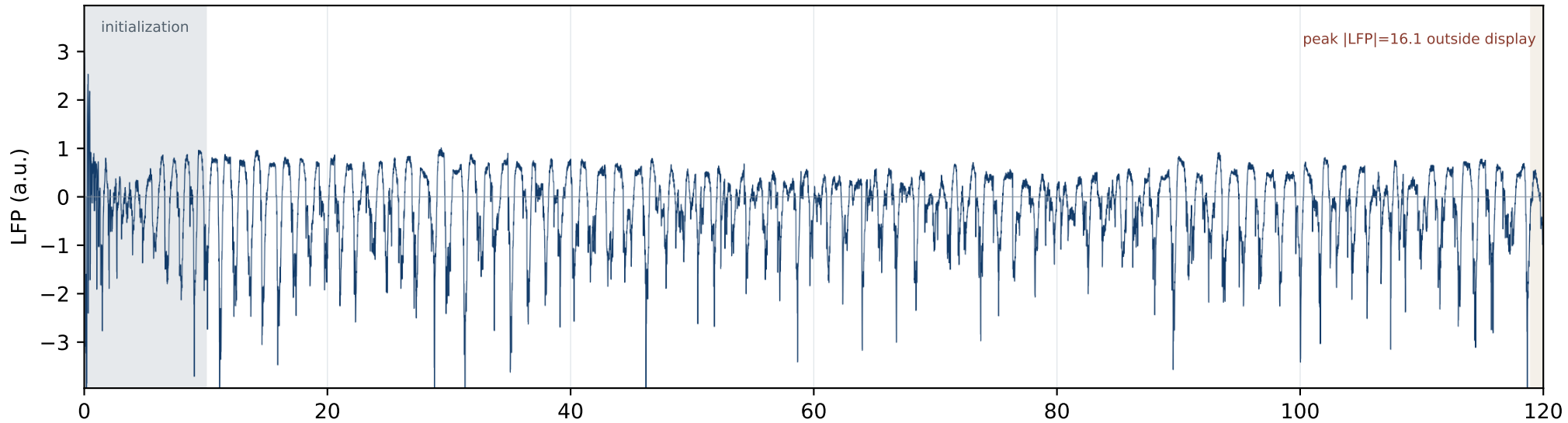
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state43_seed1_nhost=8.txt

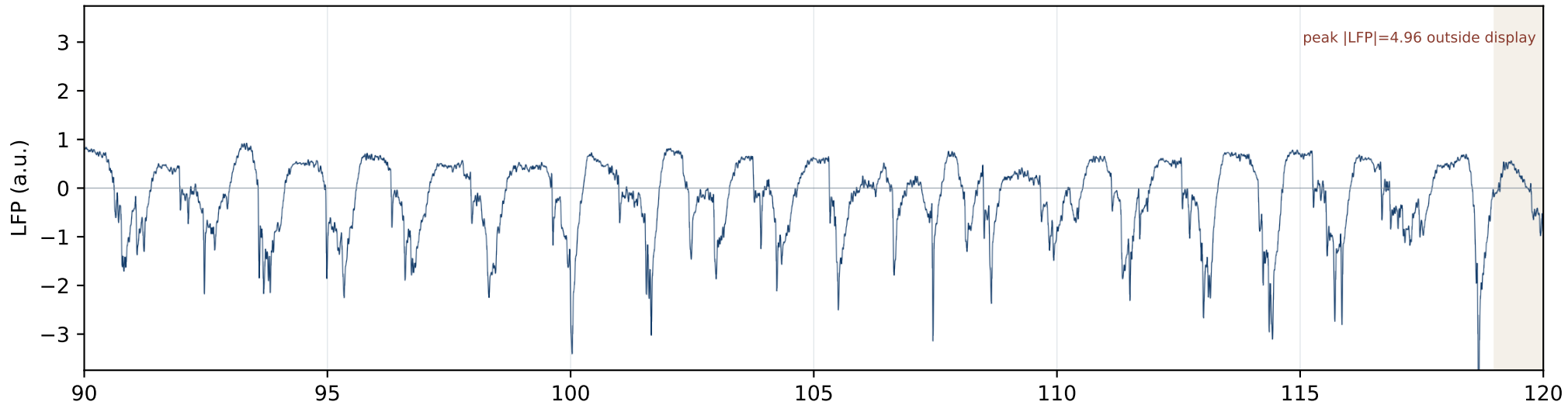
state_44

recurrence 2.20 and cortical K leak -5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.95 a.u.



Final 30 seconds independent scale: ± 3.74 a.u.



10-119 s metrics: RMS 0.778 a.u. | p99.9 |LFP| 3.59 a.u. | delta 92.2% | theta 3.2% | centroid 1.80 Hz

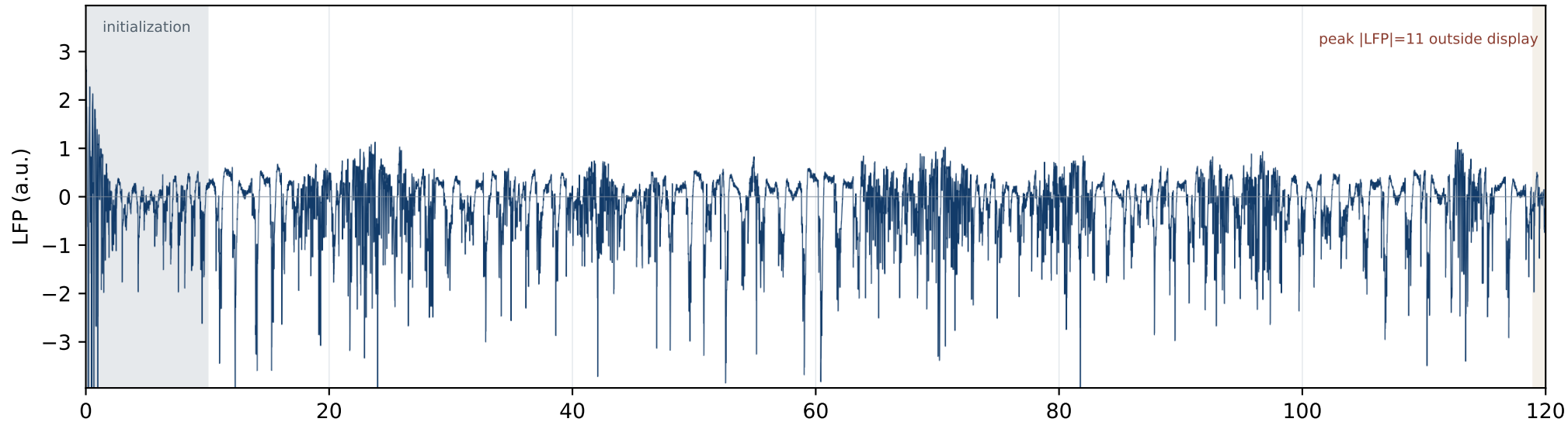
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state44_seed1_nhost=8.txt

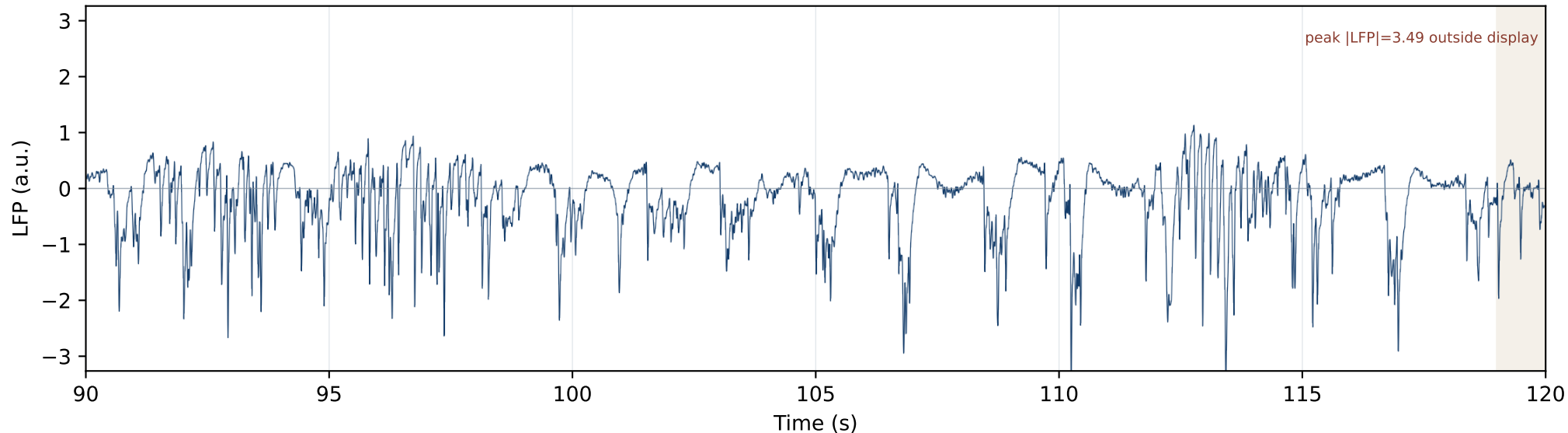
state_45

state 35 plus N2 thalamic intrinsic module

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.95 a.u.



Final 30 seconds independent scale: ± 3.26 a.u.



10-119 s metrics: RMS 0.668 a.u. | p99.9 |LFP| 3.59 a.u. | delta 63.4% | theta 18.3% | centroid 4.63 Hz

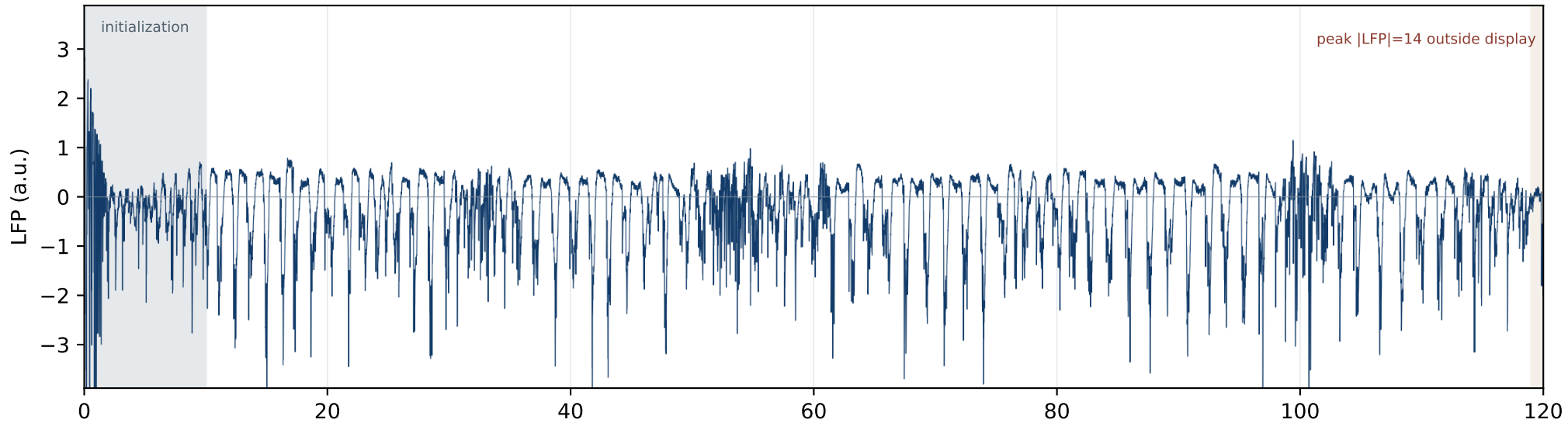
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state45_seed1_nhost=8.txt

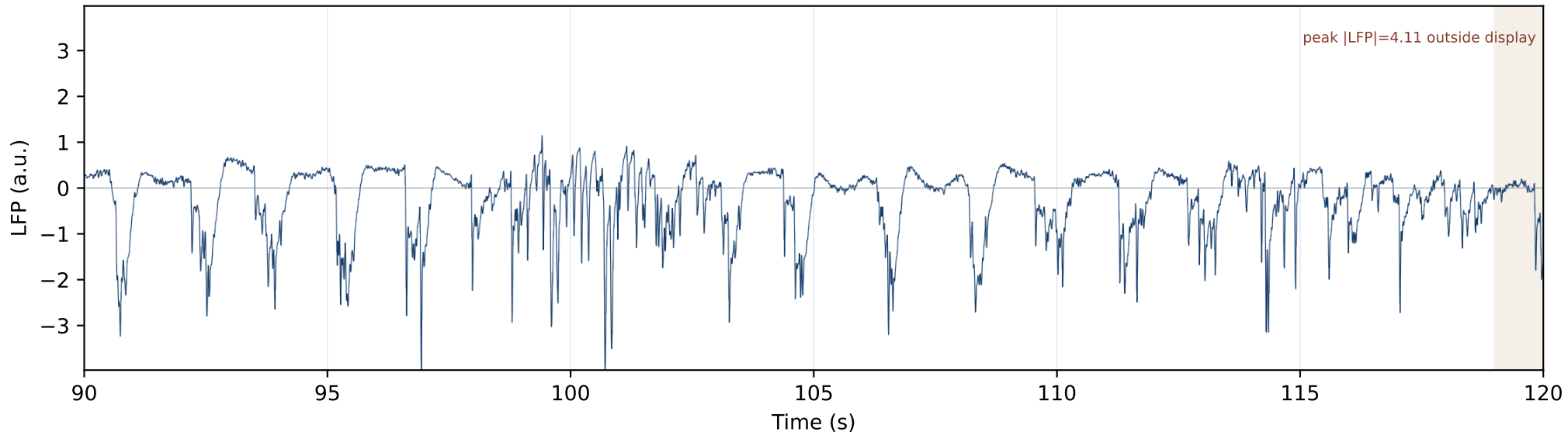
state_46

state 40 plus N2 thalamic intrinsic module

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.88 a.u.



Final 30 seconds independent scale: ± 3.97 a.u.



10-119 s metrics: RMS 0.772 a.u. | p99.9 |LFP| 3.53 a.u. | delta 82.2% | theta 7.1% | centroid 2.84 Hz

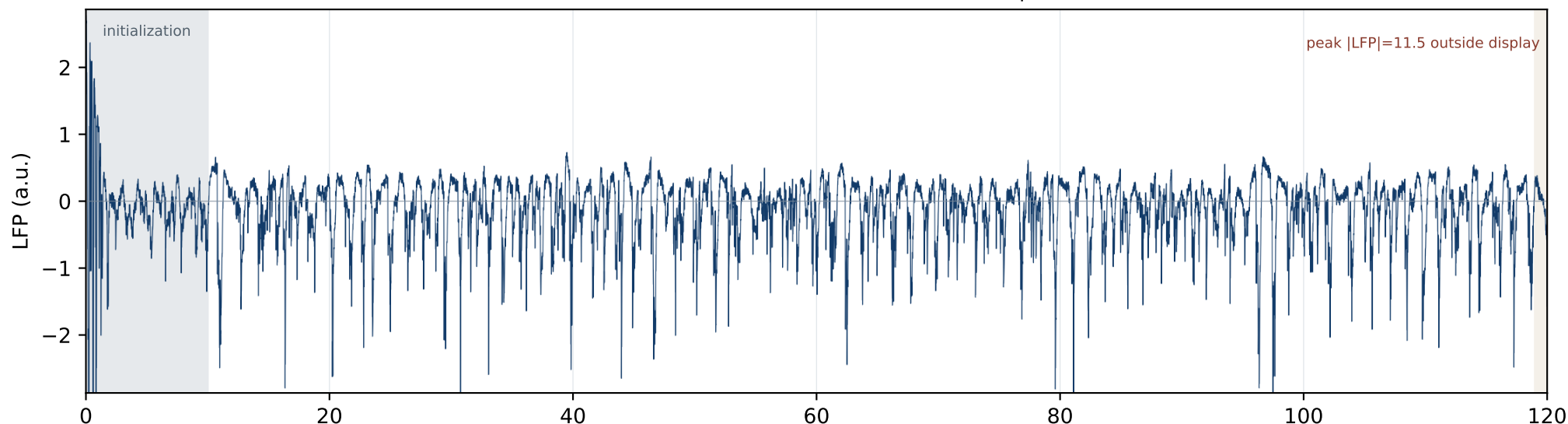
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state46_seed1_nhost=8.txt

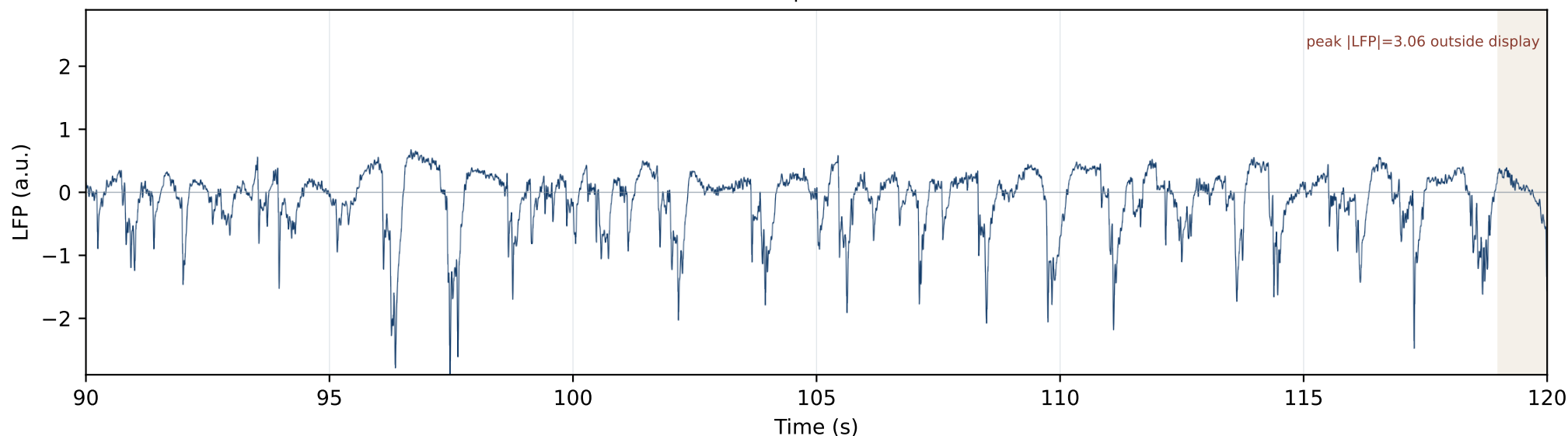
state_47

state 35 plus RE->TC GABA-A -20%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 2.87 a.u.



Final 30 seconds independent scale: ± 2.89 a.u.



10-119 s metrics: RMS 0.482 a.u. | p99.9 |LFP| 2.61 a.u. | delta 82.3% | theta 7.6% | centroid 2.97 Hz

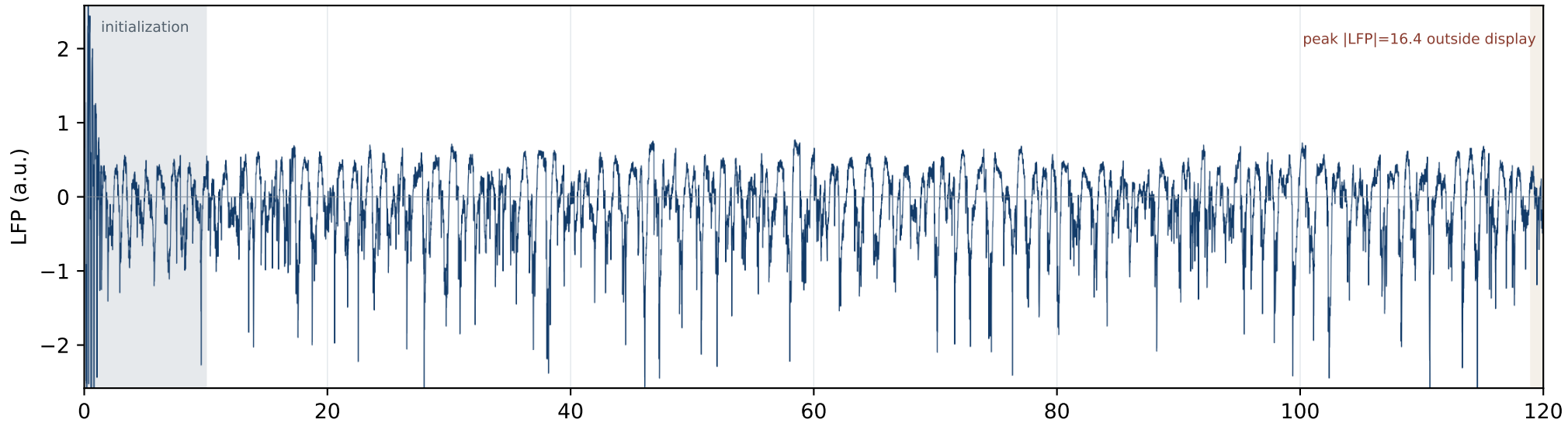
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state47_seed1_nhost=8.txt

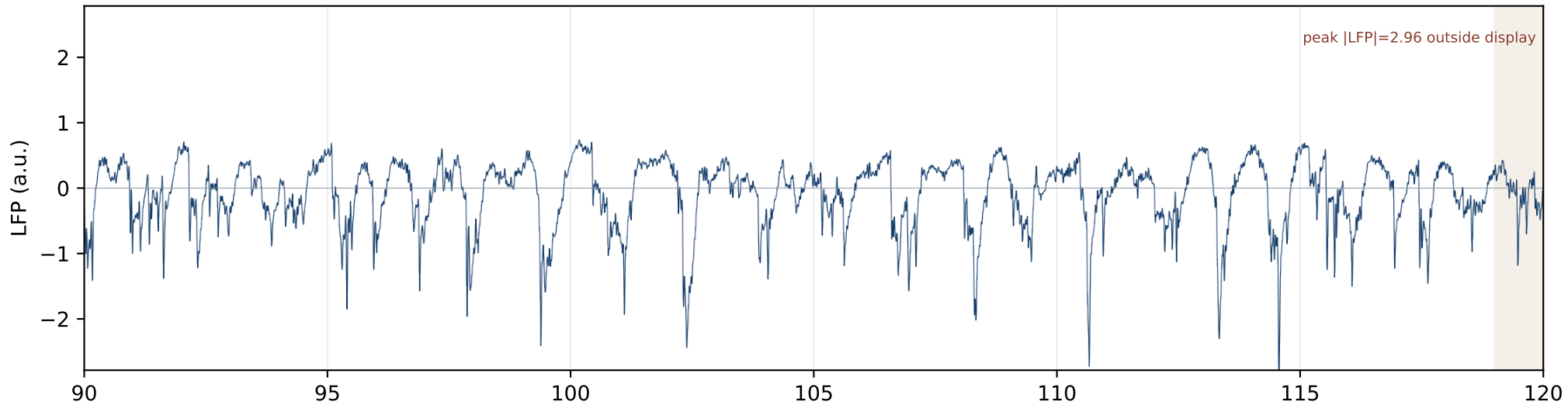
state_48

state 40 plus RE->TC GABA-A -20%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 2.58 a.u.



Final 30 seconds independent scale: ± 2.78 a.u.



10-119 s metrics: RMS 0.502 a.u. | p99.9 |LFP| 2.35 a.u. | delta 85.4% | theta 5.9% | centroid 2.61 Hz

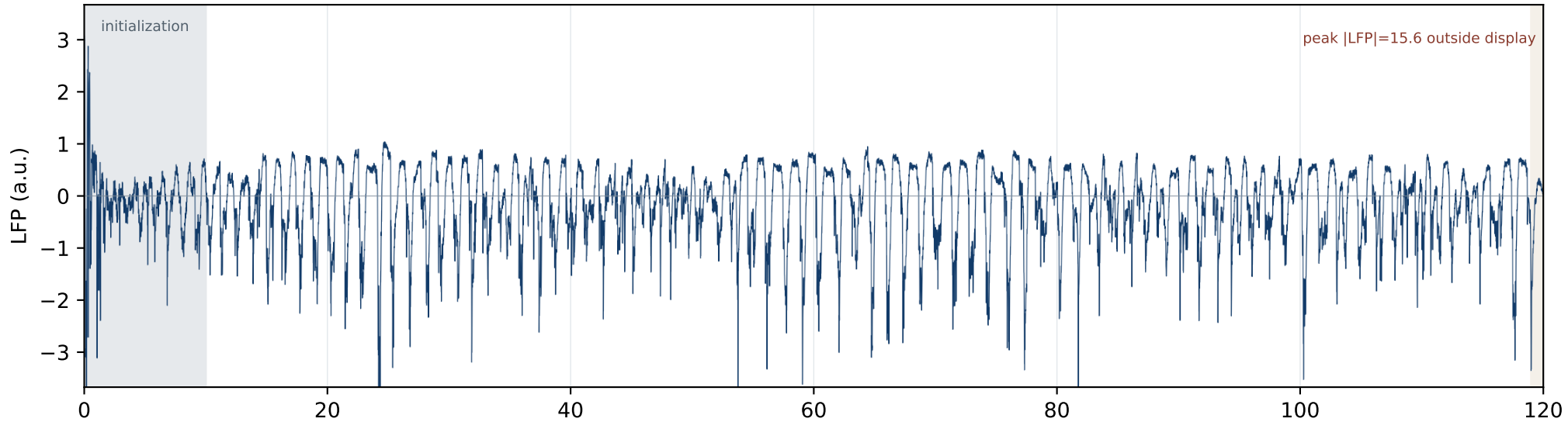
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state48_seed1_nhost=8.txt

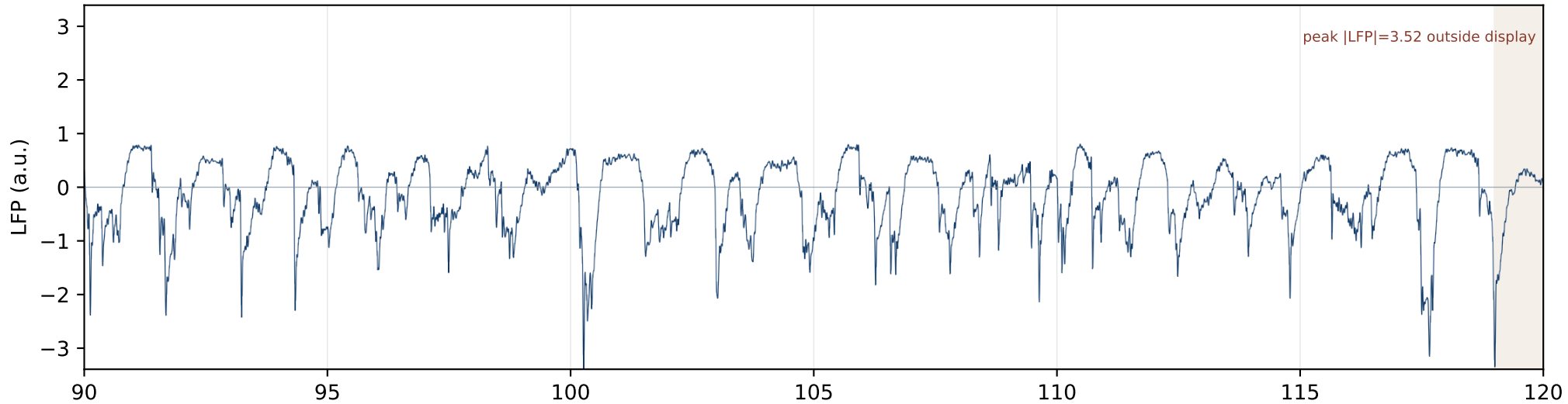
state_49

state 42 plus PYR/INH cortical K leak -2.5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.67 a.u.



Final 30 seconds independent scale: ± 3.39 a.u.



10-119 s metrics: RMS 0.746 a.u. | p99.9 |LFP| 3.34 a.u. | delta 91.6% | theta 3.5% | centroid 1.91 Hz

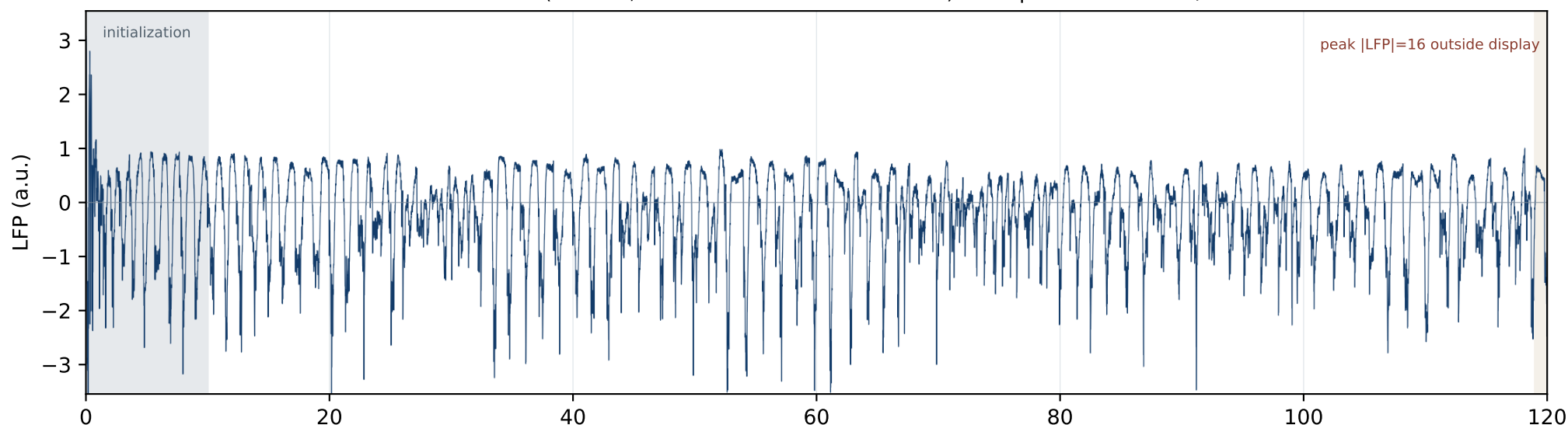
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state49_seed1_nhost=8.txt

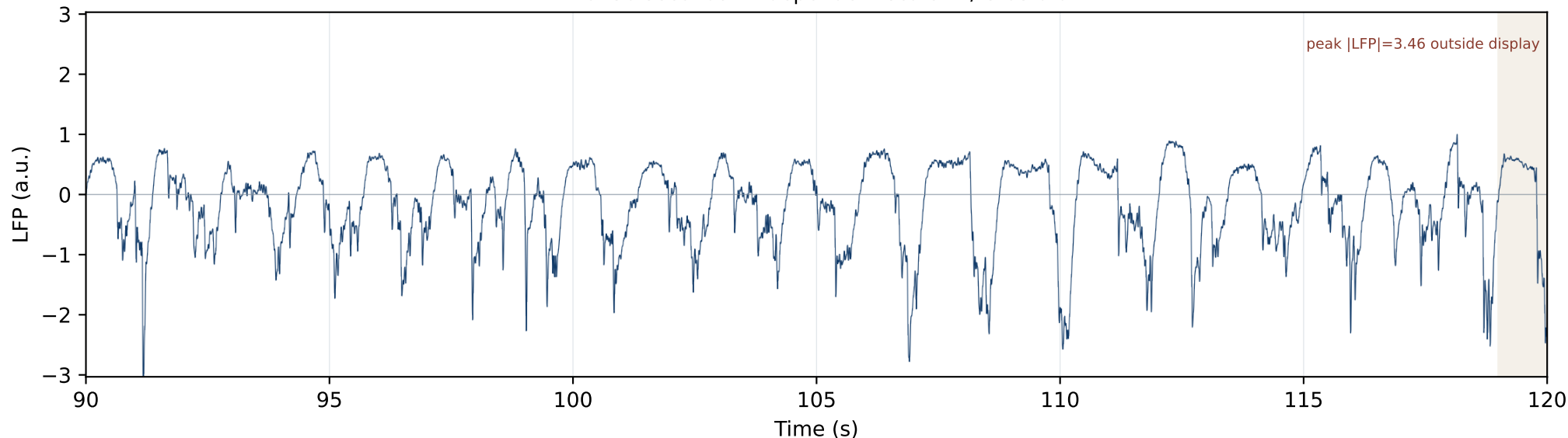
state_50

state 42 plus PYR K leak -5% only

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.54 a.u.



Final 30 seconds independent scale: ± 3.03 a.u.



10-119 s metrics: RMS 0.769 a.u. | p99.9 |LFP| 3.22 a.u. | delta 92.3% | theta 3.2% | centroid 1.83 Hz

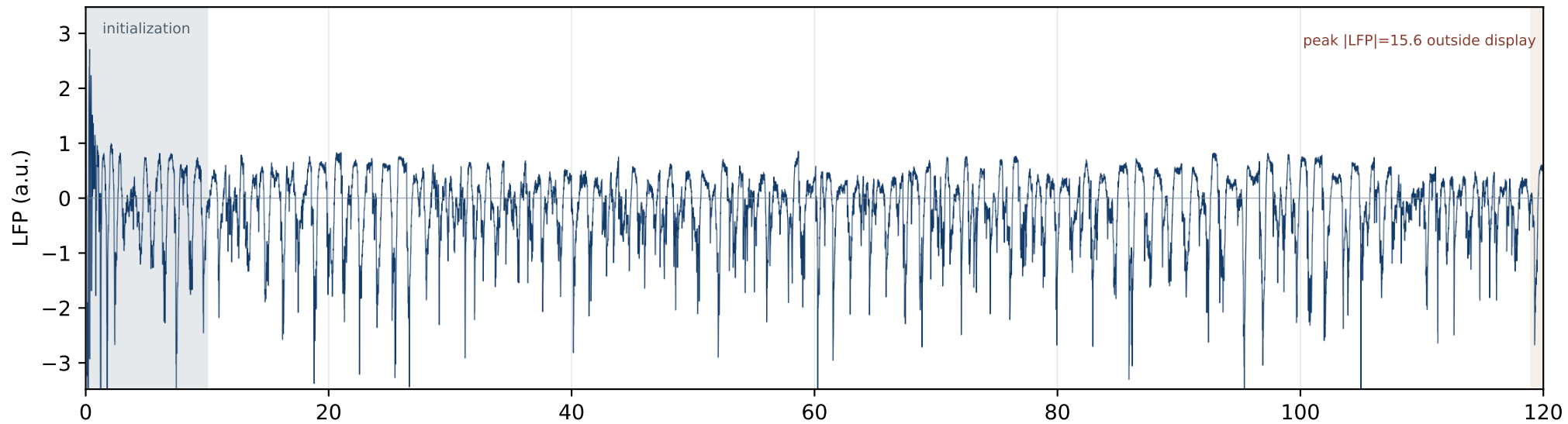
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state50_seed1_nhost=8.txt

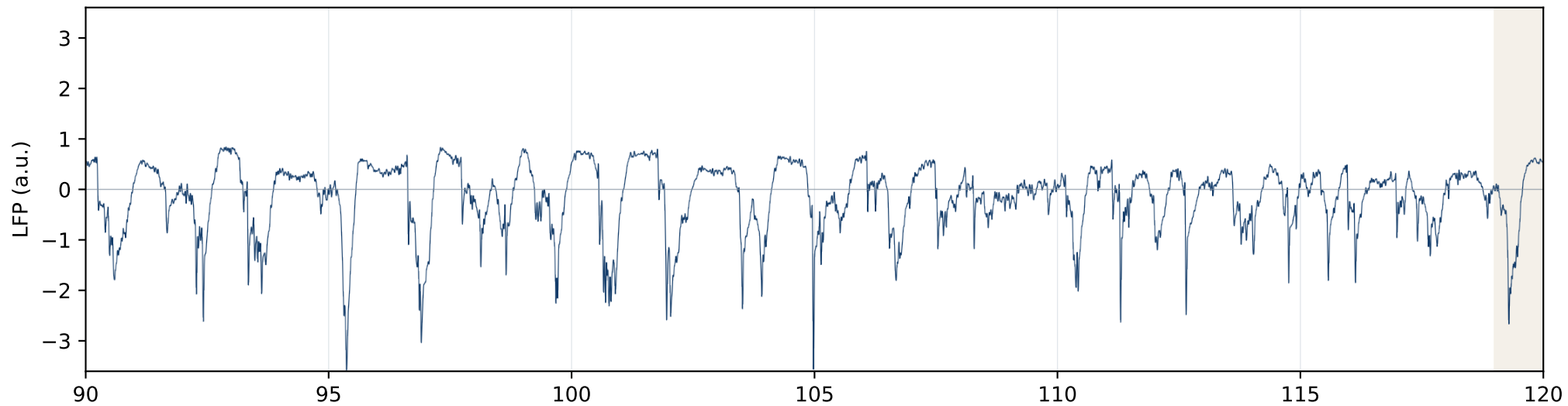
state_51

state 42 plus INH K leak -5% only

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.48 a.u.



Final 30 seconds independent scale: ± 3.6 a.u.



10-119 s metrics: RMS 0.66 a.u. | p99.9 |LFP| 3.16 a.u. | delta 88.5% | theta 4.8% | centroid 2.28 Hz

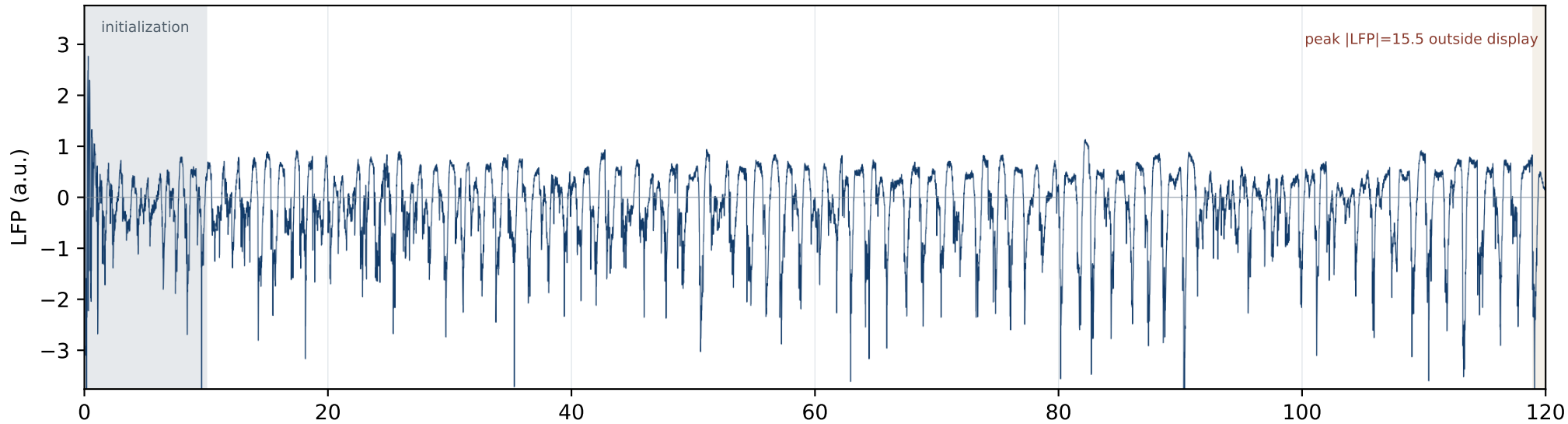
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state51_seed1_nhost=8.txt

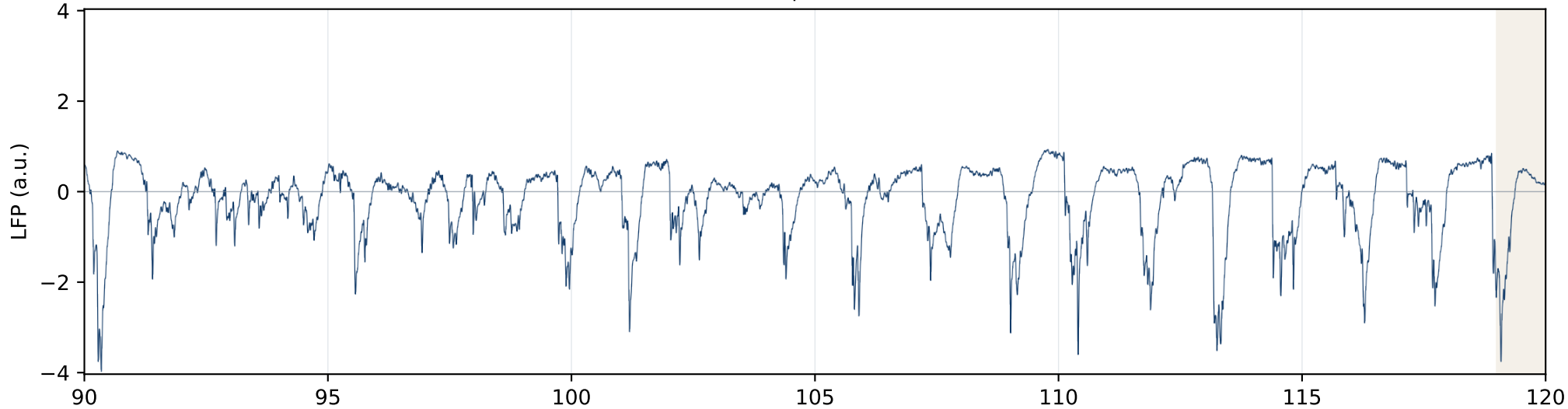
state_52

state 42 plus PYR K leak -2.5% only

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.76 a.u.



Final 30 seconds independent scale: ± 4.03 a.u.



10-119 s metrics: RMS 0.758 a.u. | p99.9 |LFP| 3.42 a.u. | delta 91.3% | theta 3.8% | centroid 1.92 Hz

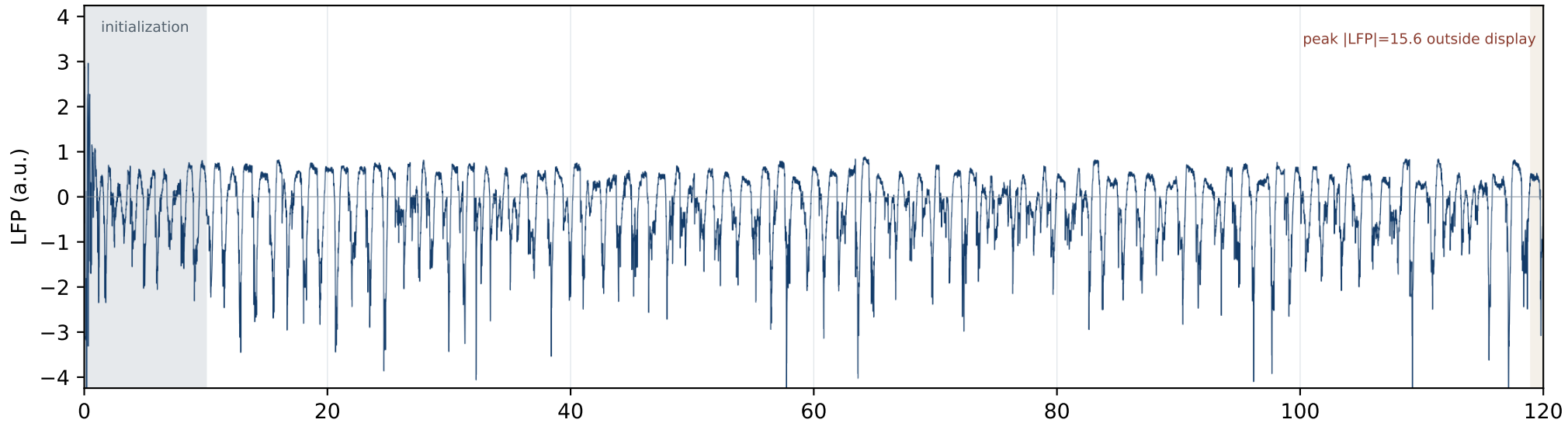
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state52_seed1_nhost=8.txt

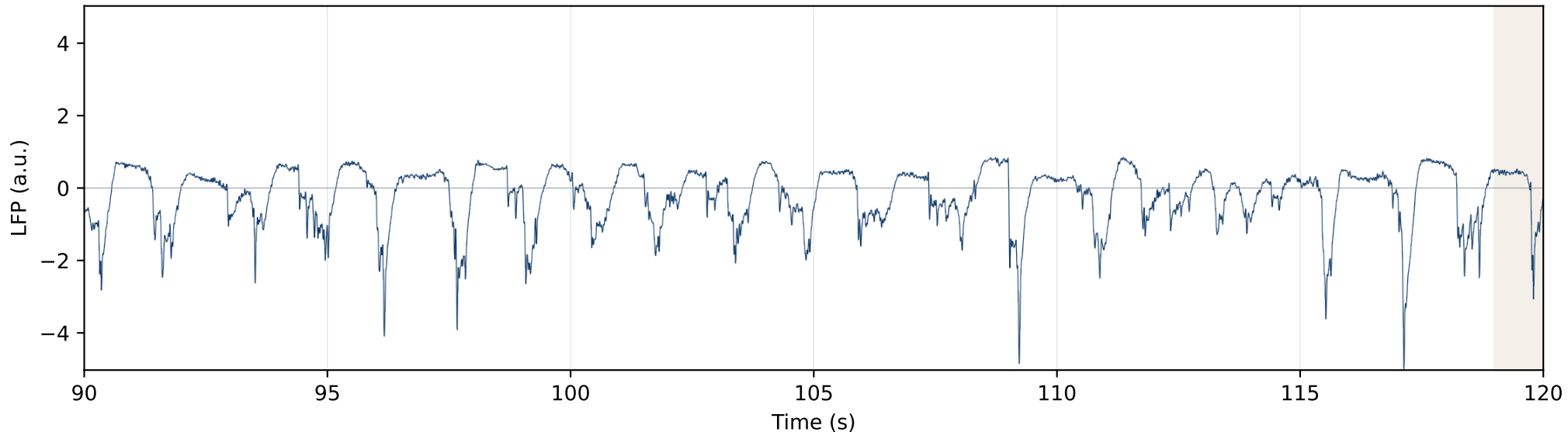
state_53

state 42 plus INH K leak -2.5% only

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 4.24 a.u.



Final 30 seconds independent scale: ± 5.03 a.u.



10-119 s metrics: RMS 0.808 a.u. | p99.9 |LFP| 3.86 a.u. | delta 91.8% | theta 3.5% | centroid 1.89 Hz

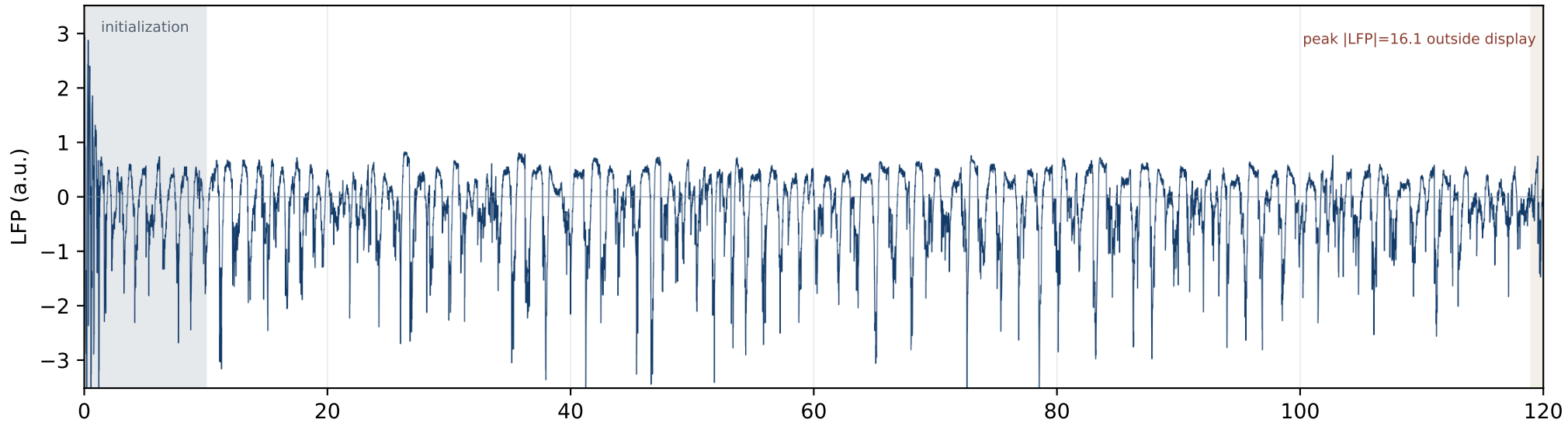
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state53_seed1_nhost=8.txt

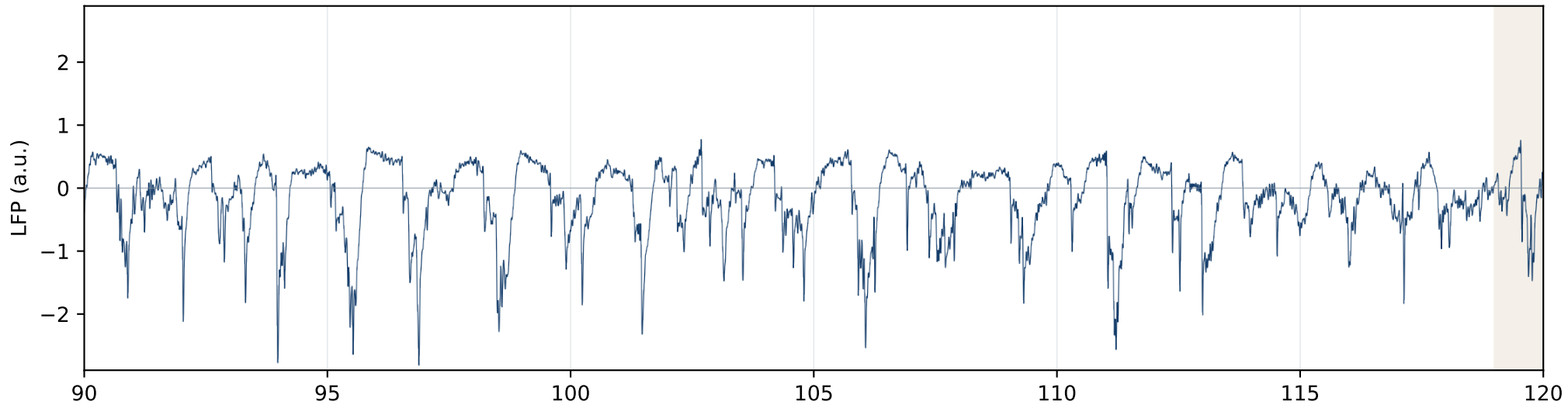
state_54

recurrence 2.10 and cortical K leak -2.5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.51 a.u.



Final 30 seconds independent scale: ± 2.89 a.u.



10-119 s metrics: RMS 0.691 a.u. | p99.9 |LFP| 3.2 a.u. | delta 87.8% | theta 5.4% | centroid 2.31 Hz

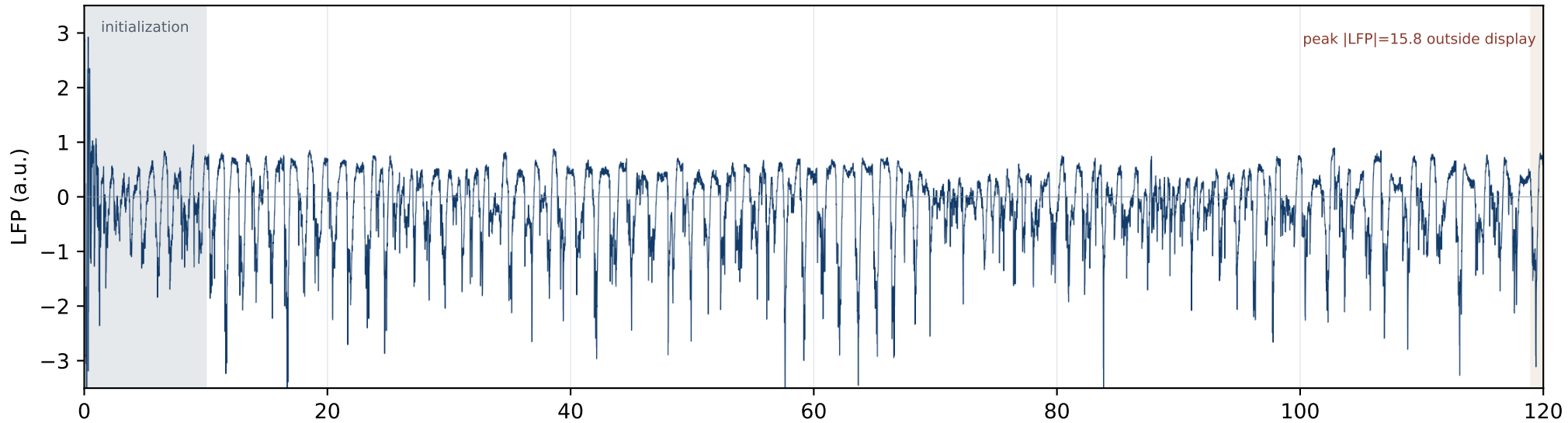
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state54_seed1_nhost=8.txt

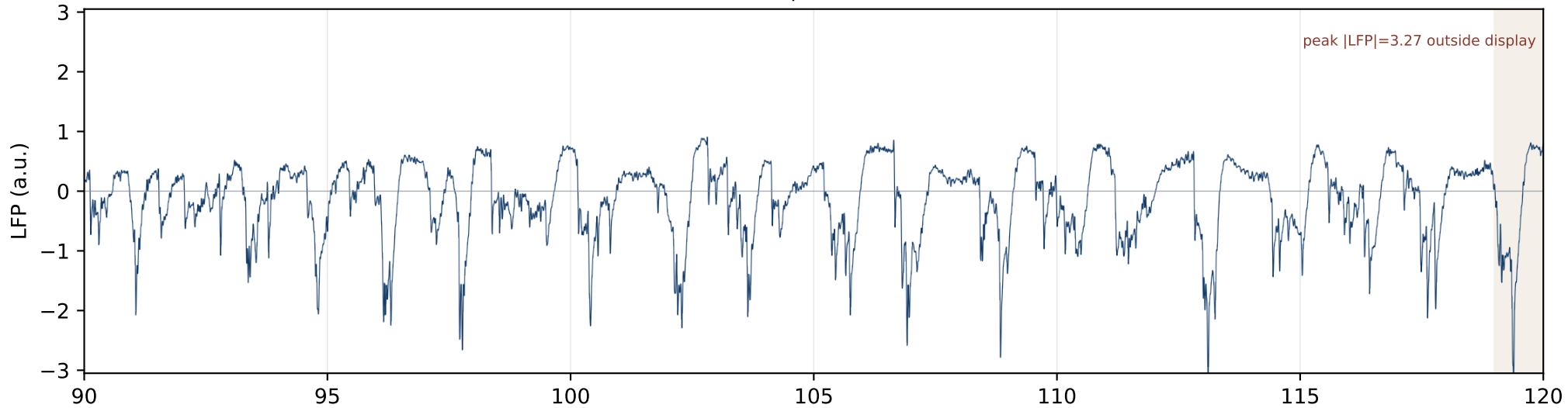
state_55

recurrence 2.15 and cortical K leak -2.5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: +/-3.5 a.u.



Final 30 seconds independent scale: +/-3.05 a.u.



10-119 s metrics: RMS 0.681 a.u. | p99.9 |LFP| 3.18 a.u. | delta 89.5% | theta 4.4% | centroid 2.13 Hz

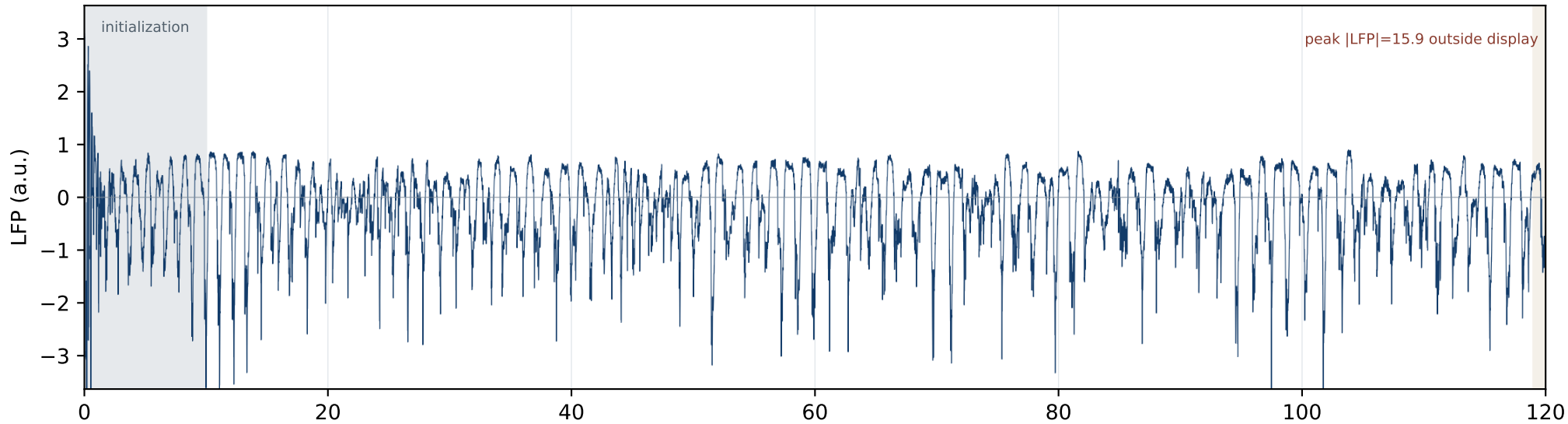
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state55_seed1_nhost=8.txt

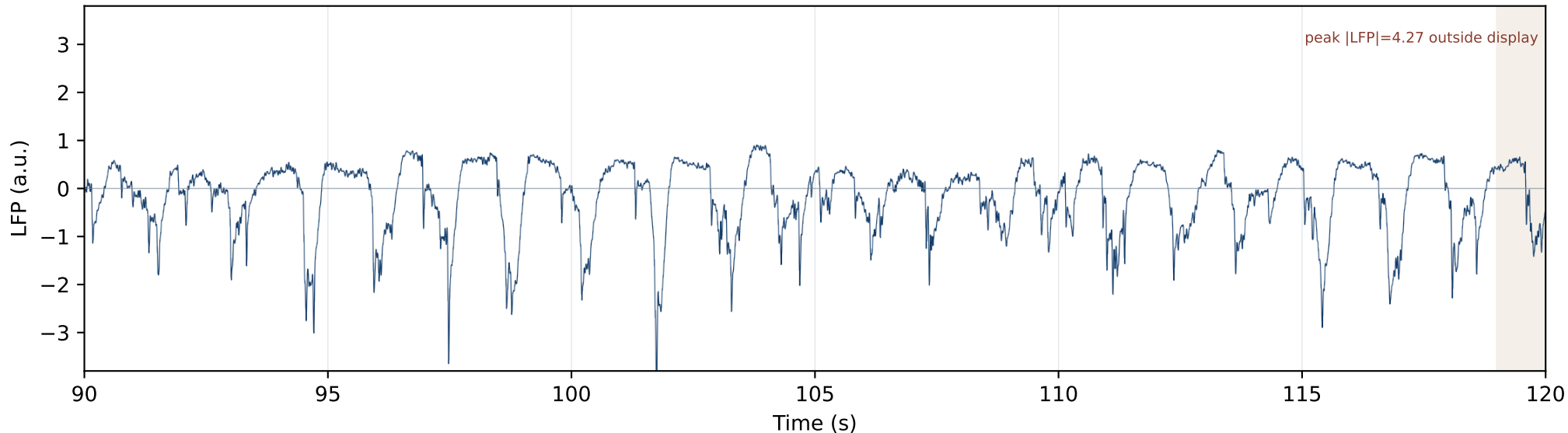
state_56

state 49 plus TC Ih activation shift -3 mV

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.63 a.u.



Final 30 seconds independent scale: ± 3.8 a.u.



10-119 s metrics: RMS 0.732 a.u. | p99.9 |LFP| 3.3 a.u. | delta 91.5% | theta 3.6% | centroid 1.91 Hz

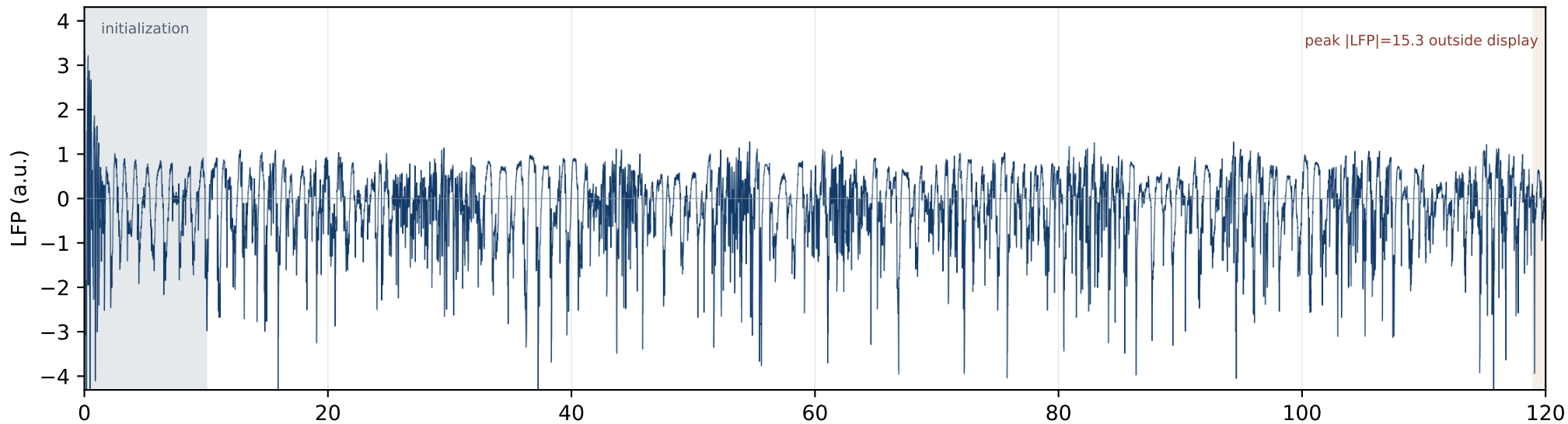
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state56_seed1_nhost=8.txt

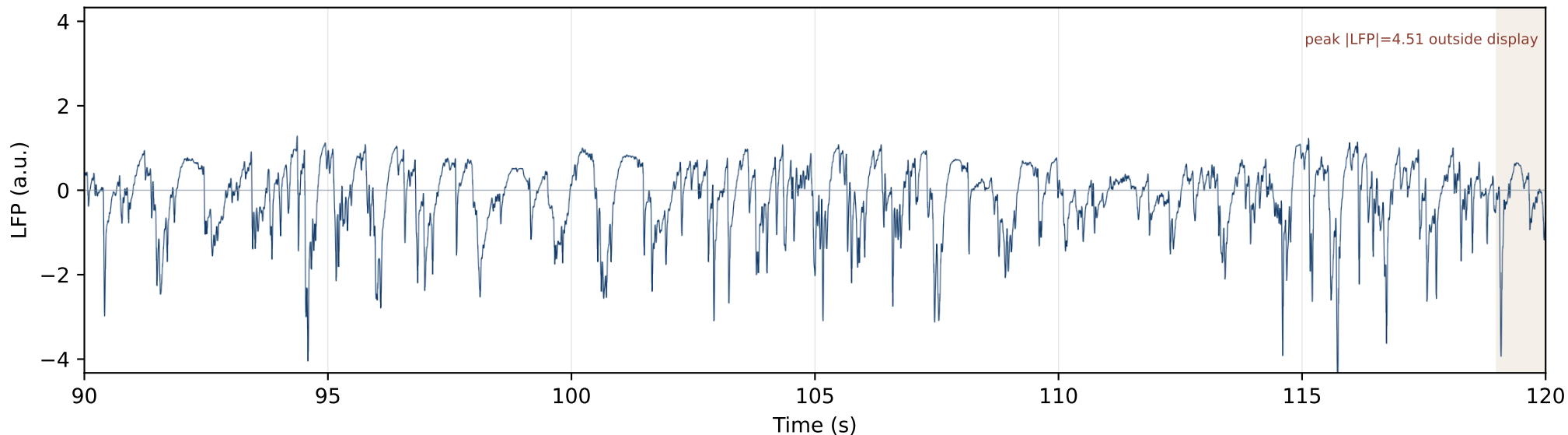
state_57

state 49 plus TC K leak halfway N3-to-N2

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 4.31 a.u.



Final 30 seconds independent scale: ± 4.33 a.u.



10-119 s metrics: RMS 0.811 a.u. | p99.9 $|LFP|$ 3.92 a.u. | delta 72.5% | theta 14.3% | centroid 3.71 Hz

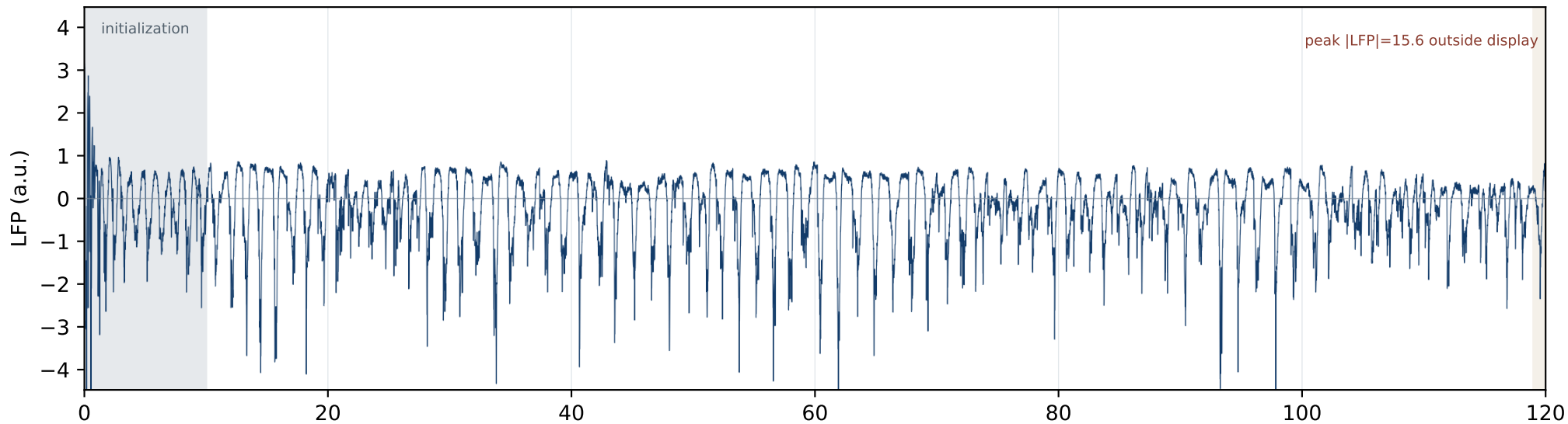
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Source: lfp_state57_seed1_nhost=8.txt

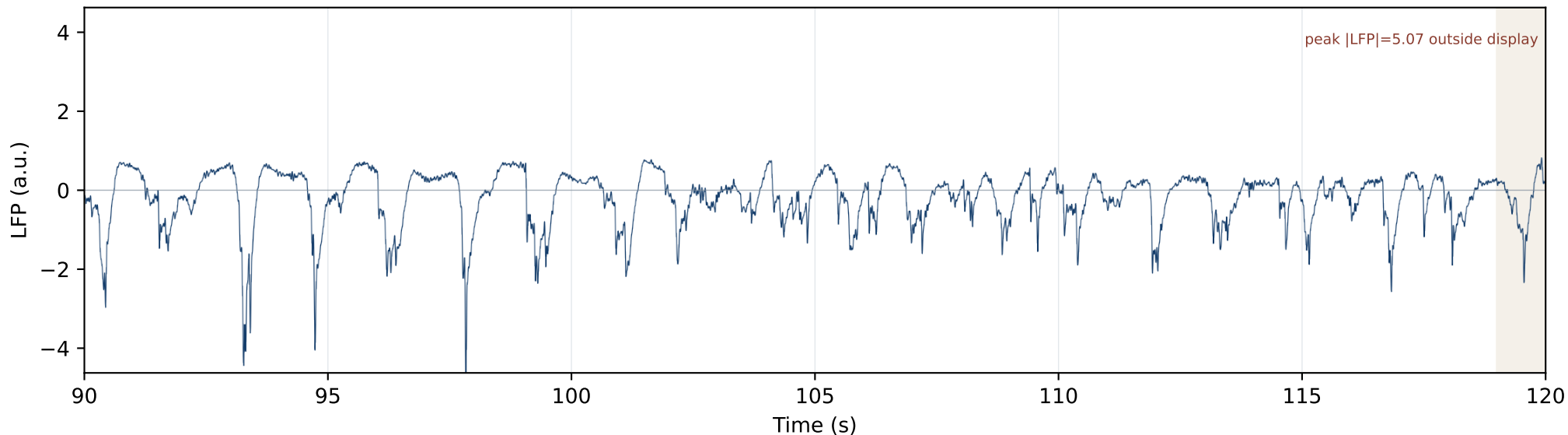
state_58

state 49 plus RE K leak halfway N3-to-N2

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 4.47 a.u.



Final 30 seconds independent scale: ± 4.63 a.u.



10-119 s metrics: RMS 0.801 a.u. | p99.9 $|LFP|$ 4.07 a.u. | delta 90.8% | theta 4.1% | centroid 1.96 Hz

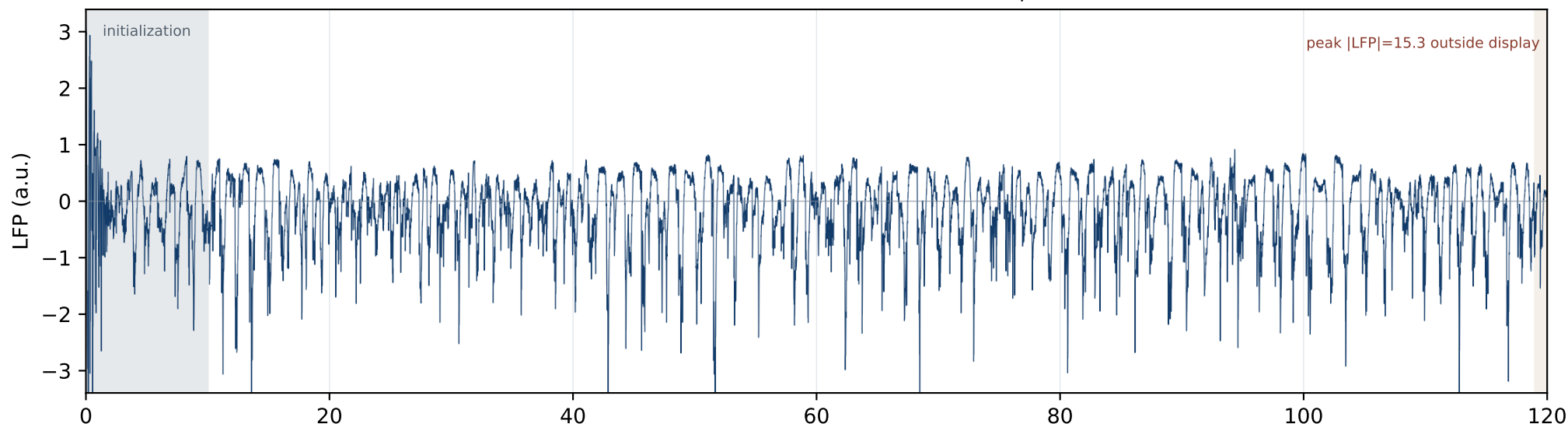
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state58_seed1_nhost=8.txt

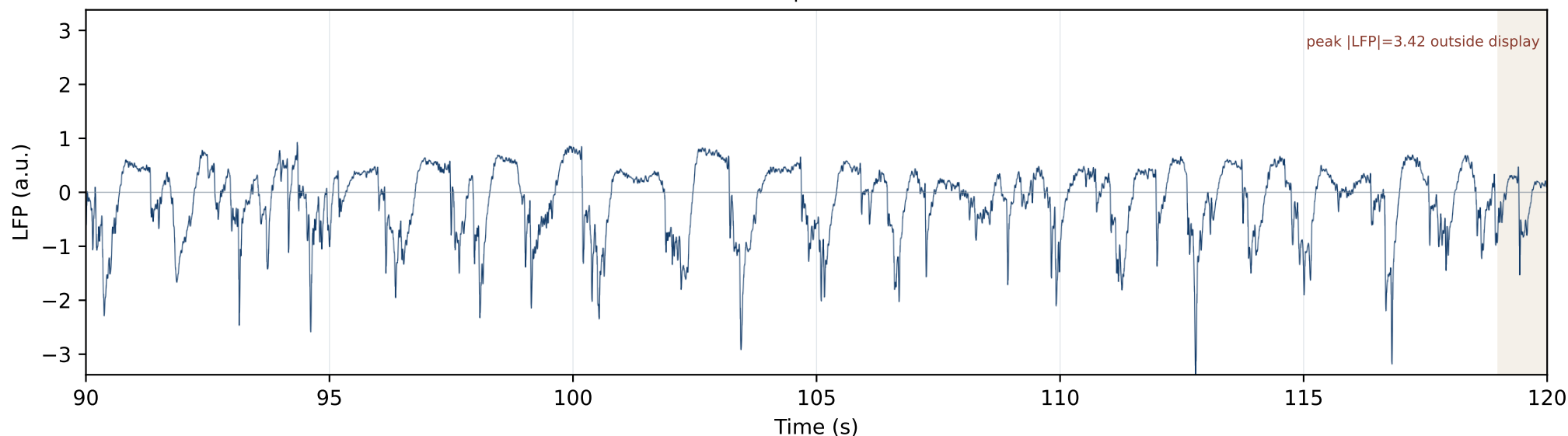
state_59

recurrence 2.15 only (matched control)

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.39 a.u.



Final 30 seconds independent scale: ± 3.38 a.u.



10-119 s metrics: RMS 0.647 a.u. | p99.9 |LFP| 3.09 a.u. | delta 86.6% | theta 5.8% | centroid 2.47 Hz

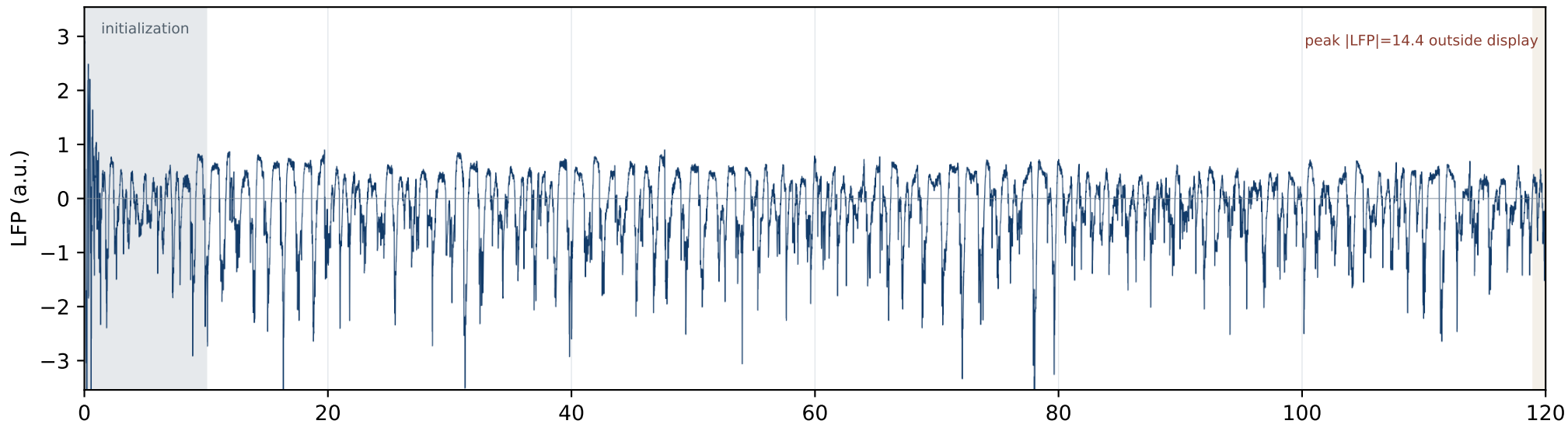
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state59_seed1_nhost=8.txt

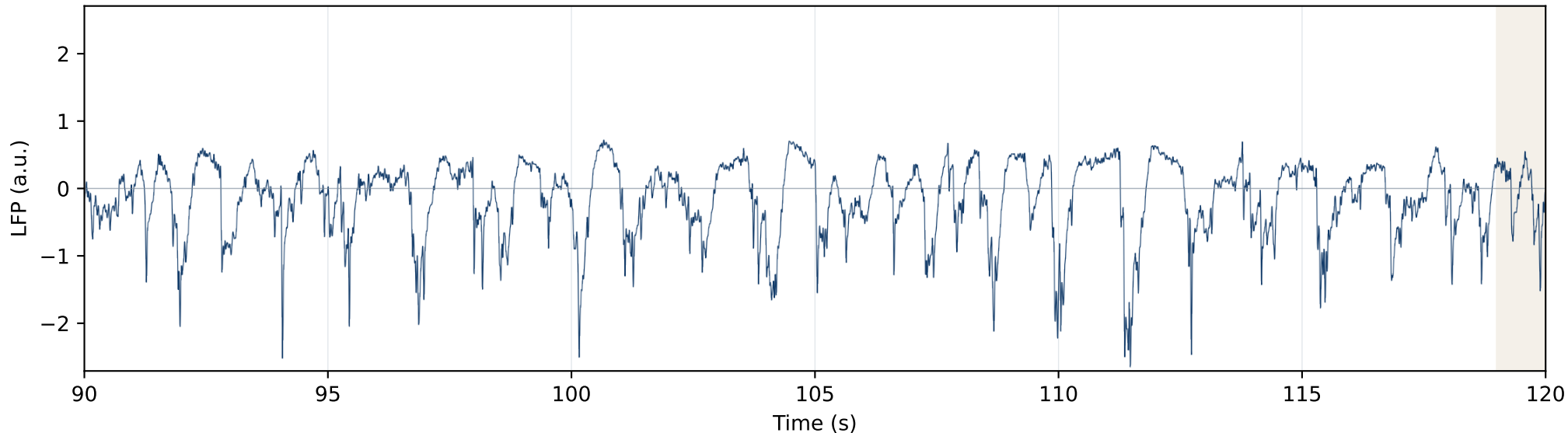
state_60

state 59 plus TC->PYR AMPA -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.54 a.u.



Final 30 seconds independent scale: ± 2.71 a.u.



10-119 s metrics: RMS 0.661 a.u. | p99.9 |LFP| 3.22 a.u. | delta 89.4% | theta 4.4% | centroid 2.16 Hz

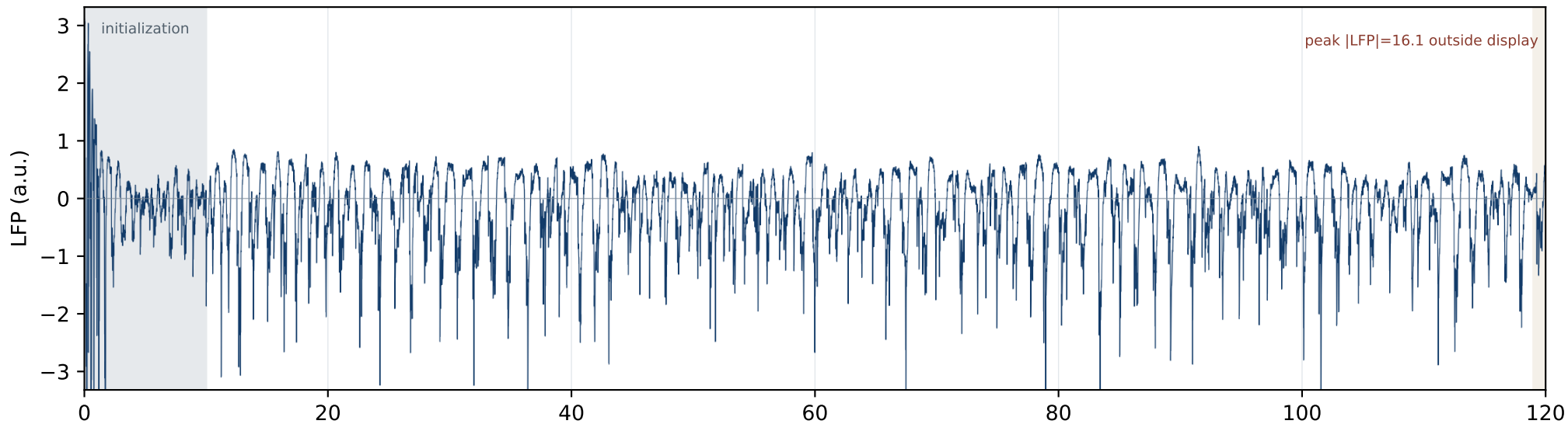
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state60_seed1_nhost=8.txt

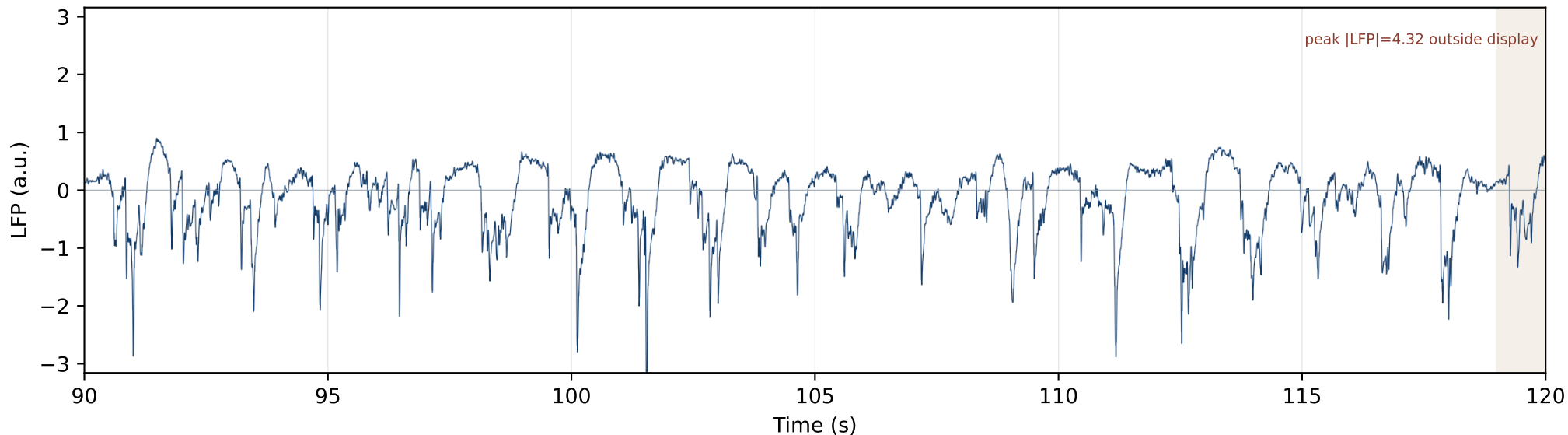
state_61

state 59 plus TC->PYR AMPA +10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.32 a.u.



Final 30 seconds independent scale: ± 3.16 a.u.



10-119 s metrics: RMS 0.641 a.u. | p99.9 |LFP| 3.02 a.u. | delta 85.4% | theta 6.3% | centroid 2.55 Hz

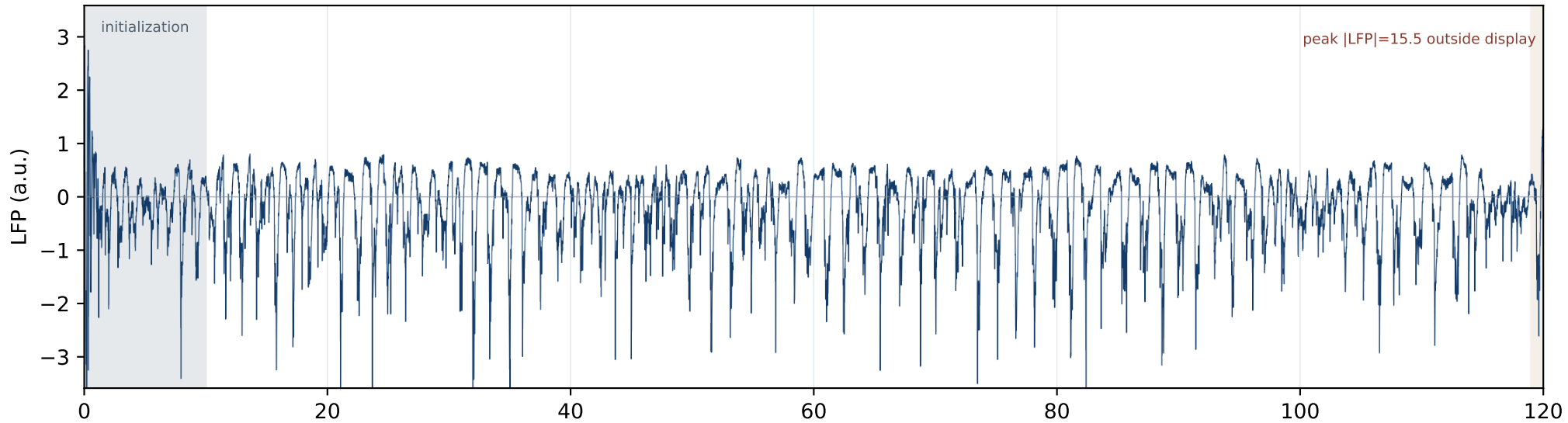
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state61_seed1_nhost=8.txt

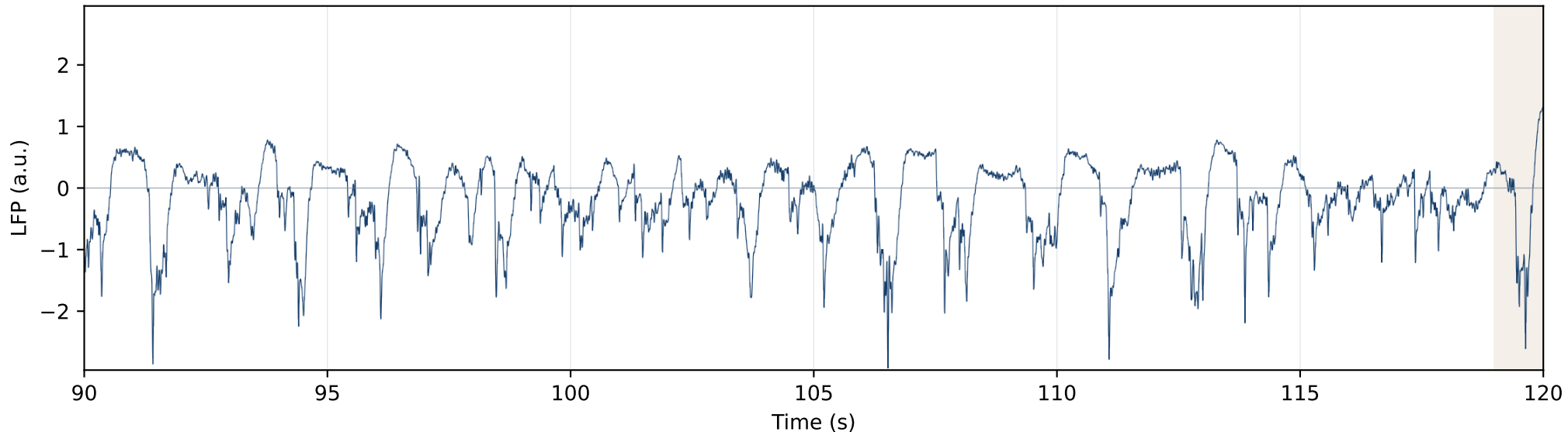
state_62

state 59 plus TC->INH AMPA -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.59 a.u.



Final 30 seconds independent scale: ± 2.96 a.u.



10-119 s metrics: RMS 0.695 a.u. | p99.9 |LFP| 3.26 a.u. | delta 88.6% | theta 4.7% | centroid 2.25 Hz

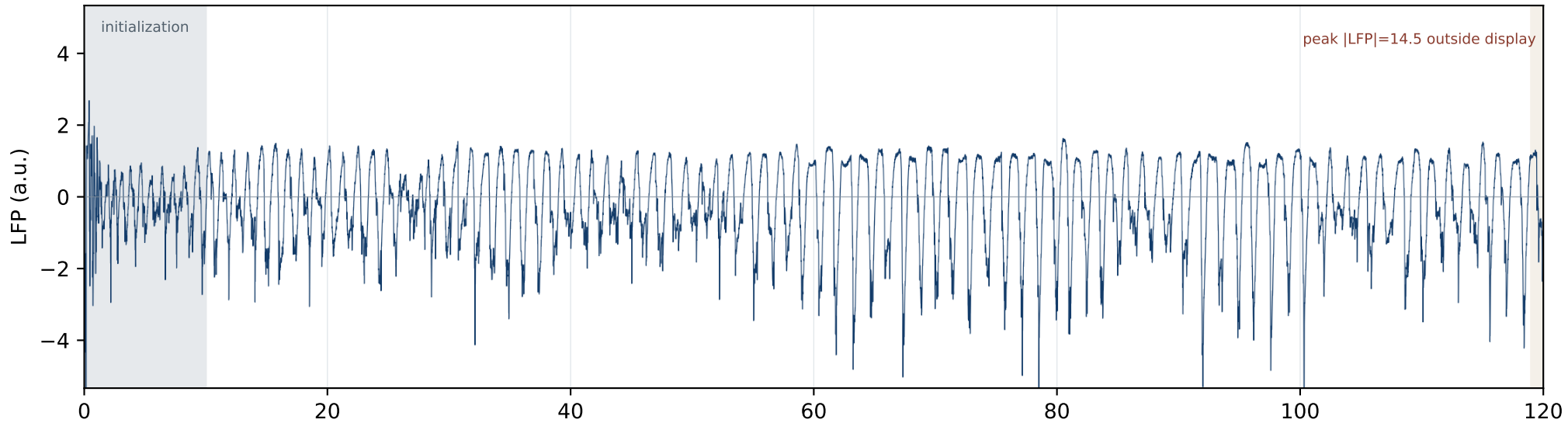
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state62_seed1_nhost=8.txt

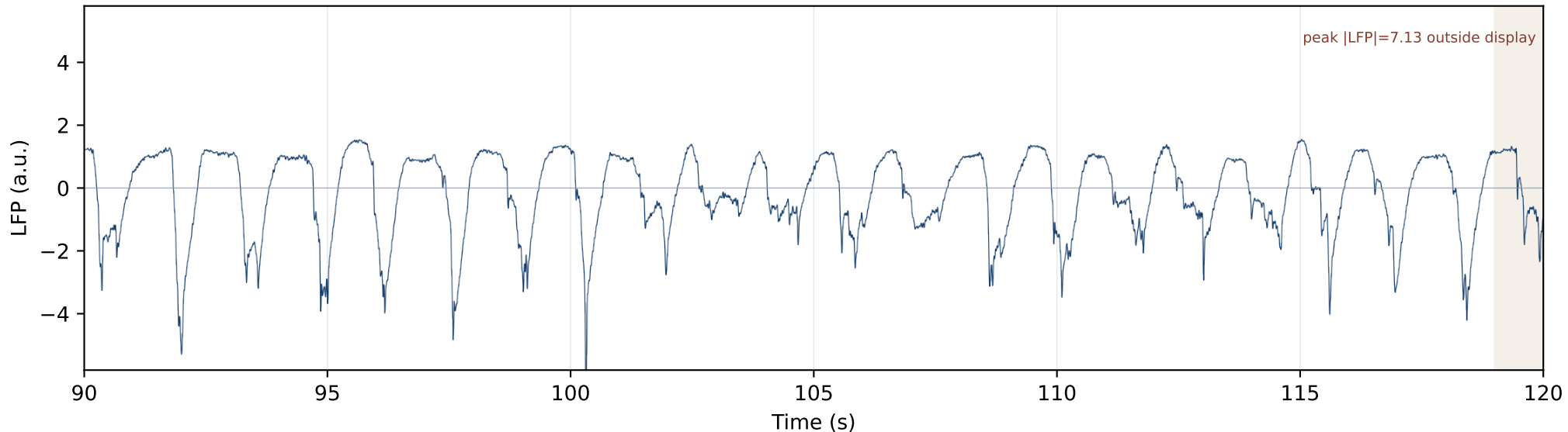
state_68

full N3 plus TC K leak -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 5.34 a.u.



Final 30 seconds independent scale: ± 5.8 a.u.



10-119 s metrics: RMS 1.2 a.u. | p99.9 |LFP| 4.85 a.u. | delta 96.8% | theta 1.5% | centroid 1.26 Hz

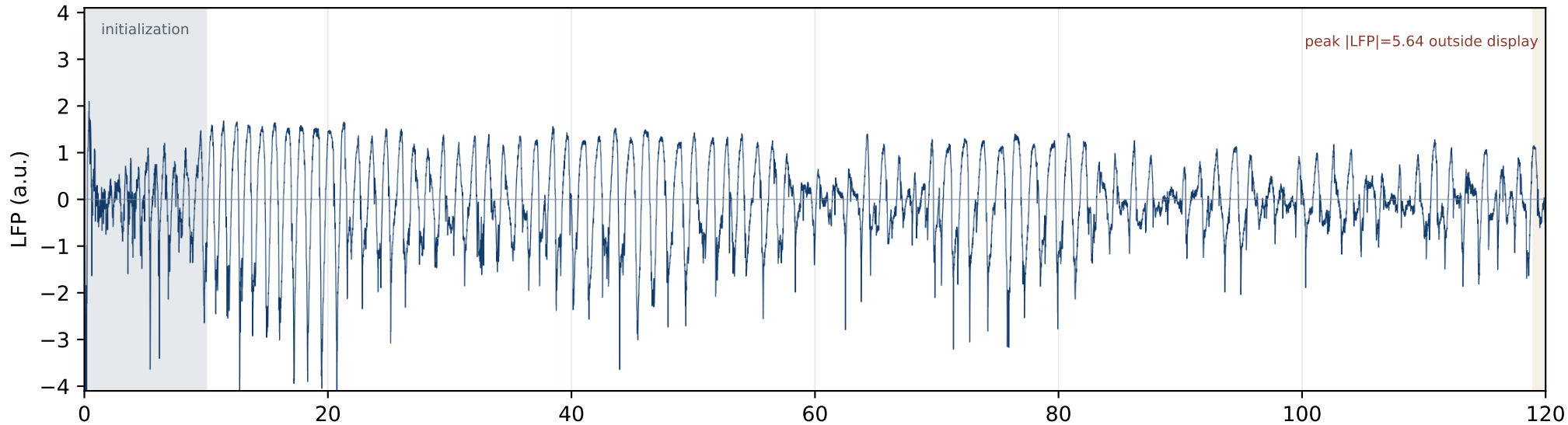
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Source: lfp_state68_seed1_nhost=8.txt

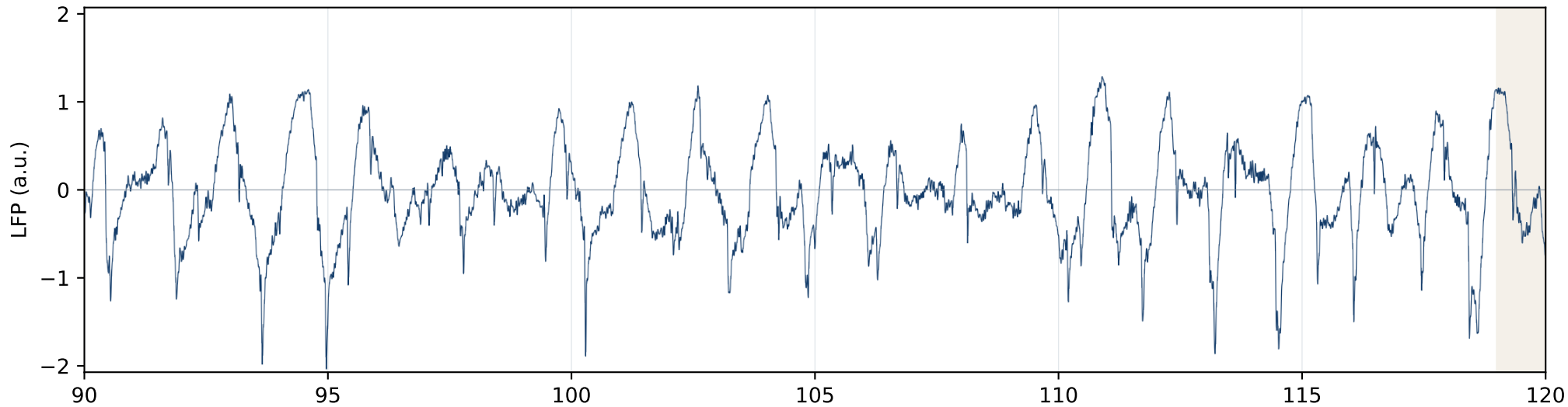
state_69

full N3 plus TC K leak +10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: +/-4.1 a.u.



Final 30 seconds independent scale: +/-2.07 a.u.



10-119 s metrics: RMS 0.897 a.u. | p99.9 |LFP| 3.73 a.u. | delta 96.8% | theta 1.4% | centroid 1.29 Hz

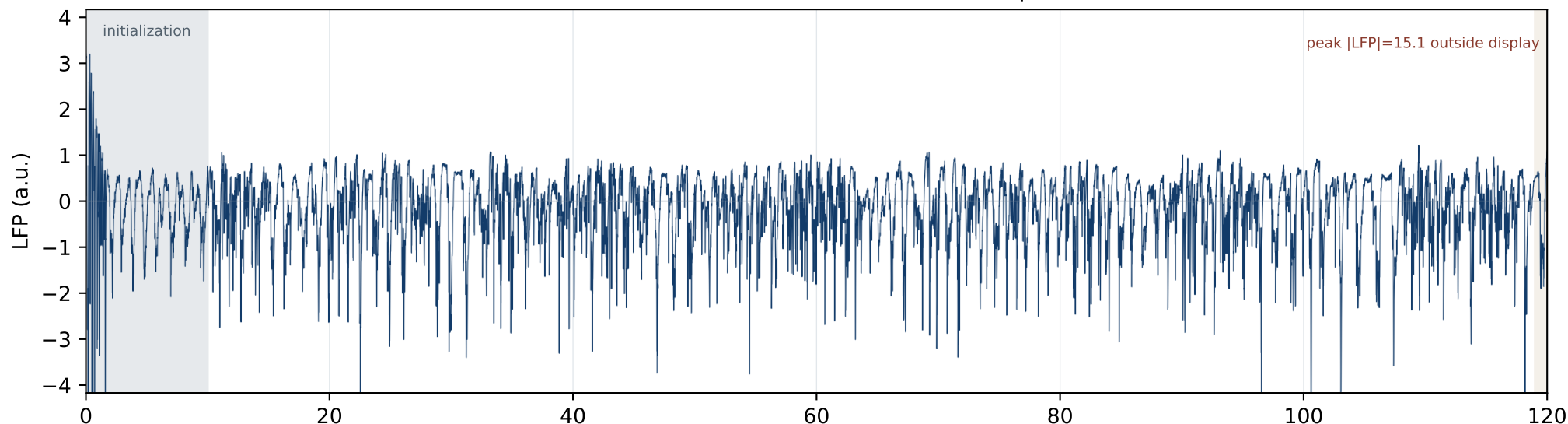
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state69_seed1_nhost=8.txt

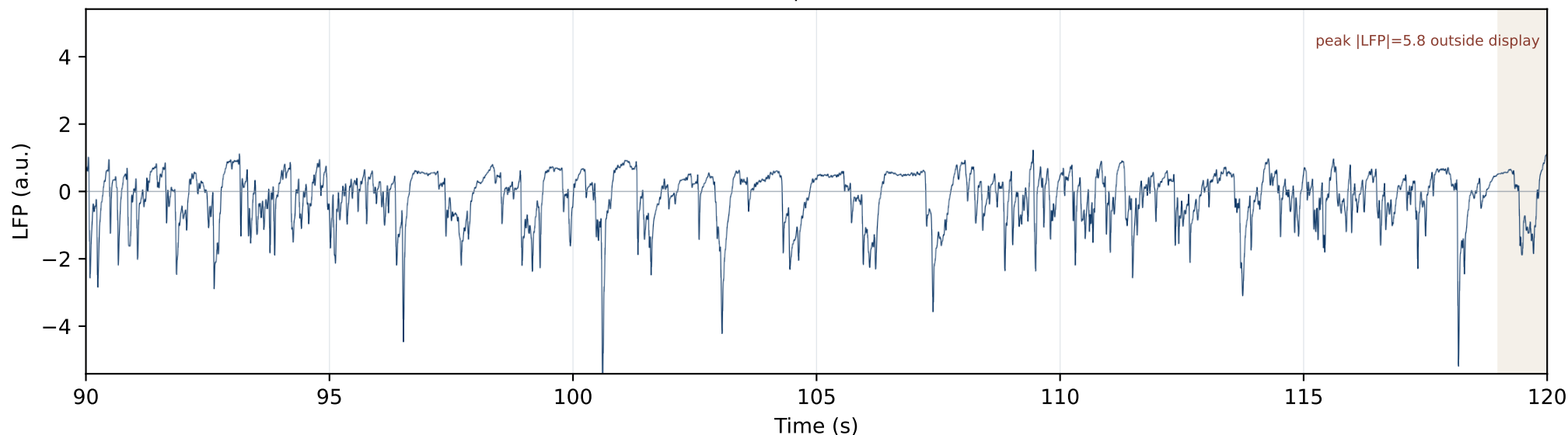
state_70

state 59 plus TC K leak -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 4.17 a.u.



Final 30 seconds independent scale: ± 5.41 a.u.



10-119 s metrics: RMS 0.764 a.u. | p99.9 |LFP| 3.79 a.u. | delta 72.3% | theta 14.4% | centroid 3.68 Hz

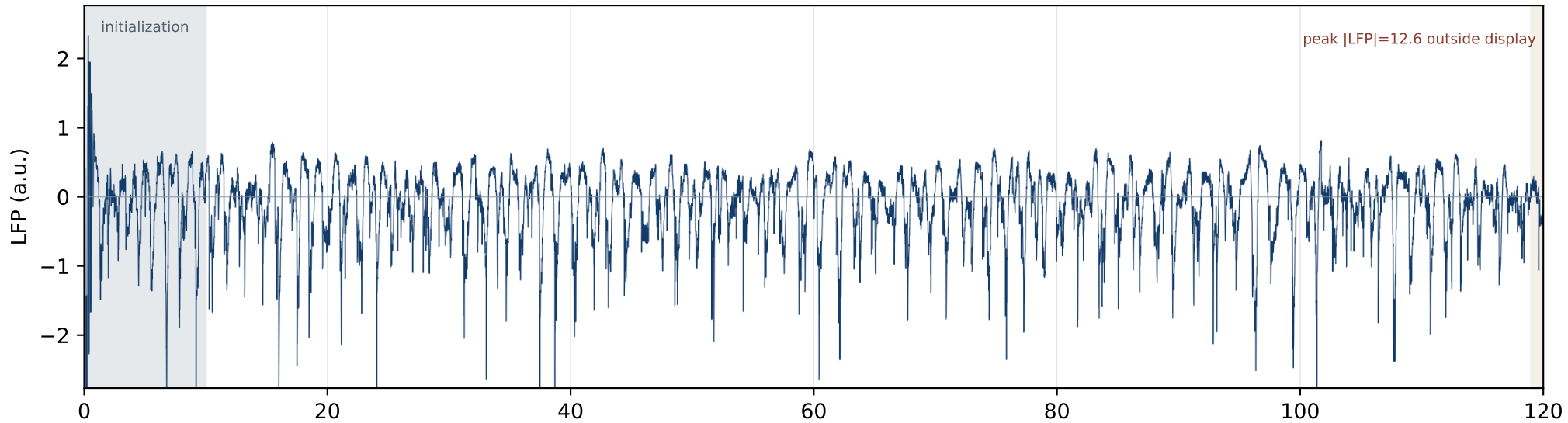
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Source: lfp_state70_seed1_nhost=8.txt

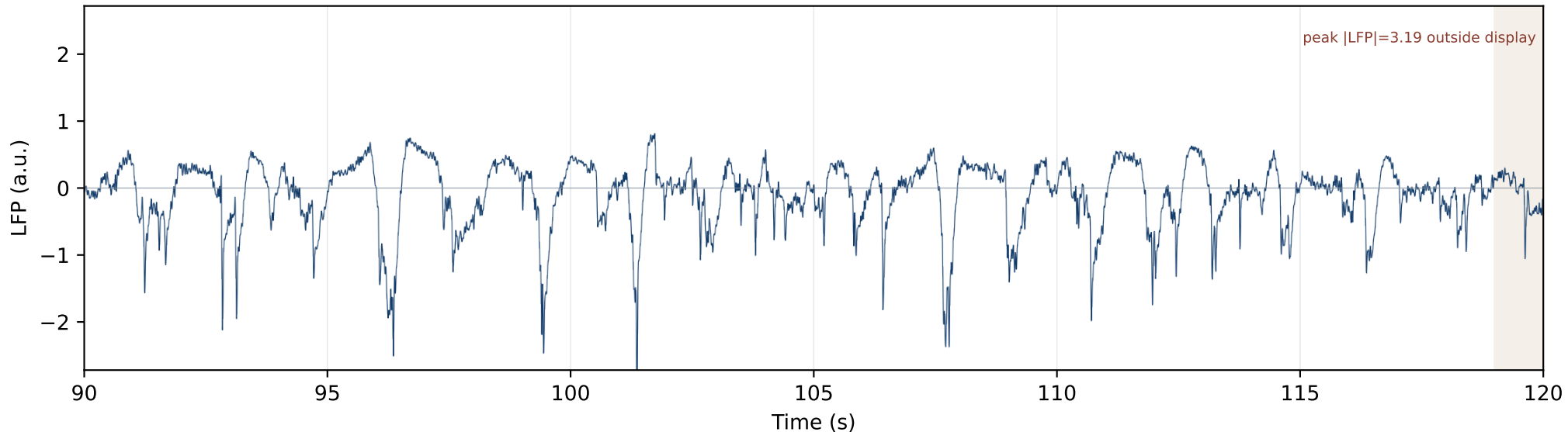
state_71

state 59 plus TC K leak +10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 2.77 a.u.



Final 30 seconds independent scale: ± 2.72 a.u.



10-119 s metrics: RMS 0.503 a.u. | p99.9 |LFP| 2.52 a.u. | delta 88.3% | theta 4.6% | centroid 2.34 Hz

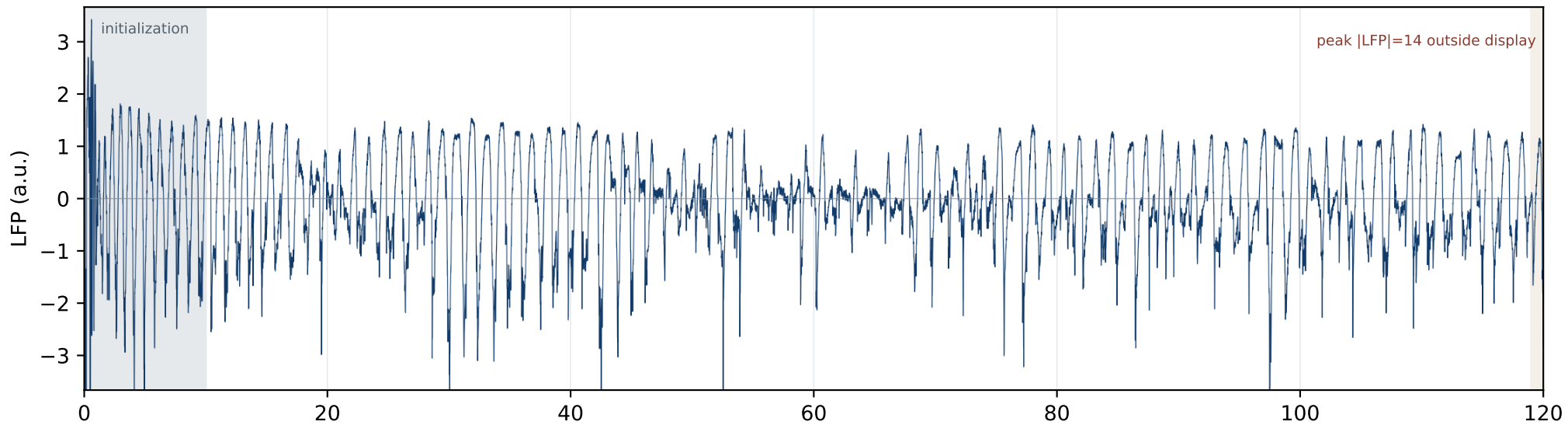
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state71_seed1_nhost=8.txt

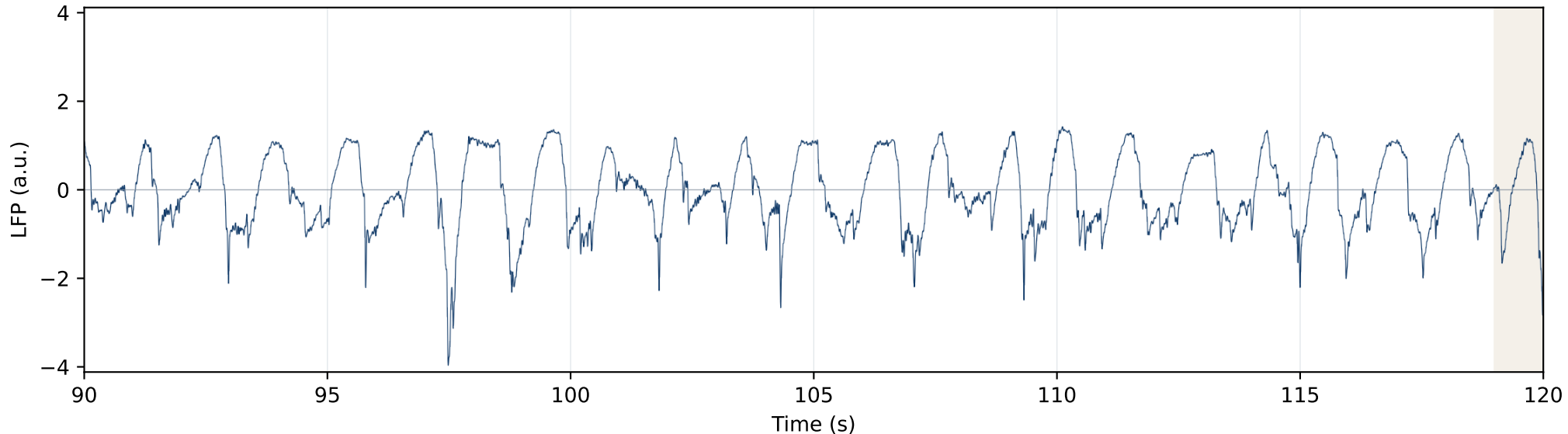
state_74

full N3 plus PYR K leak +5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.66 a.u.



Final 30 seconds independent scale: ± 4.12 a.u.



10-119 s metrics: RMS 0.848 a.u. | p99.9 |LFP| 3.33 a.u. | delta 95.7% | theta 1.9% | centroid 1.43 Hz

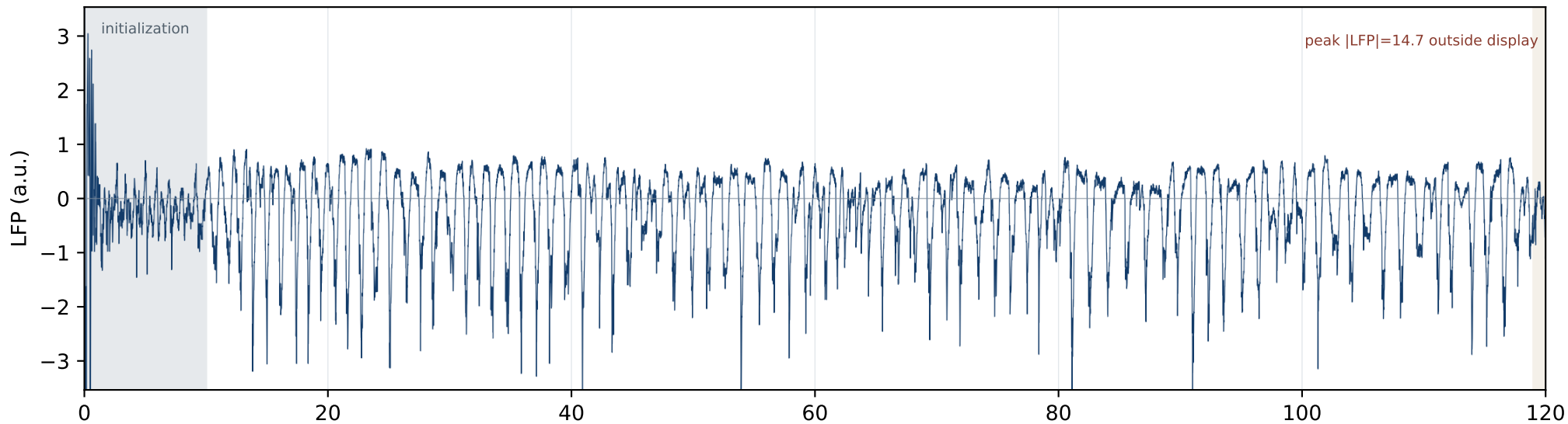
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Source: lfp_state74_seed1_nhost=8.txt

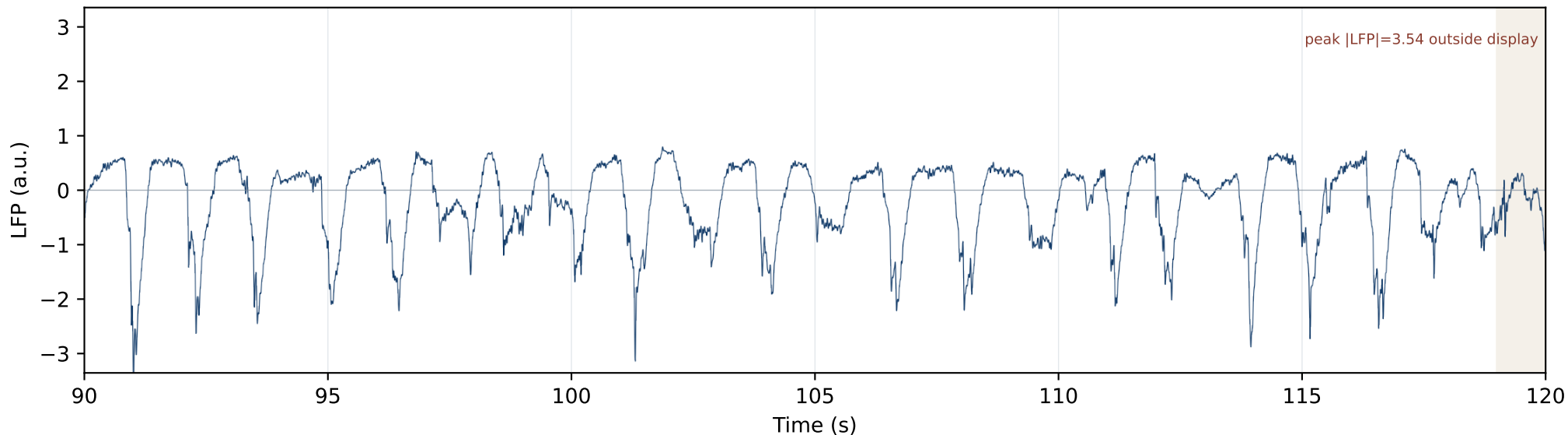
state_75

full N3 plus PYR K leak +10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.54 a.u.



Final 30 seconds independent scale: ± 3.36 a.u.



10-119 s metrics: RMS 0.788 a.u. | p99.9 |LFP| 3.21 a.u. | delta 94.9% | theta 2.5% | centroid 1.53 Hz

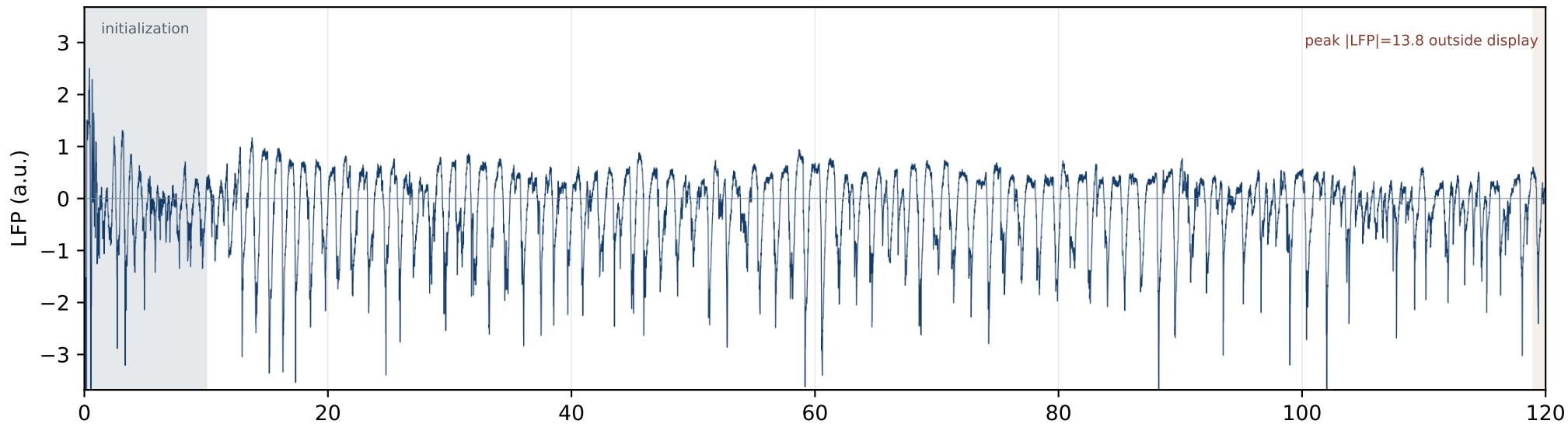
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state75_seed1_nhost=8.txt

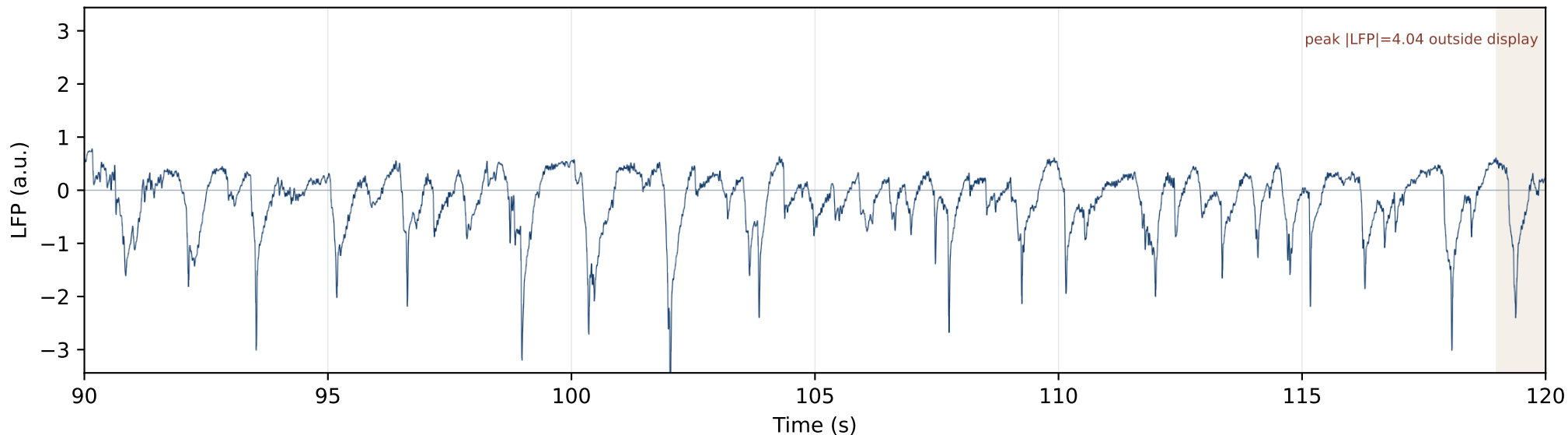
state_76

full N3 plus INH K leak +5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.68 a.u.



Final 30 seconds independent scale: ± 3.44 a.u.



10-119 s metrics: RMS 0.744 a.u. | p99.9 |LFP| 3.35 a.u. | delta 94.1% | theta 2.9% | centroid 1.61 Hz

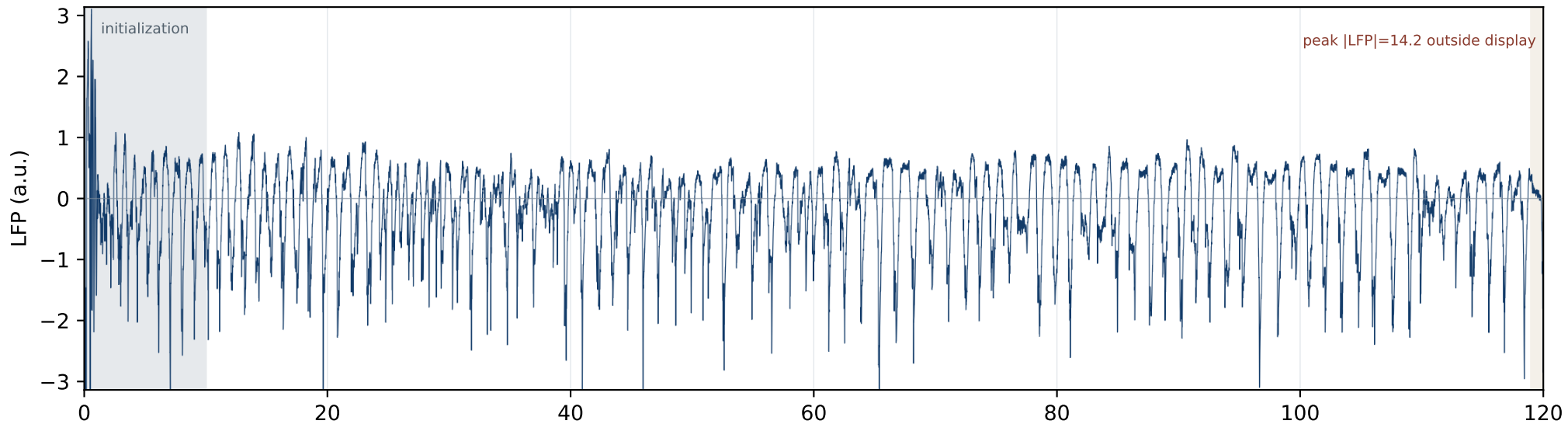
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Source: lfp_state76_seed1_nhost=8.txt

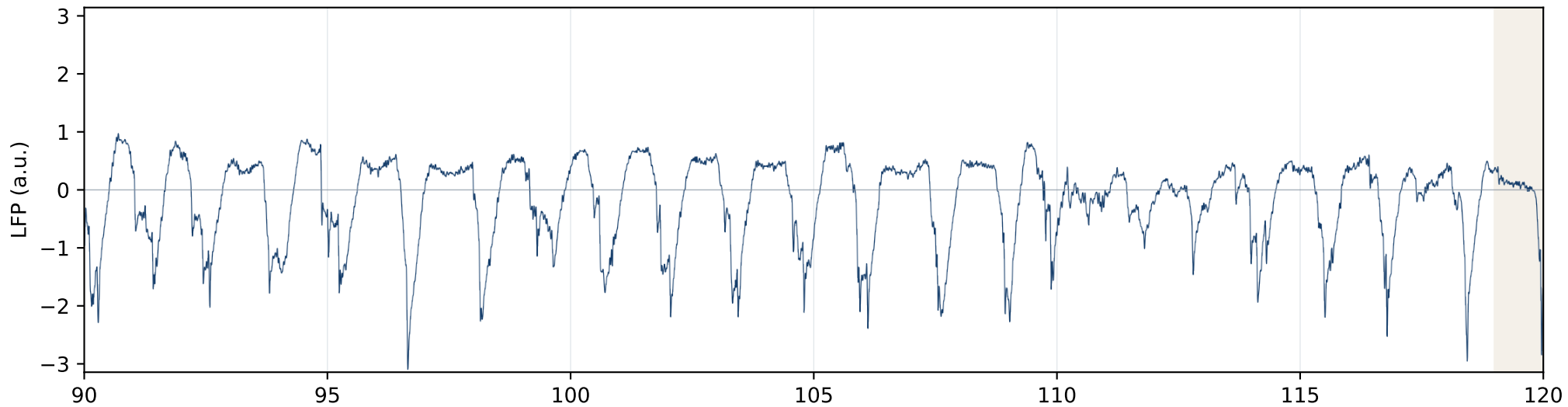
state_77

full N3 plus PYR/INH K leak +5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.14 a.u.



Final 30 seconds independent scale: ± 3.14 a.u.



10-119 s metrics: RMS 0.708 a.u. | p99.9 |LFP| 2.85 a.u. | delta 94.0% | theta 2.6% | centroid 1.63 Hz

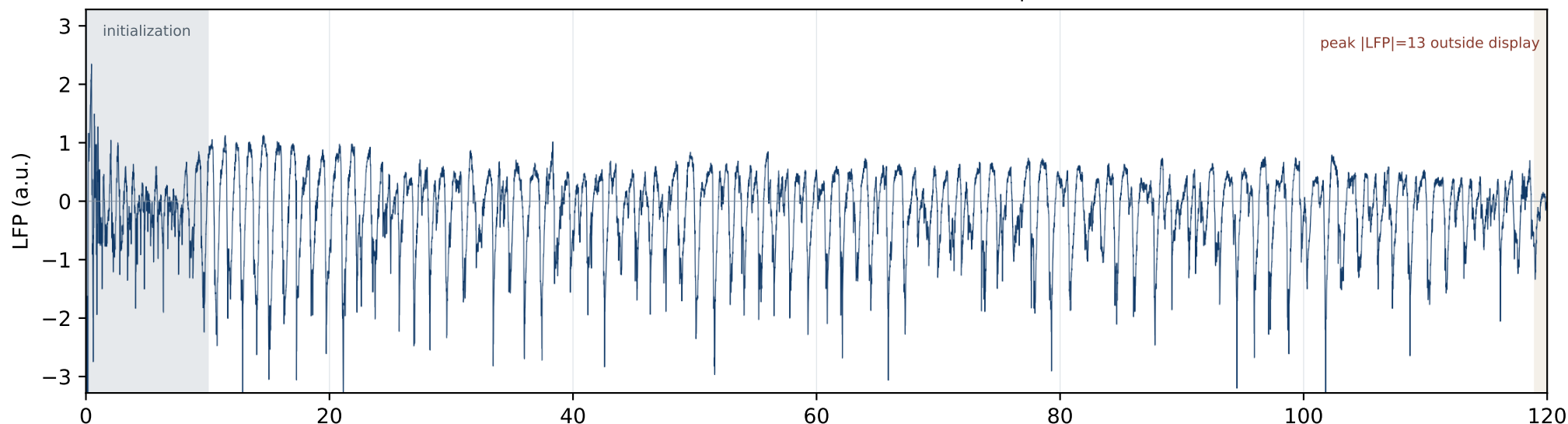
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state77_seed1_nhost=8.txt

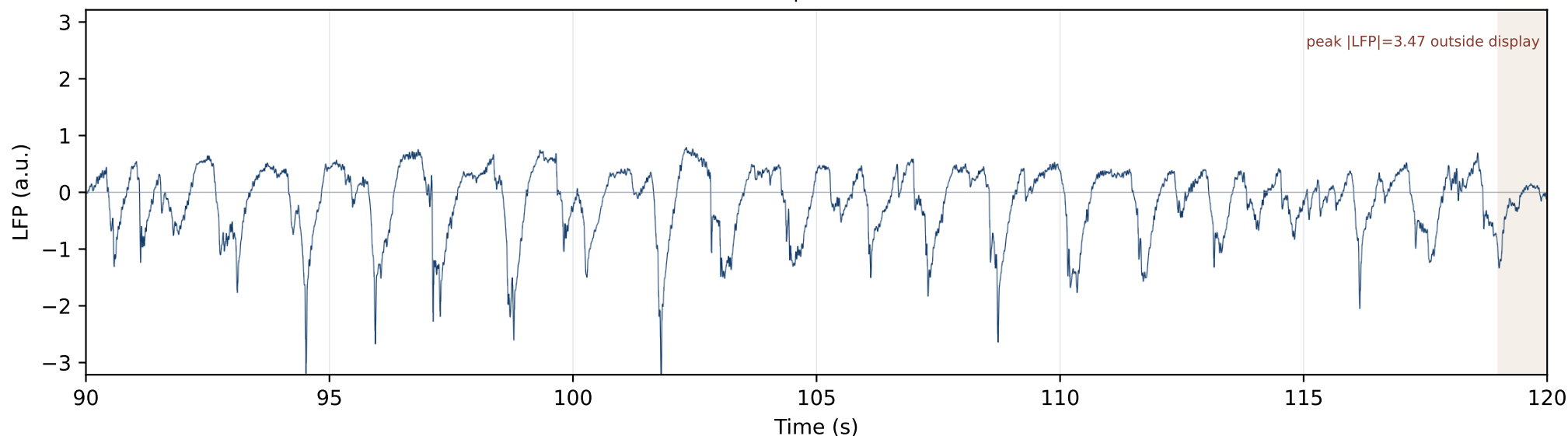
state_78

full N3 plus TC->PYR/INH AMPA -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.28 a.u.



Final 30 seconds independent scale: ± 3.21 a.u.



10-119 s metrics: RMS 0.708 a.u. | p99.9 |LFP| 2.98 a.u. | delta 94.8% | theta 2.3% | centroid 1.58 Hz

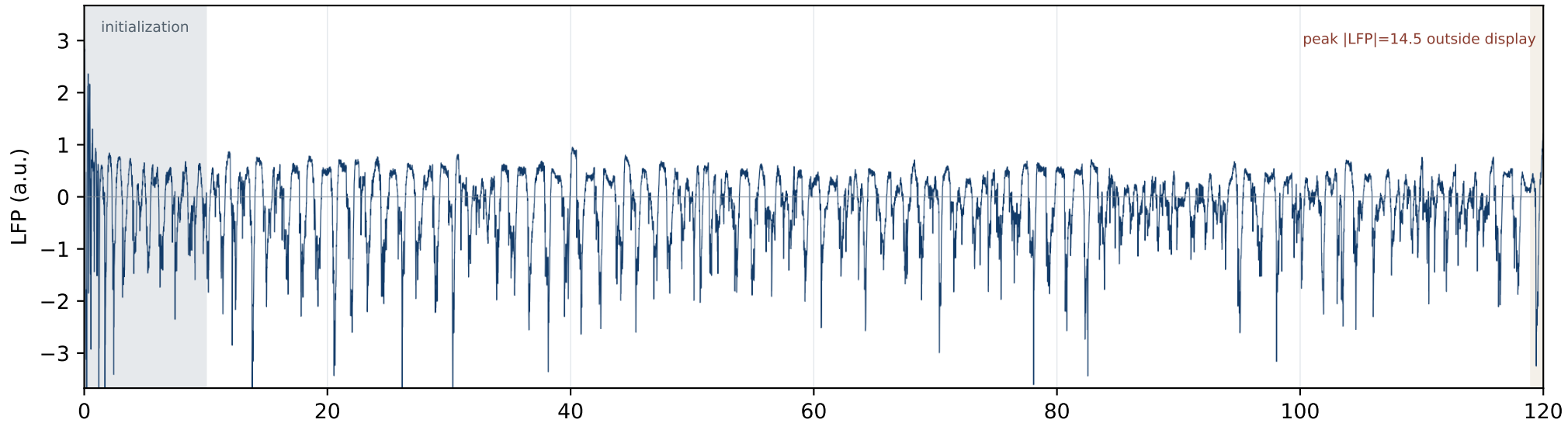
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state78_seed1_nhost=8.txt

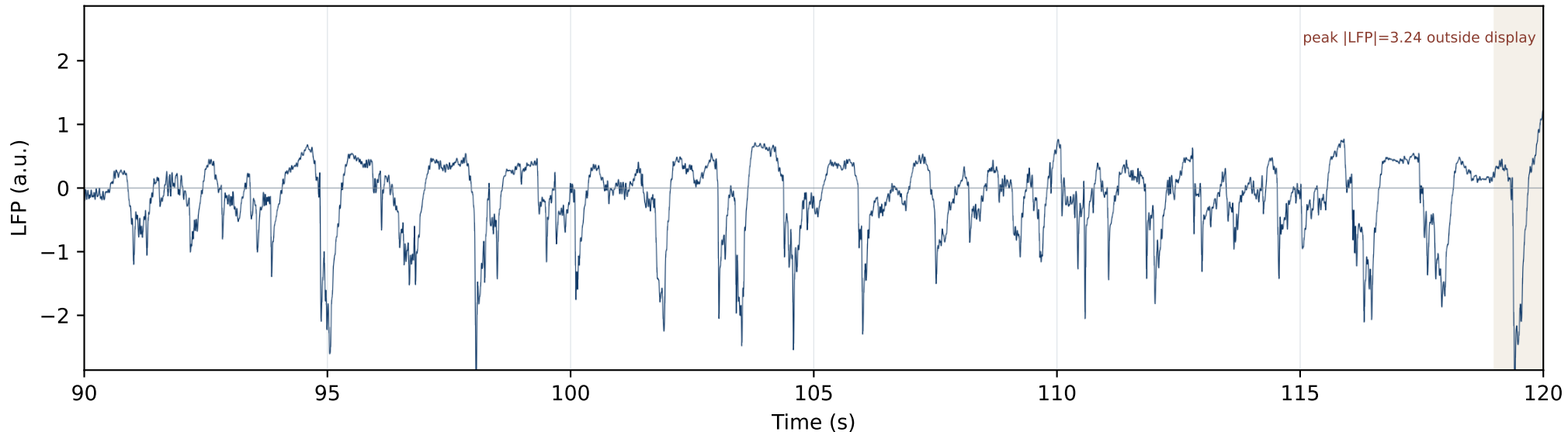
state_79

state 59 plus TC->PYR/INH AMPA -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.67 a.u.



Final 30 seconds independent scale: ± 2.86 a.u.



10-119 s metrics: RMS 0.668 a.u. | p99.9 |LFP| 3.34 a.u. | delta 89.1% | theta 4.5% | centroid 2.17 Hz

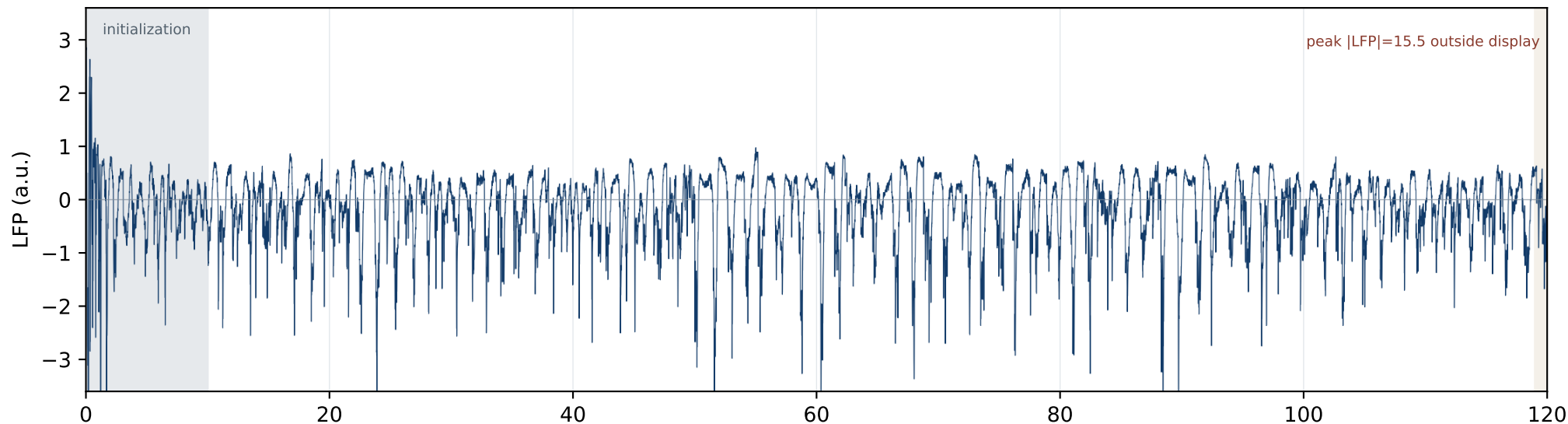
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state79_seed1_nhost=8.txt

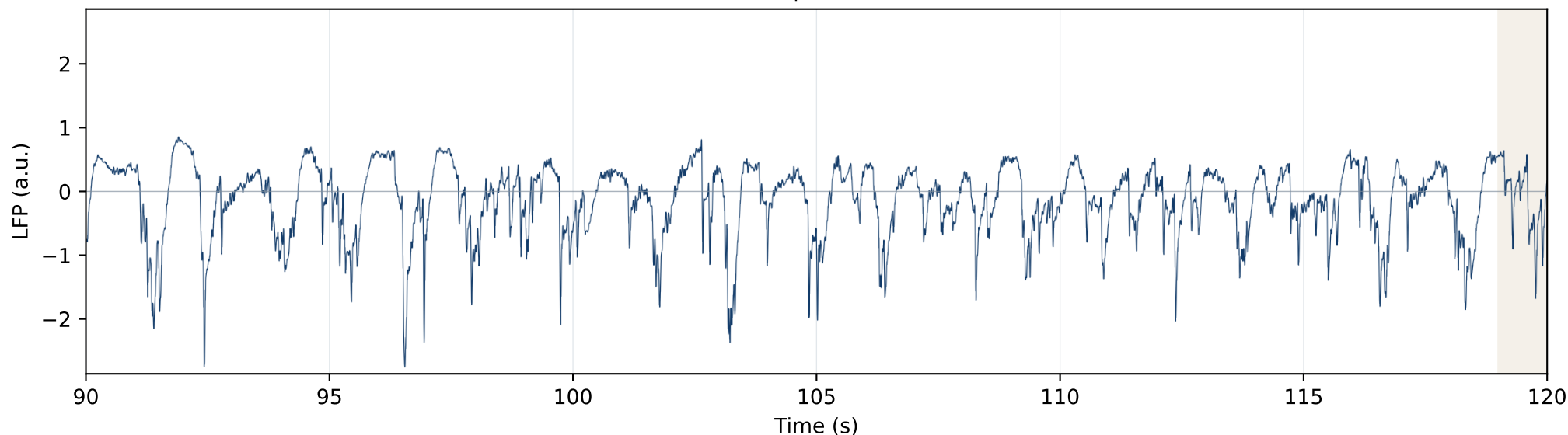
state_80

state 59 plus RE->TC GABA-A +10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: +/-3.6 a.u.



Final 30 seconds independent scale: +/-2.86 a.u.



10-119 s metrics: RMS 0.672 a.u. | p99.9 |LFP| 3.27 a.u. | delta 86.8% | theta 5.9% | centroid 2.40 Hz

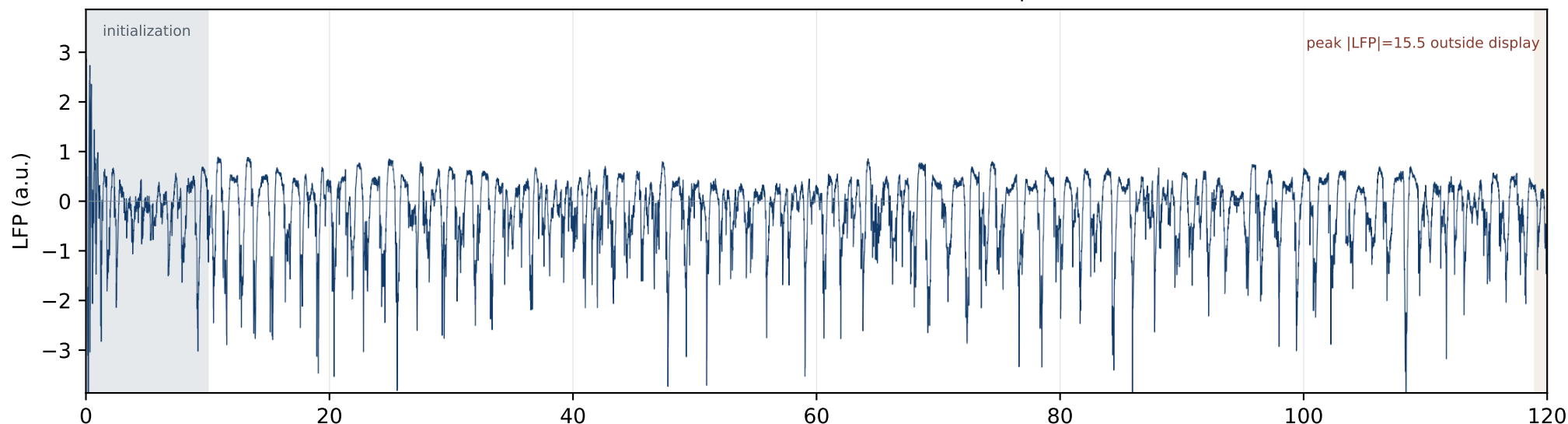
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state80_seed1_nhost=8.txt

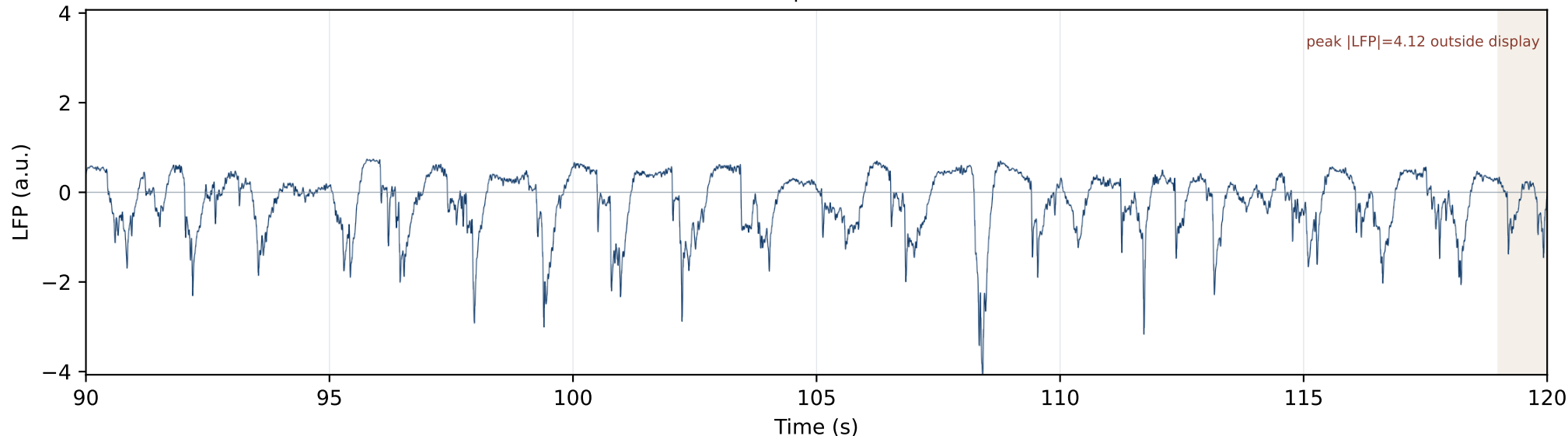
state_82

state 59 plus RE->TC GABA-B +10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.86 a.u.



Final 30 seconds independent scale: ± 4.07 a.u.



10-119 s metrics: RMS 0.727 a.u. | p99.9 |LFP| 3.51 a.u. | delta 88.5% | theta 5.6% | centroid 2.19 Hz

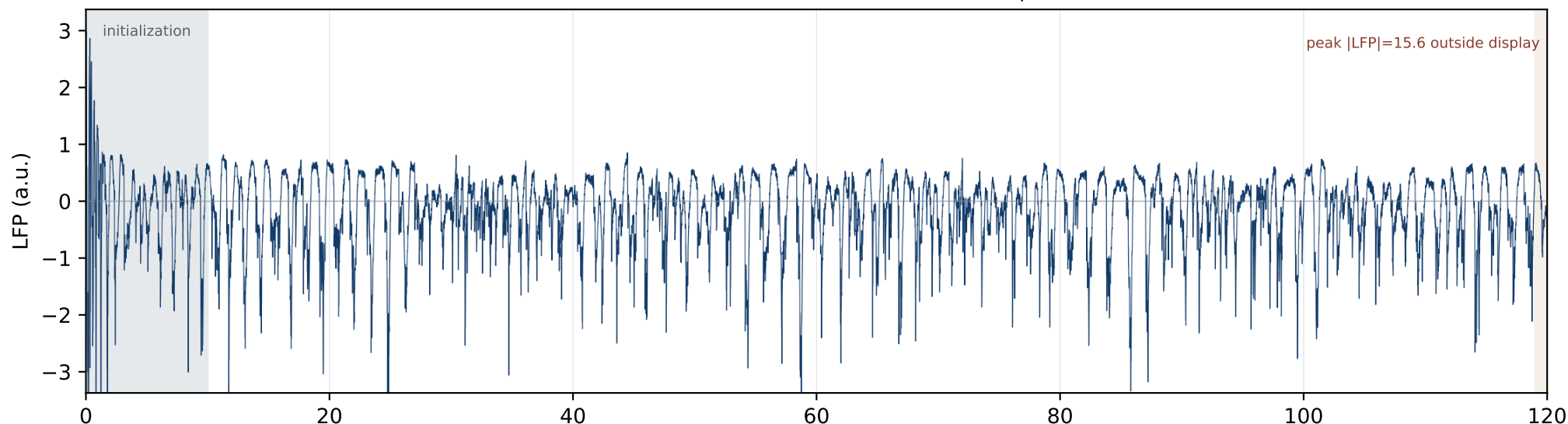
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state82_seed1_nhost=8.txt

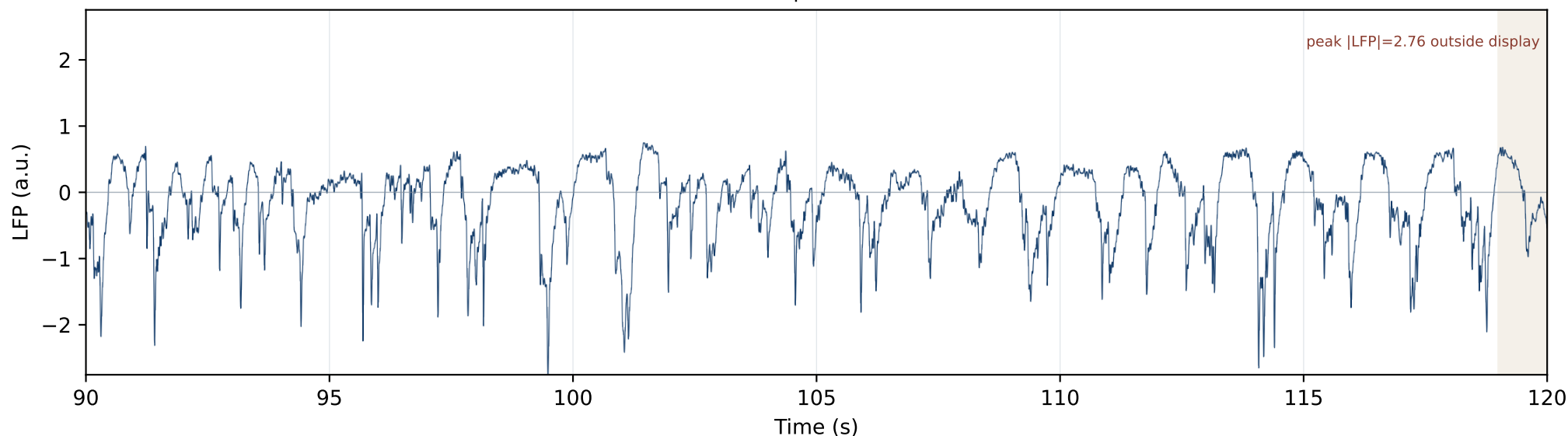
state_84

state 59 plus RE->RE GABA-A +10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.37 a.u.



Final 30 seconds independent scale: ± 2.75 a.u.



10-119 s metrics: RMS 0.64 a.u. | p99.9 |LFP| 3.07 a.u. | delta 86.9% | theta 5.5% | centroid 2.43 Hz

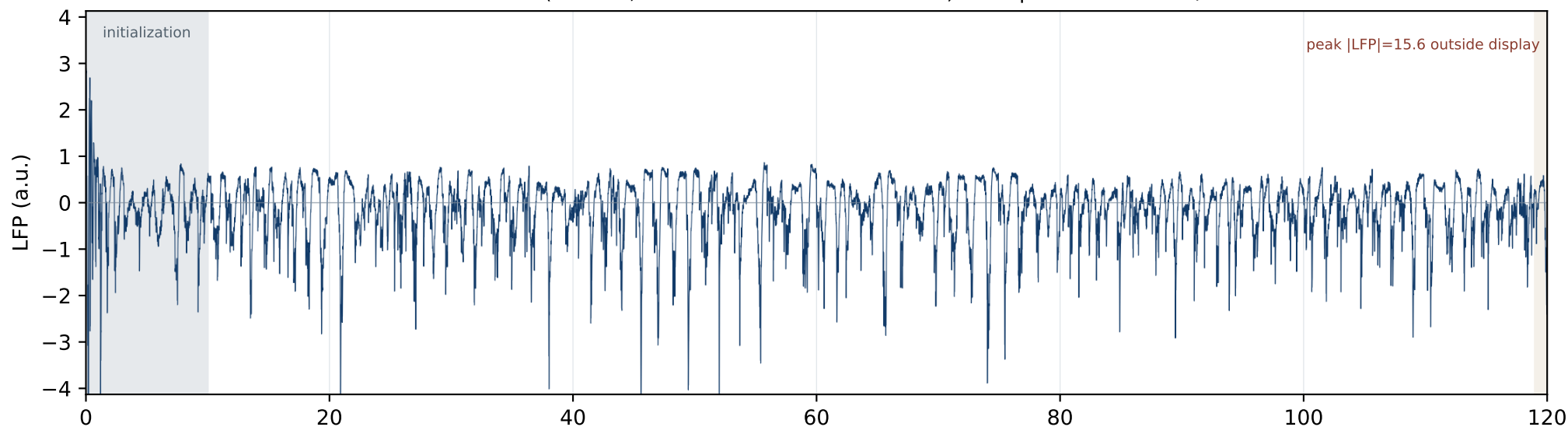
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state84_seed1_nhost=8.txt

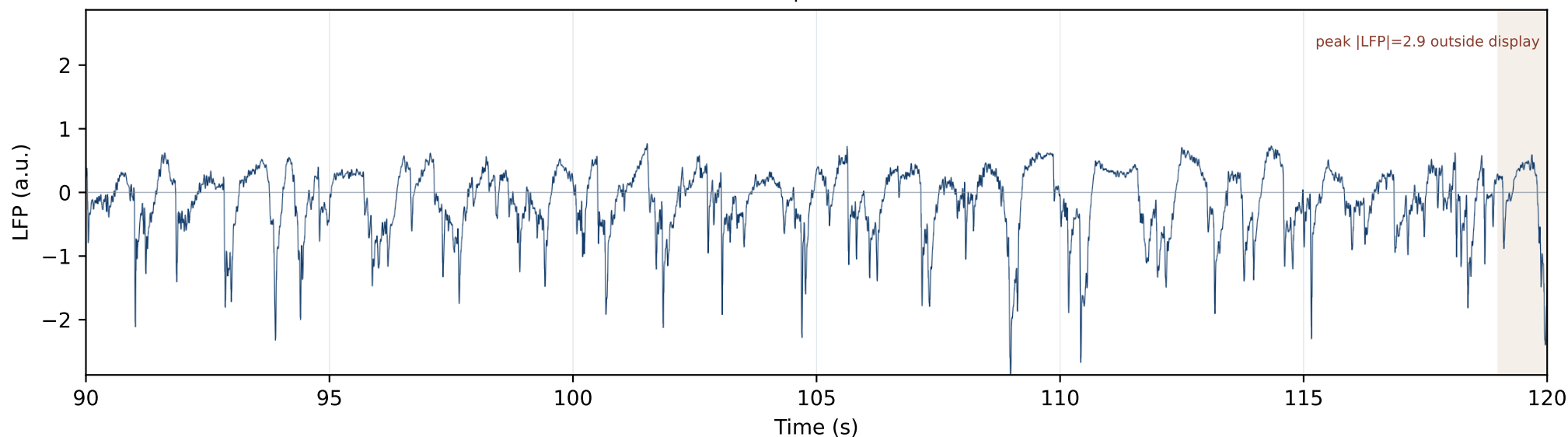
state_86

state 59 plus INH->PYR GABA-A-D2 +5%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 4.13 a.u.



Final 30 seconds independent scale: ± 2.87 a.u.



10-119 s metrics: RMS 0.639 a.u. | p99.9 |LFP| 3.75 a.u. | delta 85.9% | theta 6.2% | centroid 2.55 Hz

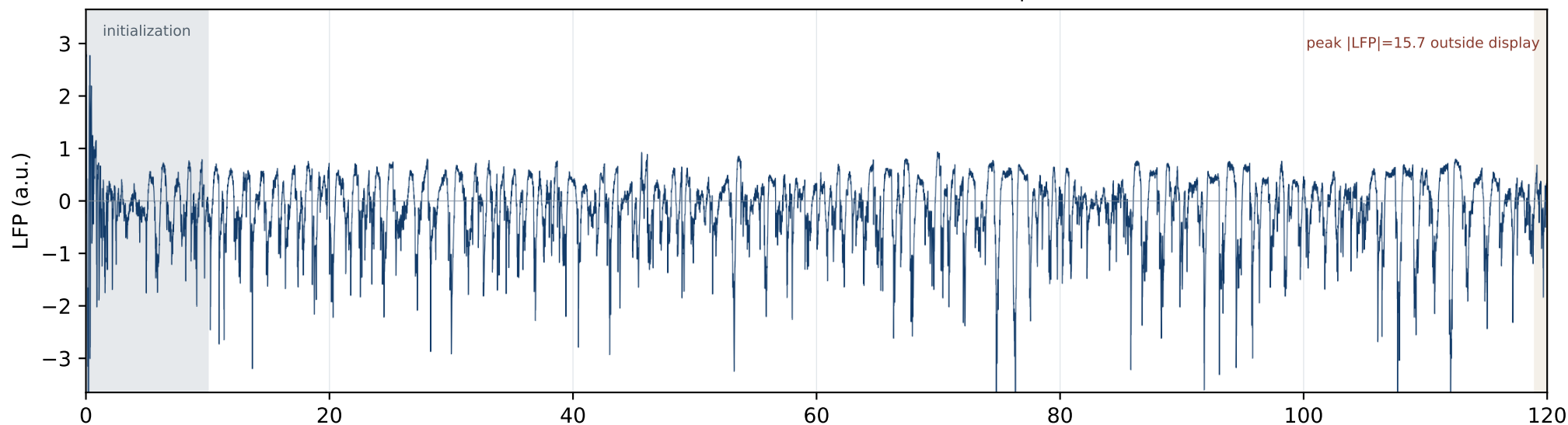
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state86_seed1_nhost=8.txt

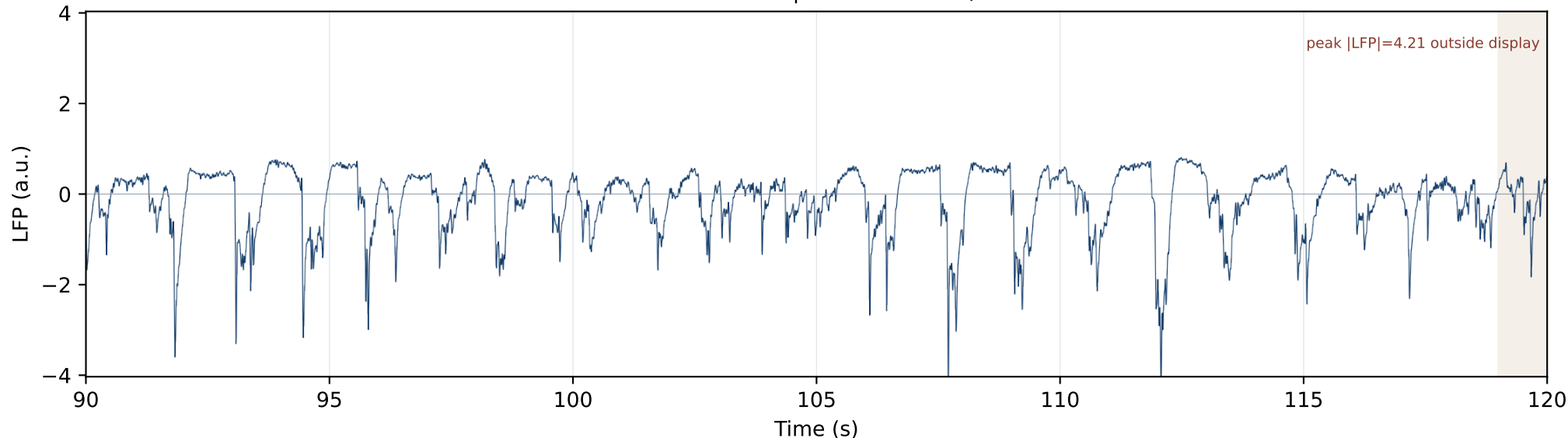
state_88

state 59 plus TC Ih maximal conductance -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.65 a.u.



Final 30 seconds independent scale: ± 4.03 a.u.



10-119 s metrics: RMS 0.655 a.u. | p99.9 |LFP| 3.32 a.u. | delta 86.0% | theta 6.2% | centroid 2.53 Hz

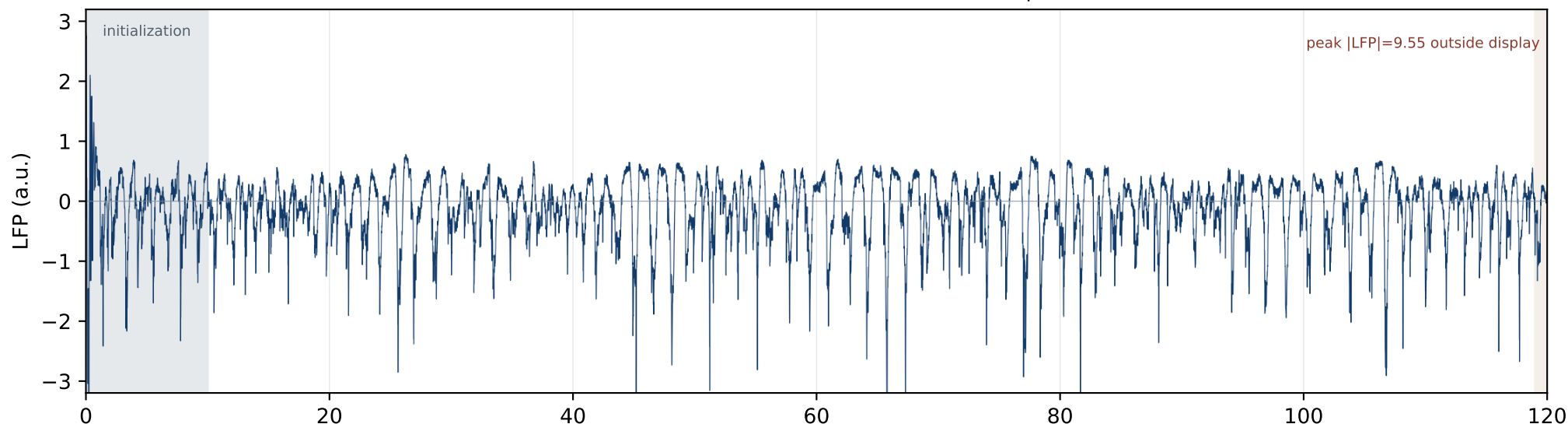
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state88_seed1_nhost=8.txt

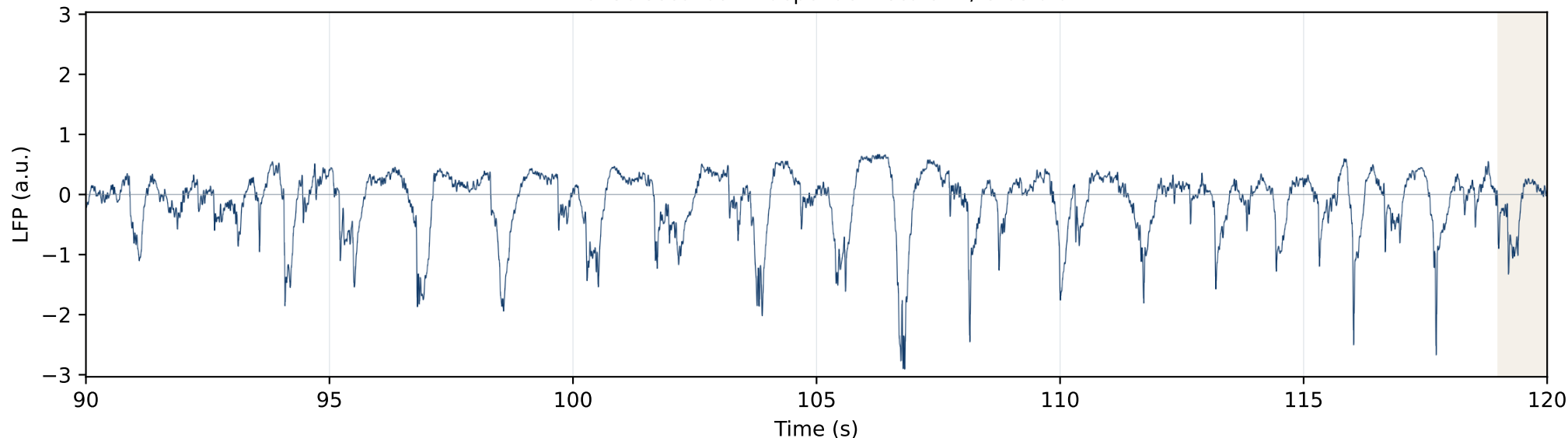
state_89

state 59 plus TC K leak +10% and TC Ih conductance -10%

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.2 a.u.



Final 30 seconds independent scale: ± 3.03 a.u.



10-119 s metrics: RMS 0.55 a.u. | p99.9 |LFP| 2.91 a.u. | delta 89.7% | theta 4.1% | centroid 2.19 Hz

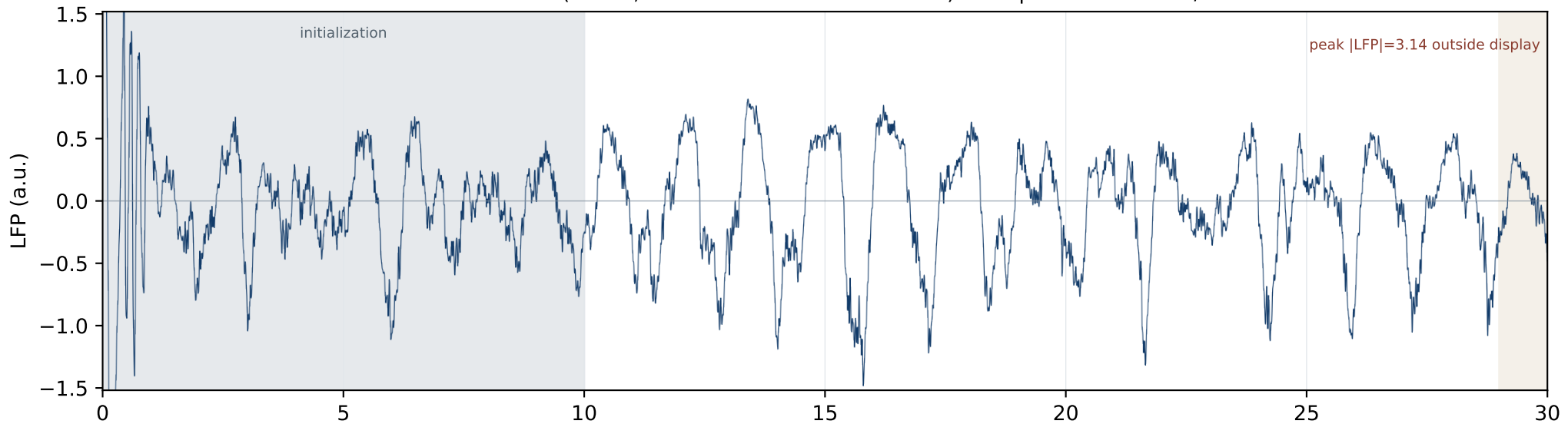
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state89_seed1_nhost=8.txt

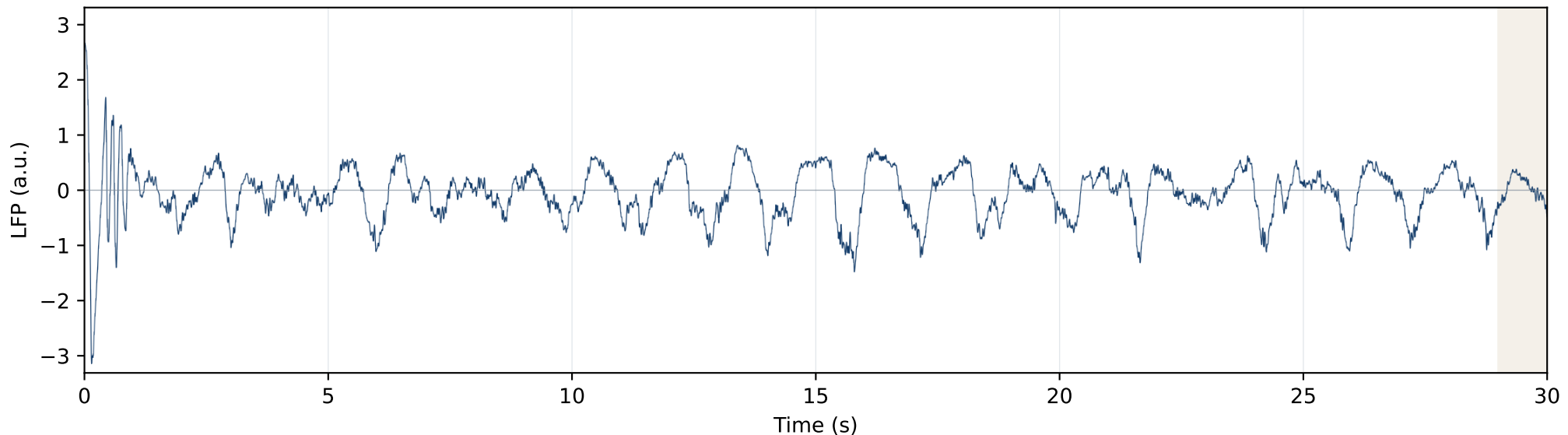
state_90

state 59 causal test: TC T-current off

Whole available run (0-30 s; 0-10 s excluded from metrics) independent scale: ± 1.52 a.u.



Final 30 seconds (same interval as the 30-s pilot above) independent scale: ± 3.31 a.u.



10-29 s metrics: RMS 0.46 a.u. | p99.9 |LFP| 1.38 a.u. | delta 96.8% | theta 1.4% | centroid 1.37 Hz

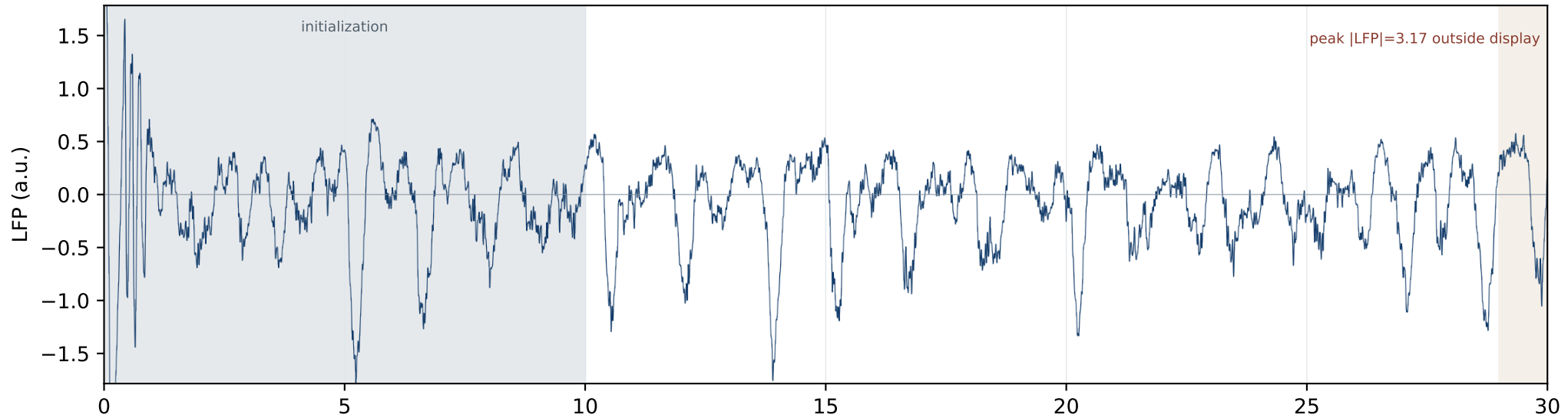
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state90_seed2_tclm0p3_r5000ms_nhost=8.txt

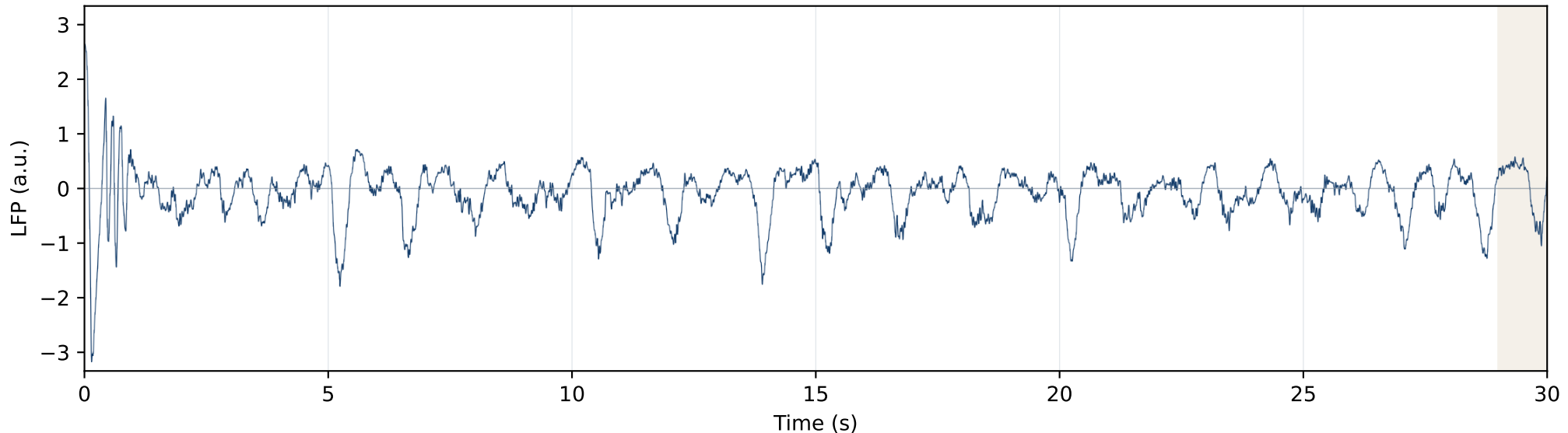
state_91

state 59 causal test: thalamus-to-cortex output off

Whole available run (0-30 s; 0-10 s excluded from metrics) independent scale: ± 1.78 a.u.



Final 30 seconds (same interval as the 30-s pilot above) independent scale: ± 3.34 a.u.



10-29 s metrics: RMS 0.404 a.u. | $p_{99.9} |LFP|$ 1.62 a.u. | Δ 95.6% | θ 1.7% | centroid 1.78 Hz

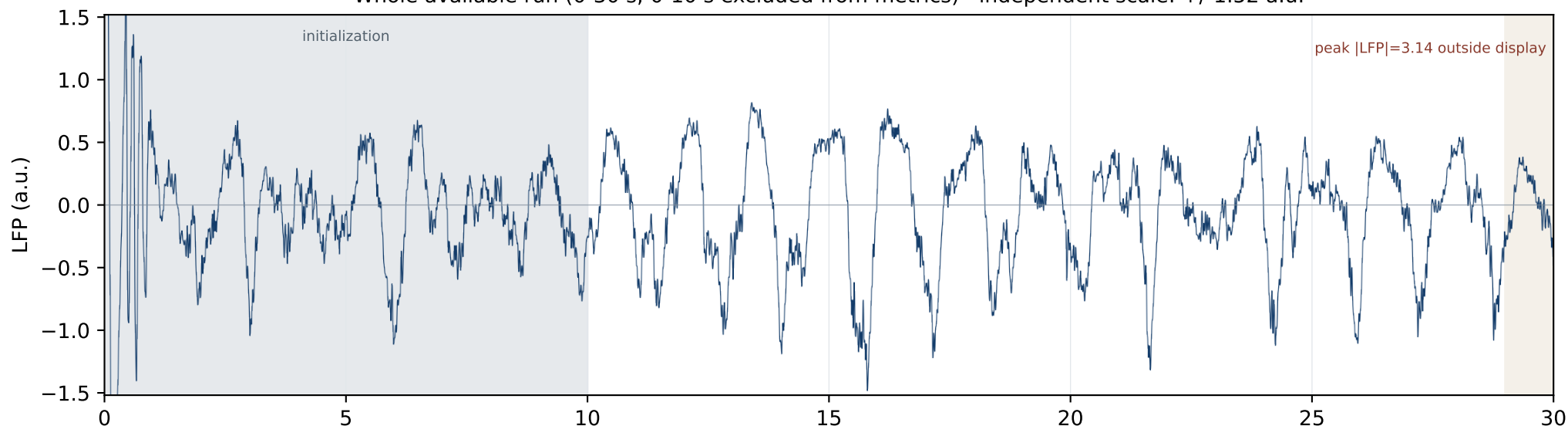
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state91_seed2_tclm0p3_r5000ms_nhost=8.txt

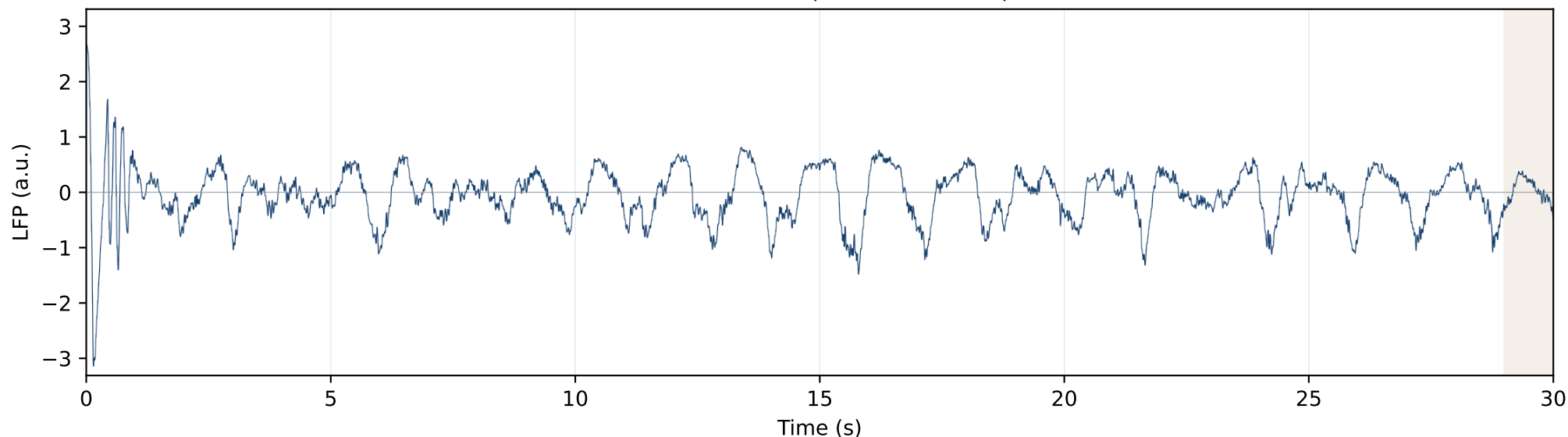
state_92

state 59 causal test: cortex-to-thalamus input off

Whole available run (0-30 s; 0-10 s excluded from metrics) independent scale: ± 1.52 a.u.



Final 30 seconds (same interval as the 30-s pilot above) independent scale: ± 3.31 a.u.



10-29 s metrics: RMS 0.46 a.u. | p99.9 |LFP| 1.38 a.u. | delta 96.8% | theta 1.4% | centroid 1.37 Hz

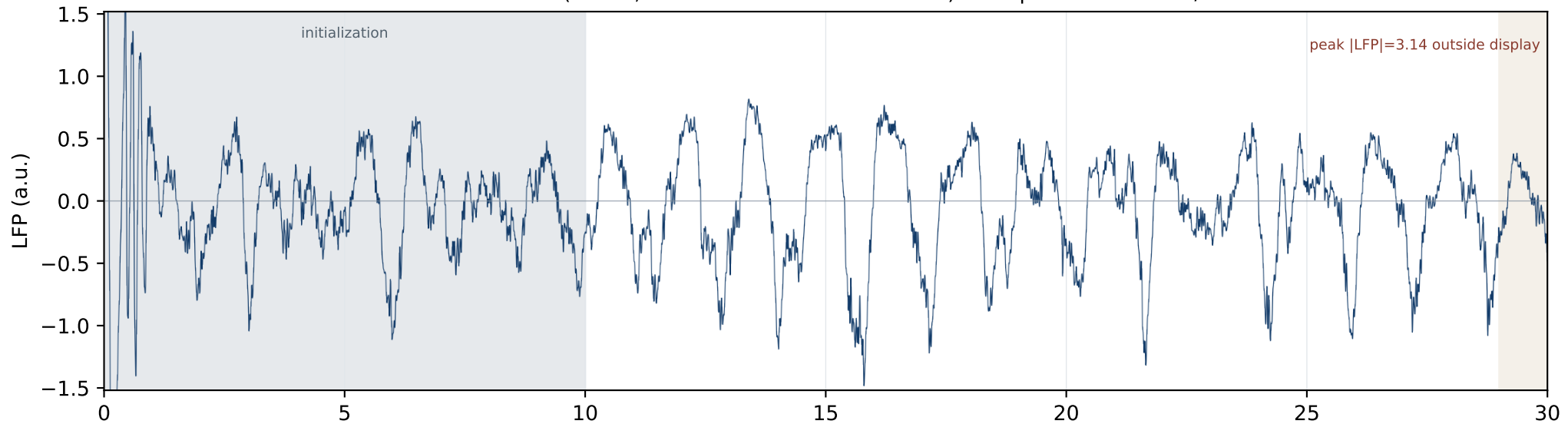
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state92_seed2_tclm0p3_r5000ms_nhost=8.txt

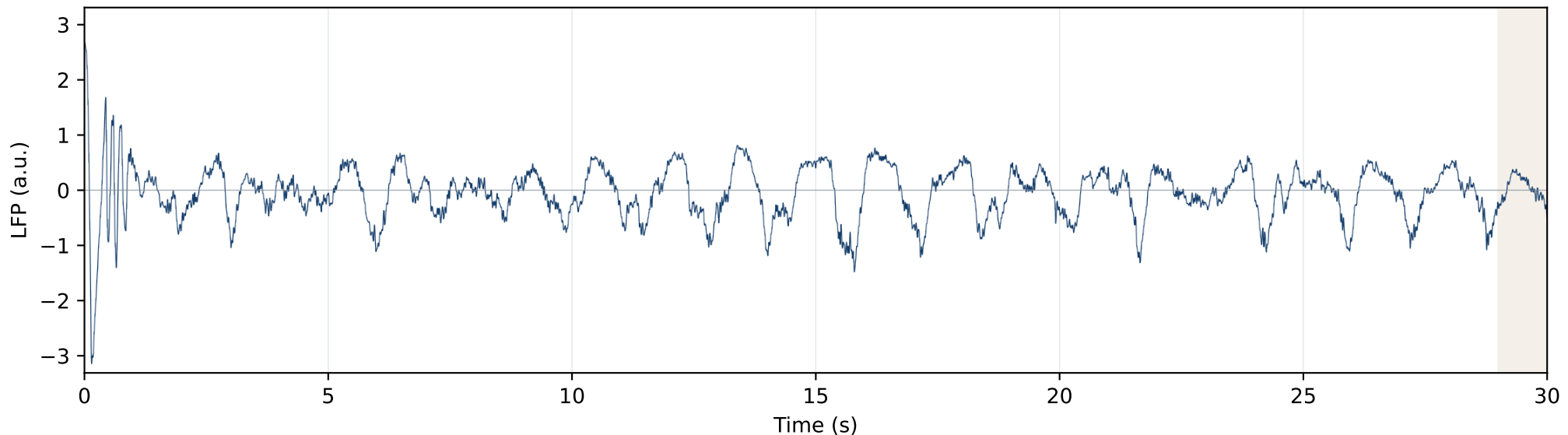
state_93

state 59 causal test: cortex and thalamus disconnected

Whole available run (0-30 s; 0-10 s excluded from metrics) independent scale: ± 1.52 a.u.



Final 30 seconds (same interval as the 30-s pilot above) independent scale: ± 3.31 a.u.



10-29 s metrics: RMS 0.46 a.u. | p99.9 $|LFP|$ 1.38 a.u. | delta 96.8% | theta 1.4% | centroid 1.37 Hz

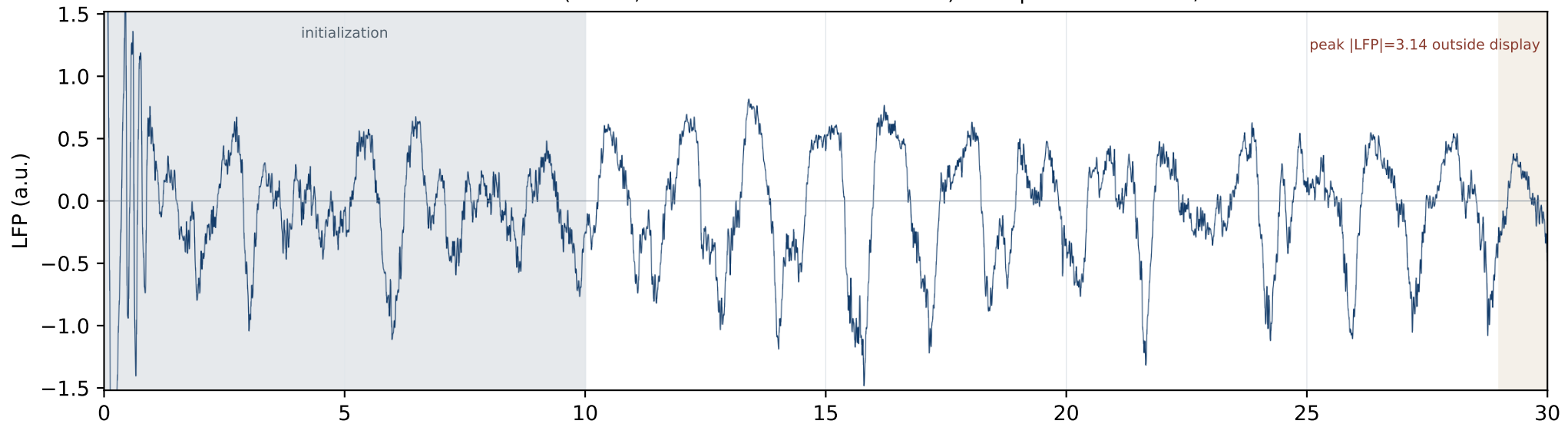
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state93_seed2_tclm0p3_r5000ms_nhost=8.txt

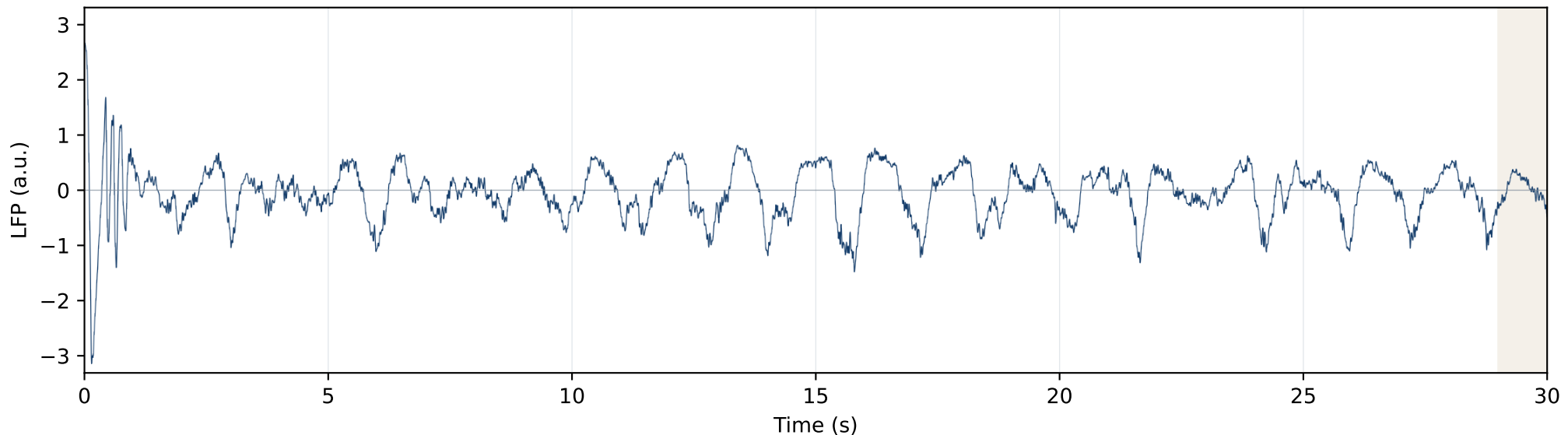
state_94

state 91 causal test: direct cortex-to-TC route only

Whole available run (0-30 s; 0-10 s excluded from metrics) independent scale: ± 1.52 a.u.



Final 30 seconds (same interval as the 30-s pilot above) independent scale: ± 3.31 a.u.



10-29 s metrics: RMS 0.46 a.u. | p99.9 |LFP| 1.38 a.u. | delta 96.8% | theta 1.4% | centroid 1.37 Hz

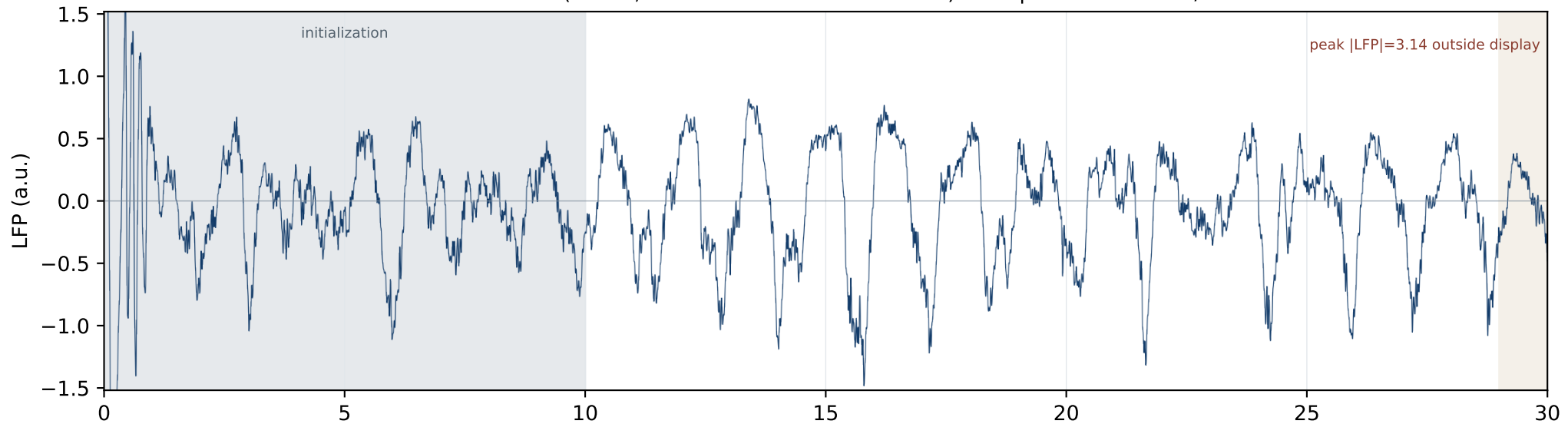
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state94_seed2_tclm0p3_r5000ms_nhost=8.txt

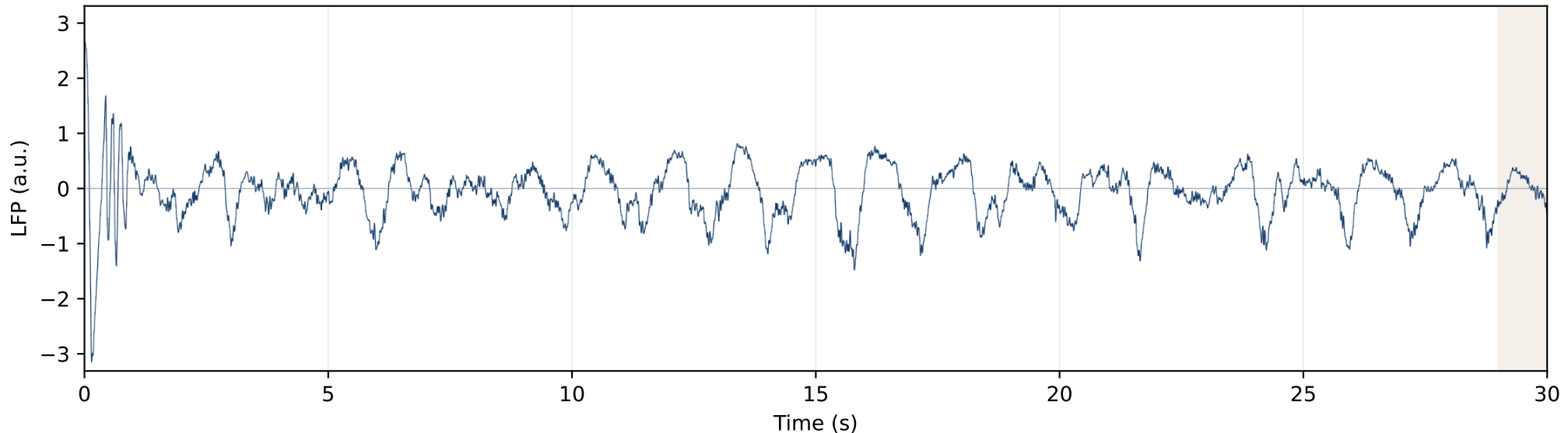
state_95

state 91 causal test: indirect cortex-to-RE route only

Whole available run (0-30 s; 0-10 s excluded from metrics) independent scale: ± 1.52 a.u.



Final 30 seconds (same interval as the 30-s pilot above) independent scale: ± 3.31 a.u.



10-29 s metrics: RMS 0.46 a.u. | p99.9 |LFP| 1.38 a.u. | delta 96.8% | theta 1.4% | centroid 1.37 Hz

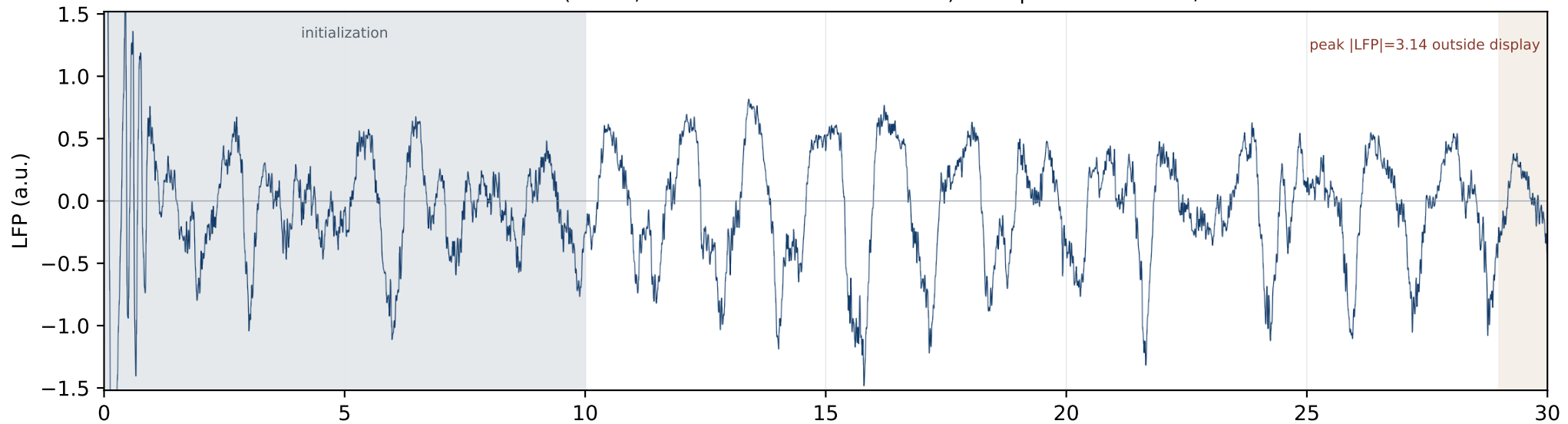
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state95_seed2_tclm0p3_r5000ms_nhost=8.txt

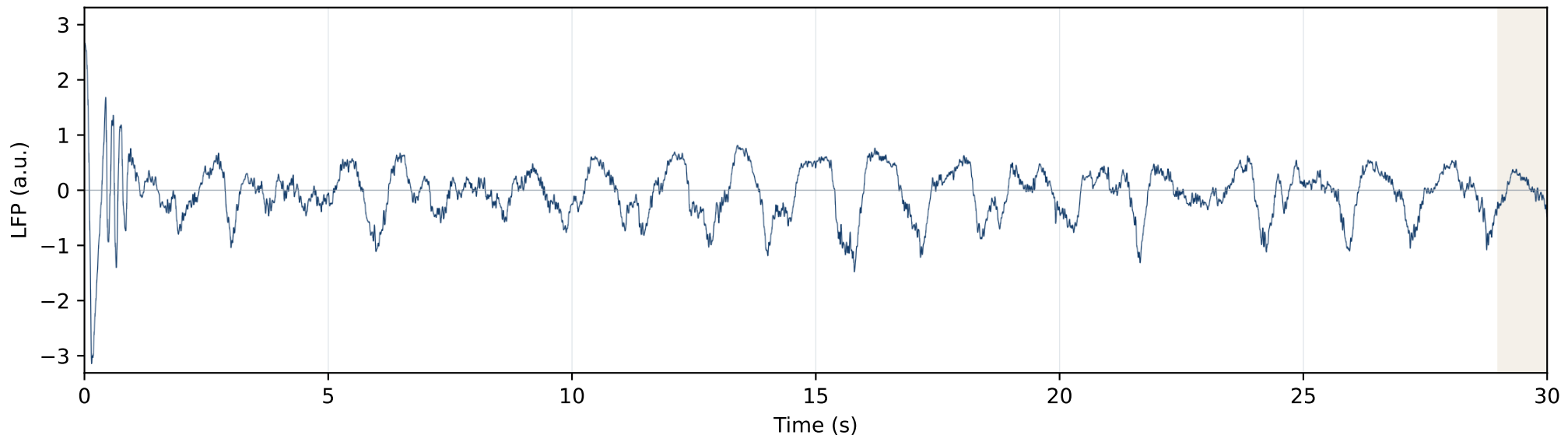
state_96

state 94 causal test: TC-to-RE feedback arm off

Whole available run (0-30 s; 0-10 s excluded from metrics) independent scale: ± 1.52 a.u.



Final 30 seconds (same interval as the 30-s pilot above) independent scale: ± 3.31 a.u.



10-29 s metrics: RMS 0.46 a.u. | p99.9 |LFP| 1.38 a.u. | delta 96.8% | theta 1.4% | centroid 1.37 Hz

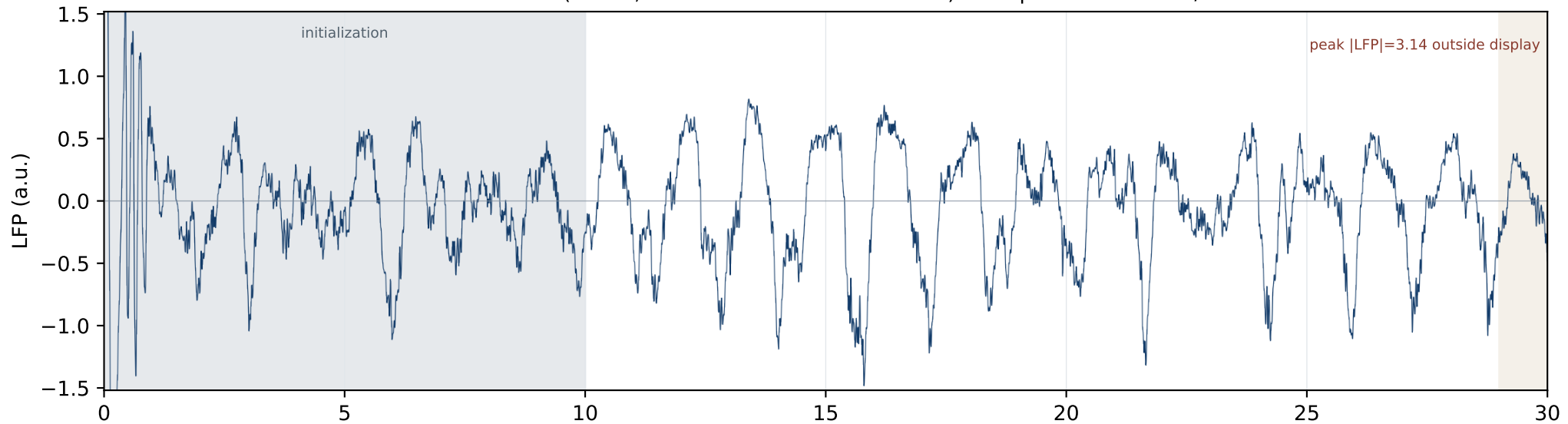
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state96_seed2_tclm0p3_r5000ms_nhost=8.txt

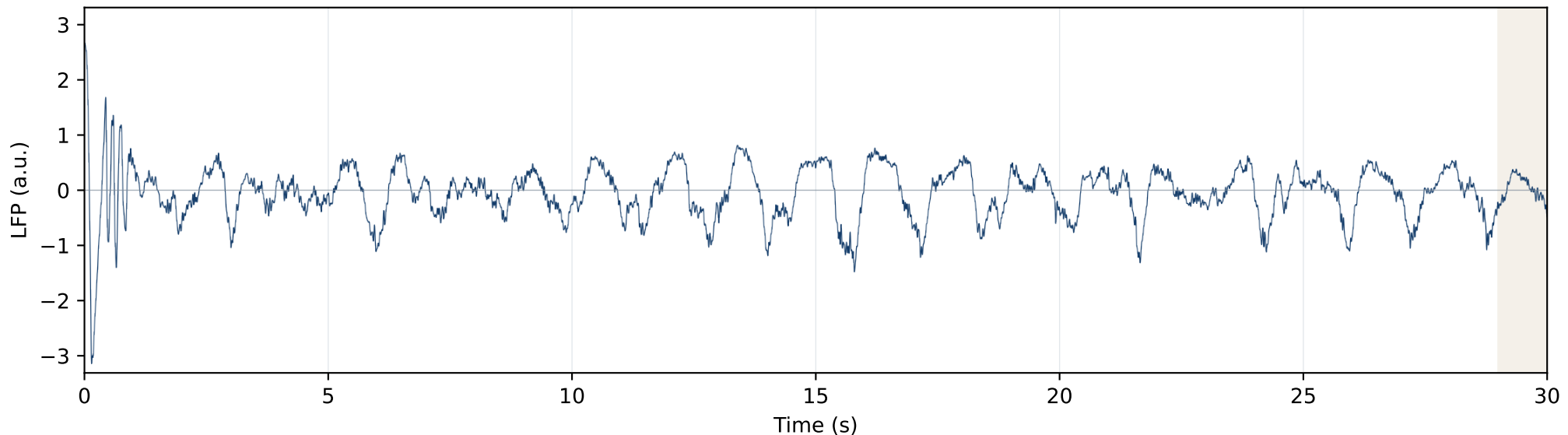
state_97

state 94 causal test: RE-to-TC GABA-A/B return arm off

Whole available run (0-30 s; 0-10 s excluded from metrics) independent scale: ± 1.52 a.u.



Final 30 seconds (same interval as the 30-s pilot above) independent scale: ± 3.31 a.u.



10-29 s metrics: RMS 0.46 a.u. | p99.9 |LFP| 1.38 a.u. | delta 96.8% | theta 1.4% | centroid 1.37 Hz

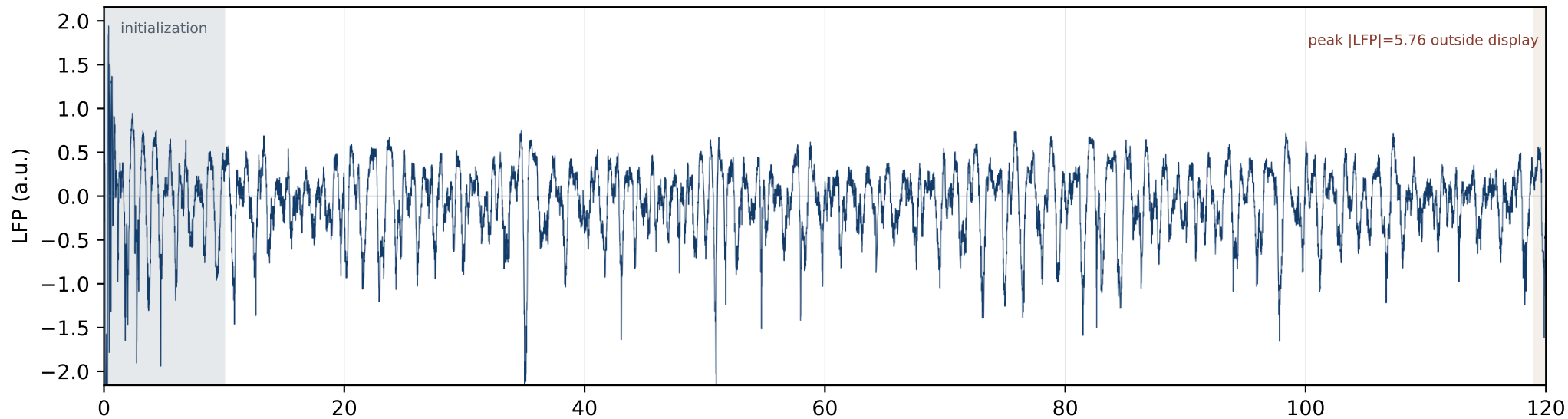
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state97_seed2_tclm0p3_r5000ms_nhost=8.txt

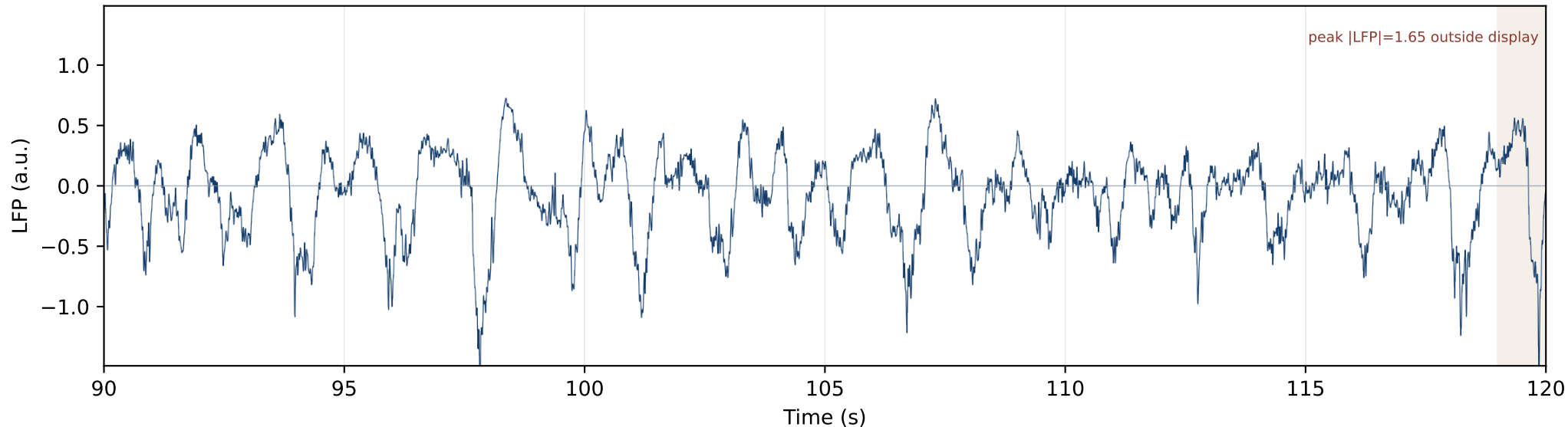
state_98

state 59 loop screen: PYR-to-TC AMPA x0.75

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: +/-2.16 a.u.



Final 30 seconds independent scale: +/-1.49 a.u.



10-119 s metrics: RMS 0.391 a.u. | p99.9 |LFP| 1.96 a.u. | delta 95.3% | theta 1.5% | centroid 1.69 Hz

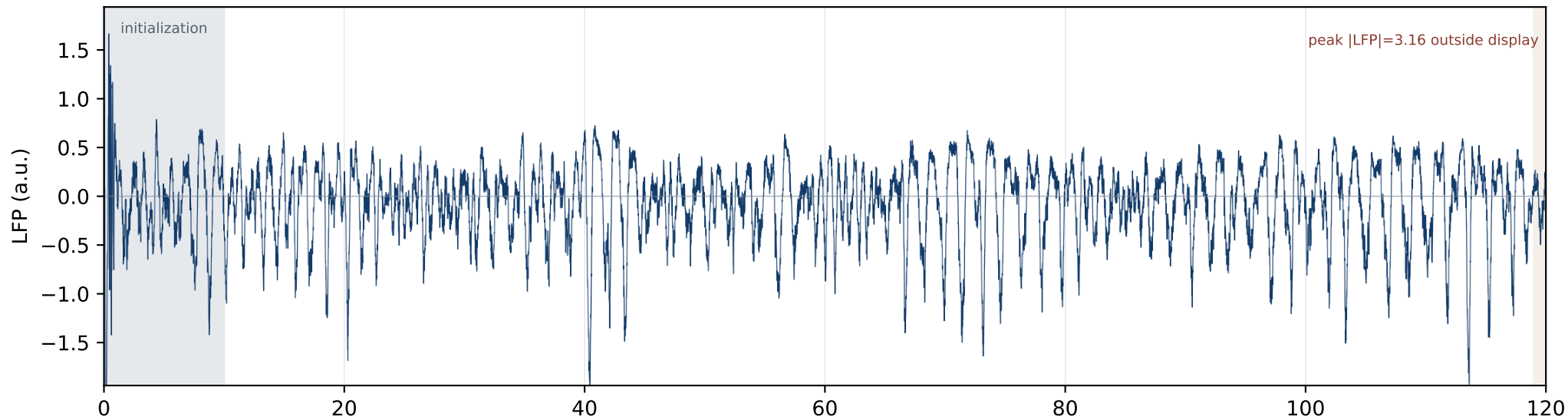
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state98_seed2_tclm0p3_r5000ms_nhost=8.txt

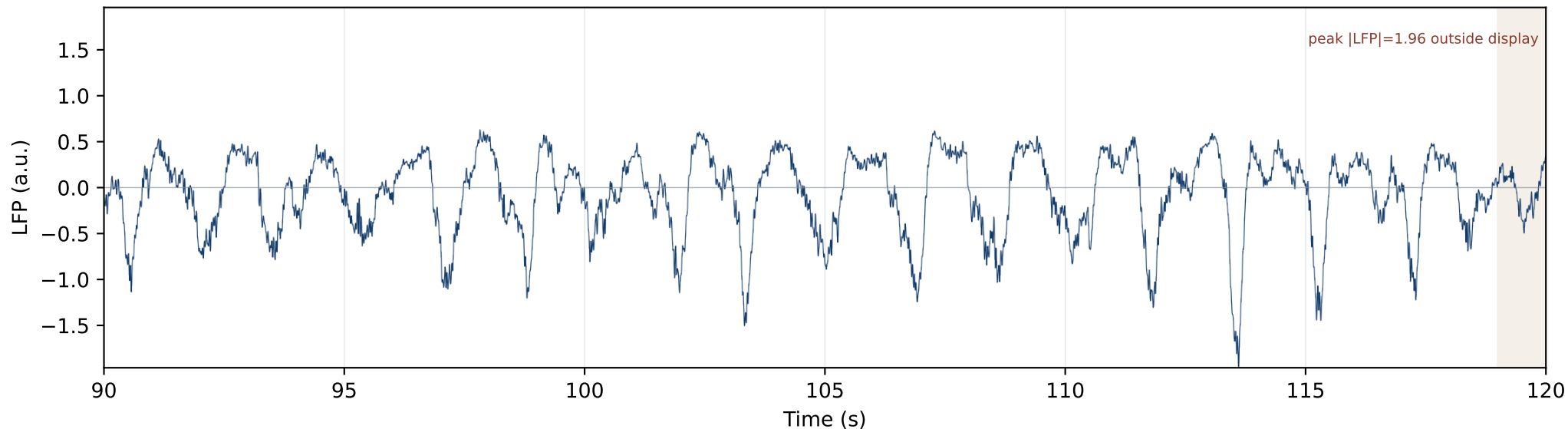
state_99

state 59 loop screen: PYR-to-TC AMPA x0.50

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 1.94 a.u.



Final 30 seconds independent scale: ± 1.96 a.u.



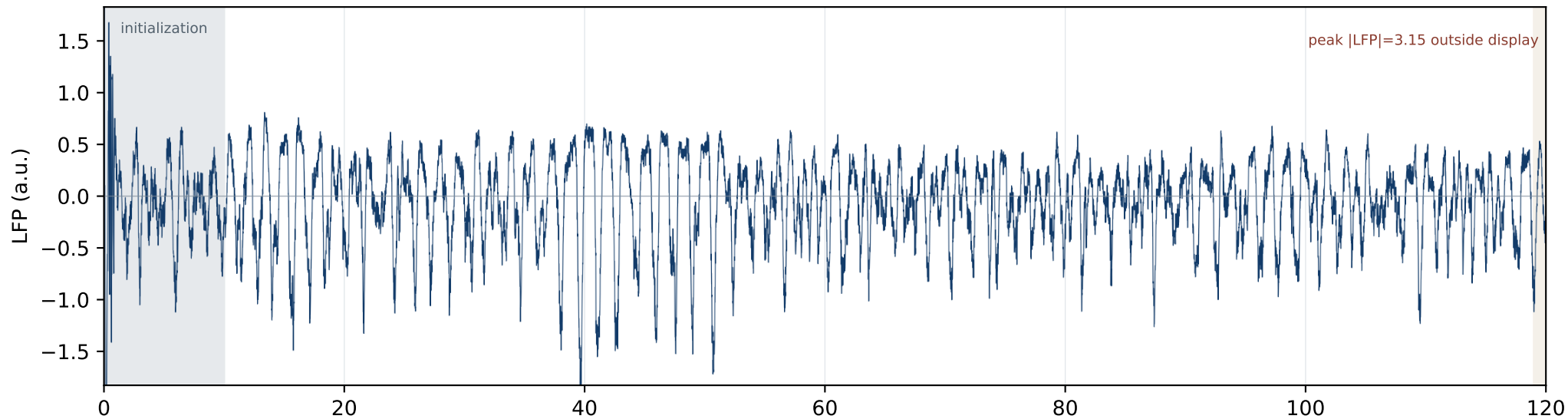
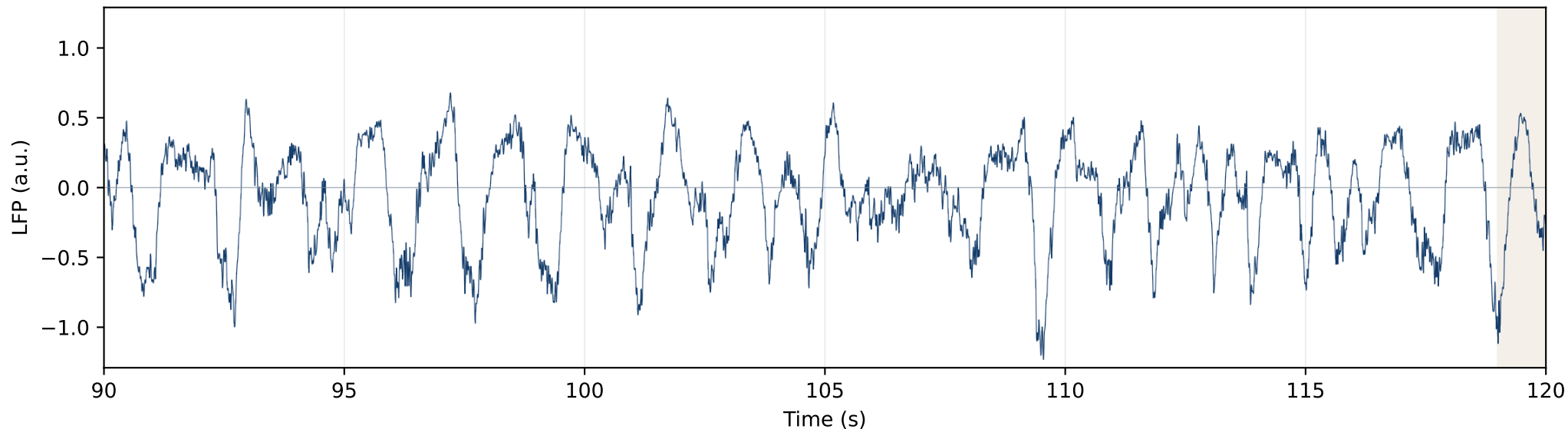
10-119 s metrics: RMS 0.405 a.u. | p99.9 |LFP| 1.76 a.u. | delta 96.2% | theta 1.3% | centroid 1.56 Hz

Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state99_seed2_tclm0p3_r5000ms_nhost=8.txt

state_100

state 59 loop screen: PYR-to-TC AMPA x0.25

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 1.83 a.u.Final 30 seconds independent scale: ± 1.29 a.u.

10-119 s metrics: RMS 0.421 a.u. | p99.9 |LFP| 1.66 a.u. | delta 96.6% | theta 1.2% | centroid 1.47 Hz

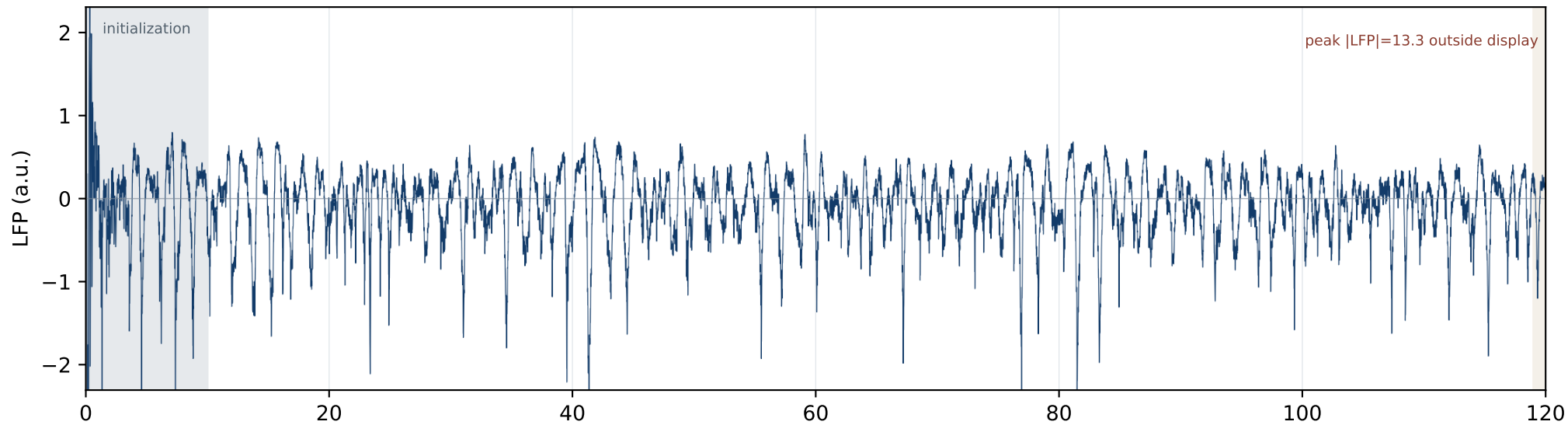
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state100_seed2_tclm0p3_r5000ms_nhost=8.txt

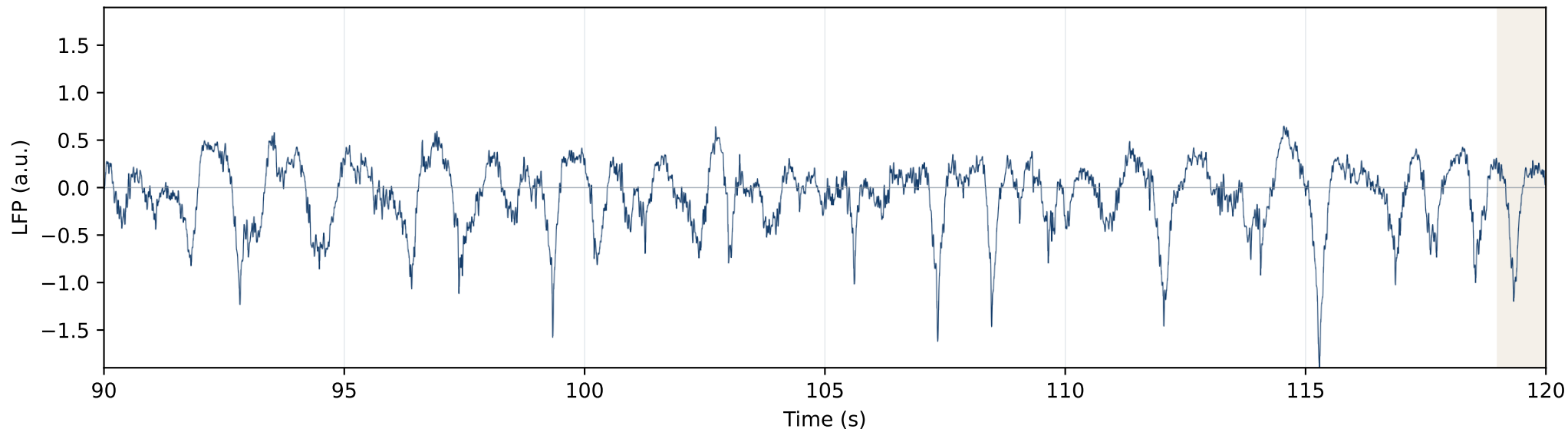
state_101

state 59 output screen: TC-to-PYR/INH AMPA x0.75

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 2.31 a.u.



Final 30 seconds independent scale: ± 1.9 a.u.



10-119 s metrics: RMS 0.41 a.u. | p99.9 |LFP| 2.1 a.u. | delta 94.0% | theta 2.3% | centroid 1.81 Hz

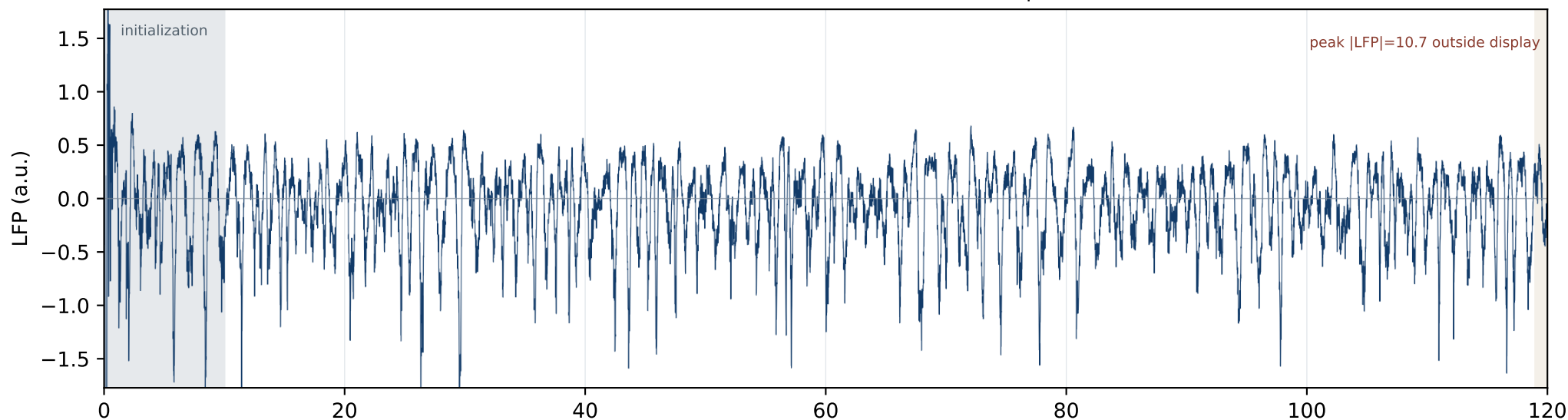
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state101_seed2_tclm0p3_r5000ms_nhost=8.txt

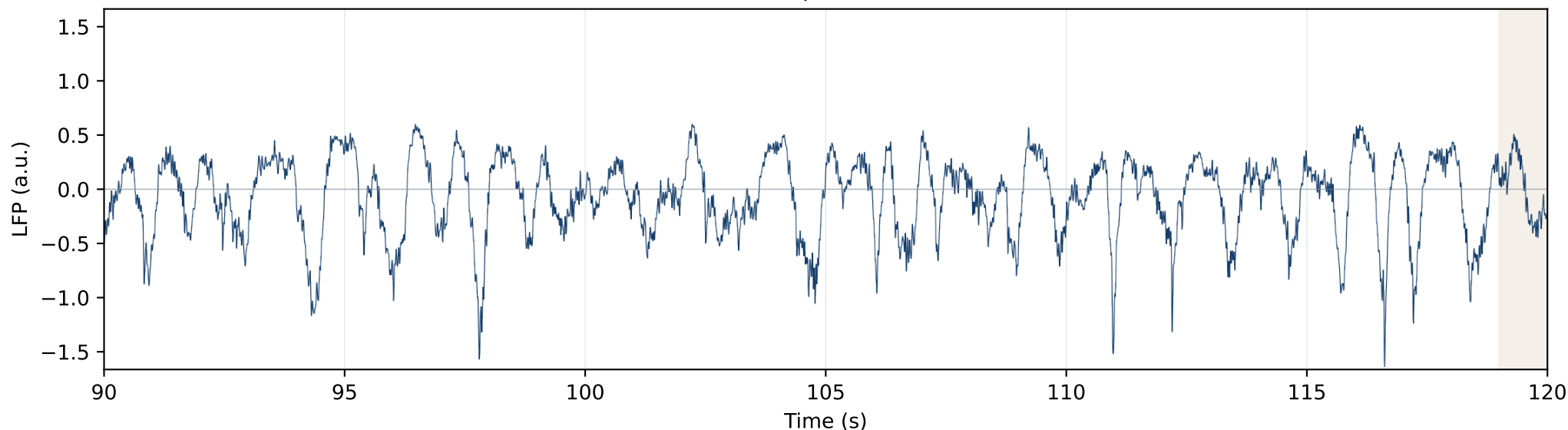
state_102

state 59 output screen: TC-to-PYR/INH AMPA x0.50

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 1.77 a.u.



Final 30 seconds independent scale: ± 1.66 a.u.



10-119 s metrics: RMS 0.381 a.u. | p99.9 |LFP| 1.61 a.u. | delta 94.1% | theta 2.1% | centroid 1.89 Hz

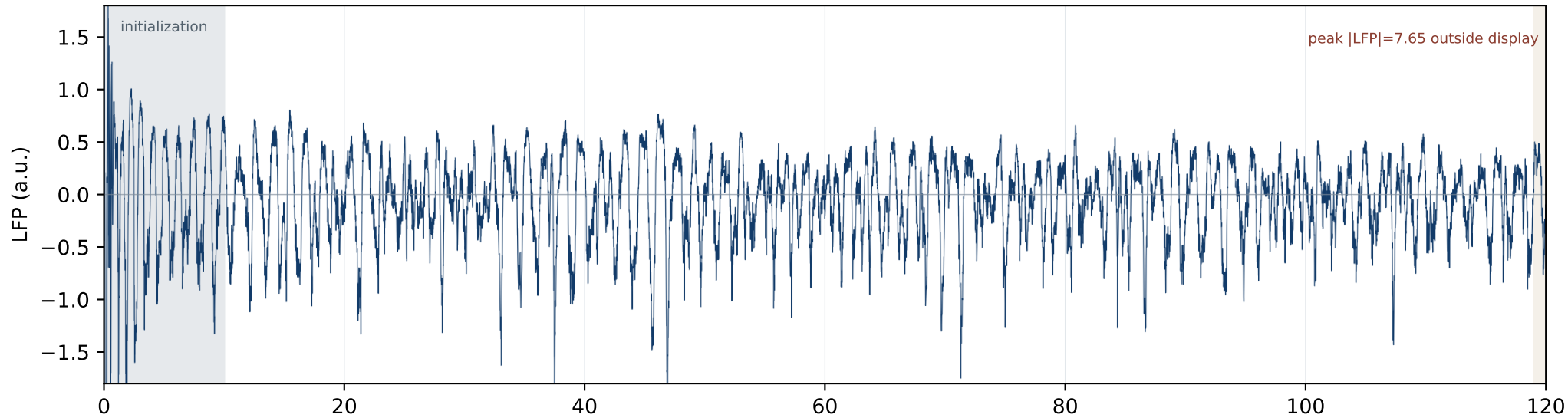
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state102_seed2_tclm0p3_r5000ms_nhost=8.txt

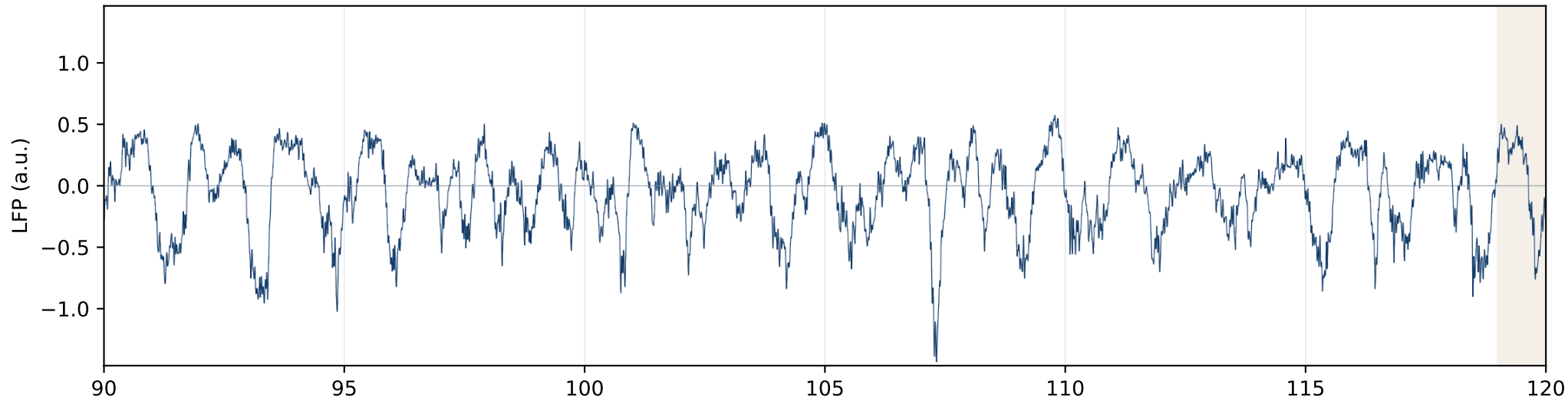
state_103

state 59 output screen: TC-to-PYR/INH AMPA x0.25

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: +/-1.8 a.u.



Final 30 seconds independent scale: +/-1.46 a.u.



10-119 s metrics: RMS 0.389 a.u. | p99.9 |LFP| 1.64 a.u. | delta 95.4% | theta 1.6% | centroid 1.67 Hz

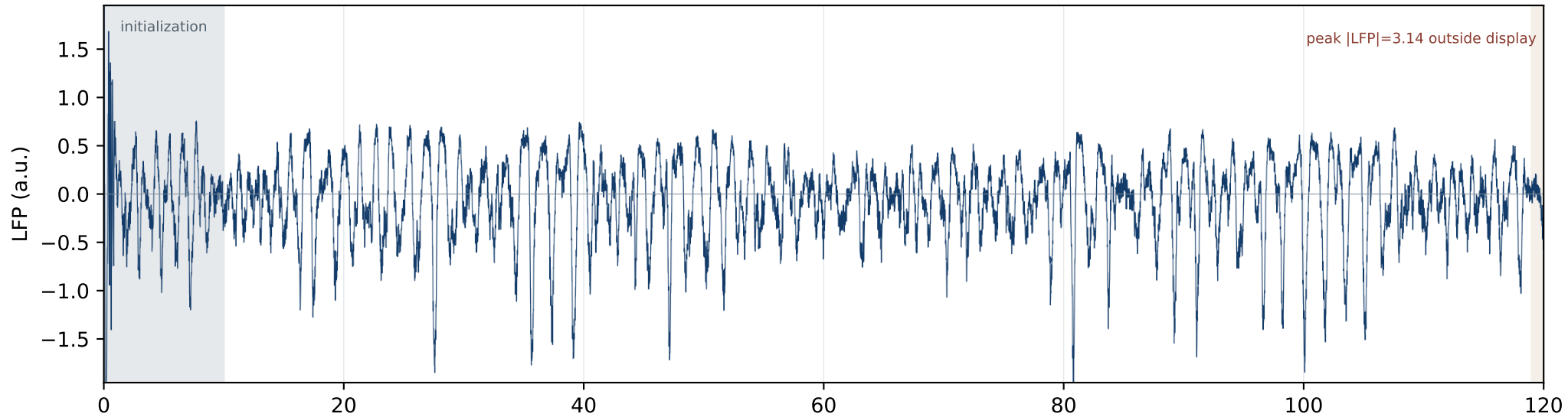
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Source: lfp_state103_seed2_tclm0p3_r5000ms_nhost=8.txt

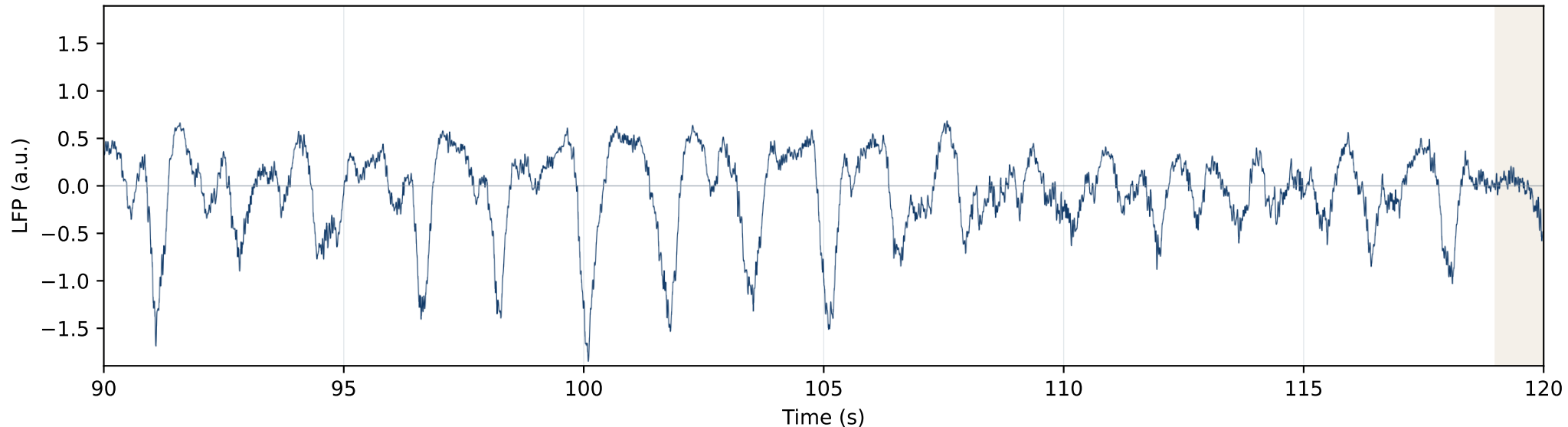
state_104

state 59 interaction: PYR-to-TC and TC-to-cortex x0.50

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 1.95 a.u.



Final 30 seconds independent scale: ± 1.9 a.u.



10-119 s metrics: RMS 0.42 a.u. | p99.9 |LFP| 1.77 a.u. | delta 96.7% | theta 1.1% | centroid 1.48 Hz

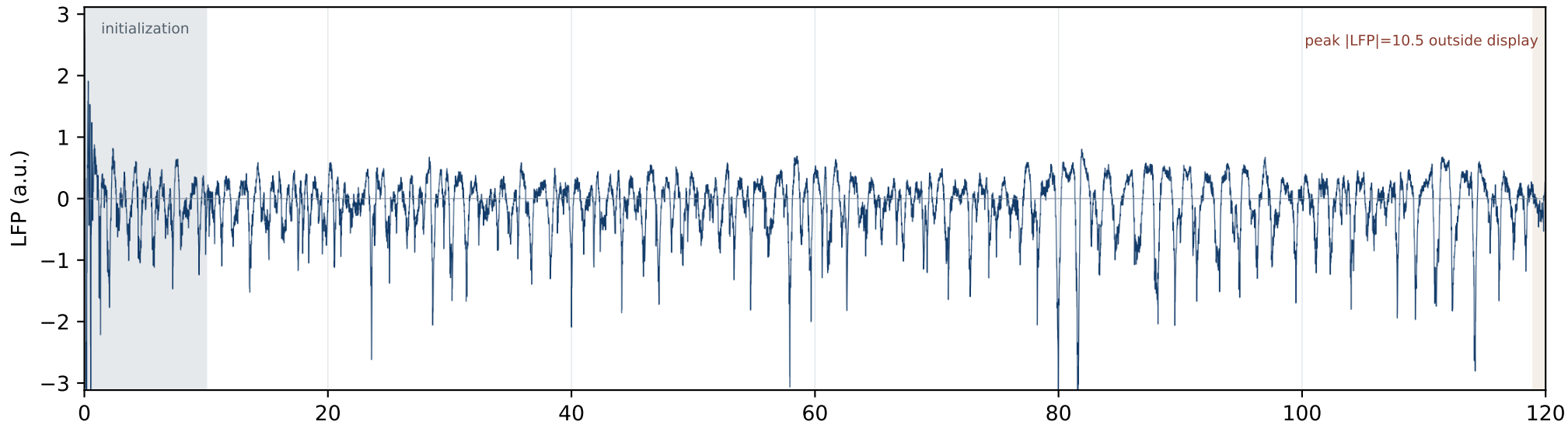
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state104_seed2_tclm0p3_r5000ms_nhost=8.txt

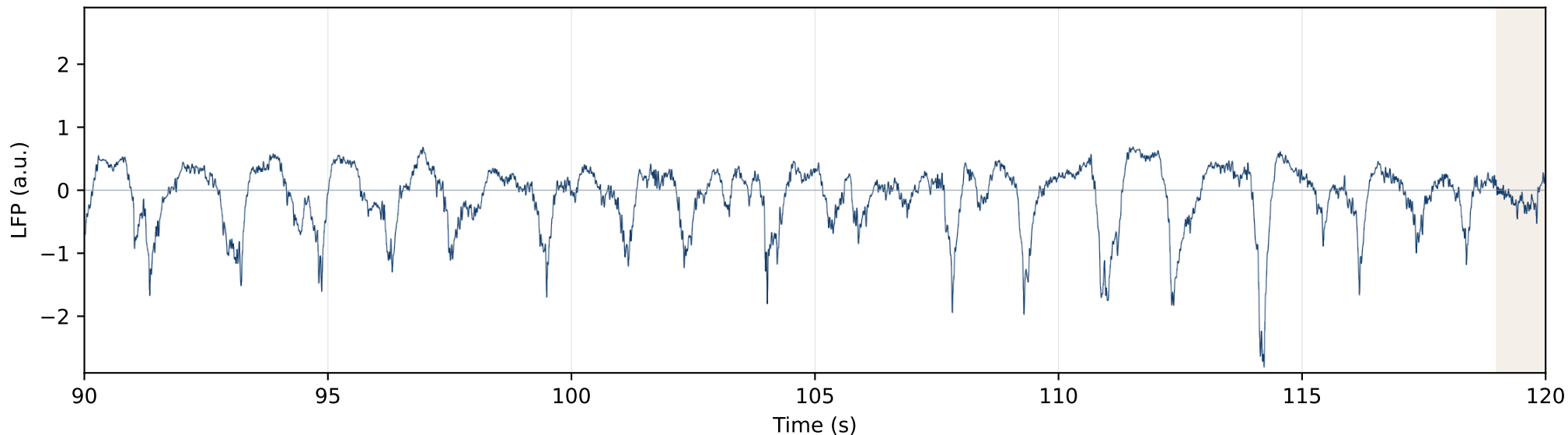
state_105

state 59 balance: PYR-to-TC x1.25, TC-to-cortex x0.50

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 3.11 a.u.



Final 30 seconds independent scale: ± 2.9 a.u.



10-119 s metrics: RMS 0.477 a.u. | p99.9 |LFP| 2.83 a.u. | delta 93.7% | theta 2.7% | centroid 1.85 Hz

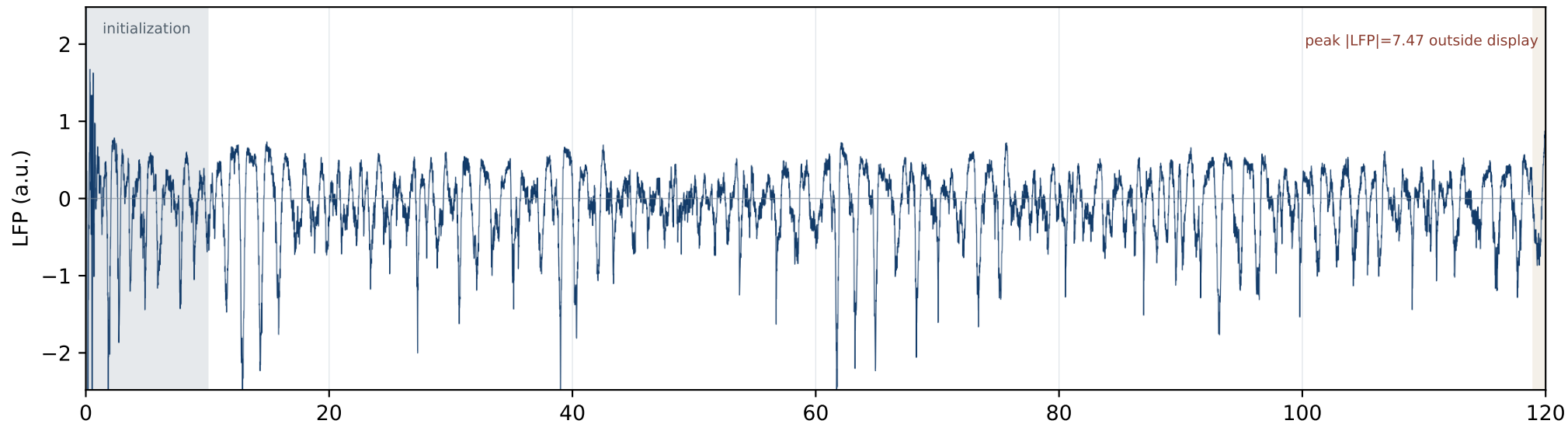
Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state105_seed2_tclm0p3_r5000ms_nhost=8.txt

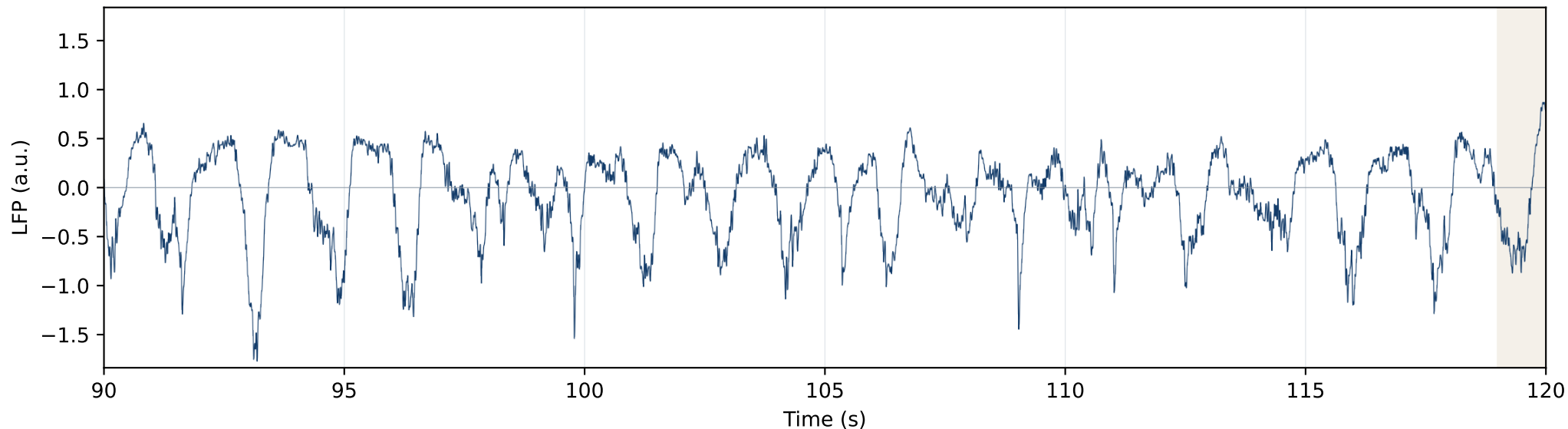
state_106

state 59 balance: PYR-to-TC x1.25, TC-to-cortex x0.25

Whole available run (0-120 s; 0-10 s excluded from metrics) independent scale: ± 2.48 a.u.



Final 30 seconds independent scale: ± 1.84 a.u.



10-119 s metrics: RMS 0.45 a.u. | p99.9 |LFP| 2.25 a.u. | delta 95.4% | theta 1.6% | centroid 1.64 Hz

Processing: median center; polyphase anti-alias 40 kHz to 200 Hz; zero-phase 0.5-30 Hz.

Panels use independent y-scales; compare amplitude using printed metrics. Tan shading marks the final filter-edge caution interval. Source: lfp_state106_seed2_tclm0p3_r5000ms_nhost=8.txt